

Adopted Levels, Gammas

$Q(\beta^-) = -5966.2$ 7; $S(n) = 12644.76$ 5; $S(p) = 6370.81$ 4; $Q(\alpha) = -6997.90$ 4 [2021Wa16](#)

$S(2n) = 24152.8$ 4, $S(2p) = 17254.1$ 11 ([2021Wa16](#)).

Isotope discovery ([2012Th10](#)): positive-ray mass spectrograph at Cambridge, UK ([1919As01](#)).

 ^{35}Cl Levels**Cross Reference (XREF) Flags**

A	^{35}S β^- decay (87.61 d)	K	^{31}P ($^6\text{Li}, d$)	U	^{35}Cl ($p, p'\gamma$)
B	^{35}Ar ε decay (1.7755 s)	L	^{32}S (α, p)	V	^{36}Ar (n, d)
C	^{36}K εp decay (341 ms)	M	^{32}S ($\alpha, p\gamma$)	W	^{36}Ar ($d, ^3\text{He}$)
D	^1H ($^{34}\text{S}, ^{35}\text{Cl}\gamma$)	N	^{33}S (α, d)	X	^{36}Ar ($\text{pol } d, ^3\text{He}$)
E	^2H ($^{34}\text{S}, n\gamma$)	O	^{34}S (p, γ)	Y	^{37}Cl (p, t)
F	^{12}C ($^{28}\text{Si}, \alpha p\gamma$)	P	^{34}S (p, p), ($p, p'\gamma$): resonances	Z	^{39}K (γ, α)
G	^{16}O ($^{24}\text{Mg}, \alpha p\gamma$)	Q	^{34}S (d, n)	Others:	
H	^{24}Mg ($^{16}\text{O}, \alpha p\gamma$)	R	^{34}S ($^3\text{He}, d$)	AA	^{40}Ca ($\mu^-, \nu\alpha n\gamma$)
I	^{27}Al ($^{14}\text{N}, \alpha p n\gamma$)	S	^{35}Cl (γ, γ')	AB	Coulomb excitation
J	^{31}P (α, p), (α, n): resonances	T	^{35}Cl ($n, n'\gamma$)		

E(level) [†]	J π [#]	T _{1/2} [‡]	XREF	Comments
0.0	3/2 ⁺	stable	ABC EFGHI KLM O QRSTU VWXYZ	XREF: Others: AA , AB $\mu = +0.821870$ 2 (2013Ja17 , 2019StZV) $Q = -0.0817$ 8 (2008Py02 , 2021StZZ) J^π : L=2 from 0 ⁺ in (d,n), (n,d), ($^3\text{He}, d$), ($d, ^3\text{He}$), and (pol d, ^3He) and L-1/2 transfer from analyzing power. Spin=3/2 from microwave spectroscopy and atomic beam methods (1948To10 , 1949Da14). μ : Nuclear Magnetic Resonance (2013Ja17). Q : Atomic beam magnetic resonance (2008Py02). Others: -0.08249 2 (1972St38), -0.076 5 (1986El09), -0.817 8 (1993Su36), 0.0819 11 (2000Ha64), 0.0850 11 (2004Al08). Evaluated rms nuclear charge radius $R = 3.365$ fm 19 (2013An02). XREF: Others: AA , AB $B(E2)^\dagger = 0.00076$ 7 E(level): weighted average of 1219.2 2 from ^{35}Ar ε decay, 1219.39 7 from (p, γ), and 1219.42 10 from (p,p' γ). Others: 1219.0 10 from ($^{16}\text{O}, \alpha p\gamma$), 1219.5 10 from ($\alpha, p\gamma$). J^π : L=0 from 0 ⁺ in (d,n), (n,d), ($^3\text{He}, d$), ($d, ^3\text{He}$), and (pol d, ^3He). T _{1/2} : from lifetime $\tau = 0.172$ ps 20: weighted average of 0.29 ps 4 from ($^{34}\text{S}, n\gamma$) (1973Wa10 , DSAM), 0.21 ps +10-8 from ($\alpha, p\gamma$) (1969In04 , DSAM), 80 fs 40 (1971Wi13 , DSAM), 175 fs 20 (1972Hu11 , DSAM), 150 fs 20 (1973Fa07 , DSAM), 145 fs 30 (1976Me12 , DSAM) from (p, γ), 0.13 ps +3-2 from (γ, γ') (1966Ho02 , resonance fluorescence), 200 fs 80 from (n,n' γ) (1989Ge09 , DSAM), 0.215 ps 75 (1972Va06 , DSAM), 0.21 ps +6-4 (1972Br45 , DSAM), 0.27 ps +19-10 (1969Du08 , DSAM) from (p,p' γ), 0.16 ps 5 (1969Ha17 , DSAM), and 0.230 ps 25 (1977Sc36 , DSAM) from Coulomb excitation. XREF: Others: AA , AB
1219.39 7	1/2 ⁺	0.119 ps 14	B E H KLM O QRSTU VWXYZ	
1763.18 10	5/2 ⁺	0.36 ps 3	B EFGHI KLM O QRSTU VWXYZ	XREF: Others: AA , AB

Continued on next page (footnotes at end of table)

Adopted Levels, Gammas (continued) ^{35}Cl Levels (continued)

<u>E(level)[†]</u>	<u>J^π#</u>	<u>T_{1/2}[‡]</u>	<u>XREF</u>	<u>Comments</u>
				T=1/2 B(E2)↑=0.0106 7 XREF: X(1785)Y(1750) E(level): weighted average of 1763.1 2 from ^{35}Ar ε decay, 1763.0 6 from ($^{28}\text{Si},\alpha\text{p}\gamma$), 1762.95 17 from ($^{24}\text{Mg},\alpha\text{p}\gamma$), 1763.3 2 from ($^{16}\text{O},\alpha\text{p}\gamma$), 1763.19 10 from ($^{14}\text{N},\alpha\text{pn}\gamma$), 1763.0 5 from (p,$\gamma$), 1763.26 10 from (p,p'$\gamma$). J^π: L=2 from 0⁺ in (d,n); spin=5/2 from pγ(θ) or γ(θ) in ($\alpha,\text{p}\gamma$), (p,γ), and (p,p'γ). T_{1/2}: lifetime τ=0.52 ps 5: weighted average of 0.54 ps 7 from ($^{34}\text{S},\text{n}\gamma$) (1973Wa10, DSAM), 0.55 ps 15 from ($\alpha,\text{p}\gamma$) (1969In04, DSAM), 0.50 ps 10 (1972Hu11, DSAM), 0.40 ps 14 (1973Fa07, DSAM), 0.53 ps 9 (1976Me12, DSAM) from (p,γ), 0.60 ps 21 from (γ,γ') (1962Bo17, resonance fluorescence), 0.205 ps 75 (1972Va06, DSAM), 0.49 ps +9-6 (1972Br45, DSAM), 1.15 ps +45-63 (1969Du08, DSAM) from (p,p'γ), 0.57 ps 7 (1969Ha17, DSAM), and 0.63 ps 5 (1977Sc36, DSAM) from Coulomb excitation.
2645.66 8	7/2 ⁺	0.159 ps 21	B FGHI LM O Qr TU y	XREF: Others: AA, AB XREF: r(2645)y(2650) E(level): weighted average of 2645.81 18 from ($^{24}\text{Mg},\alpha\text{p}\gamma$), 2645.8 3 from ($^{16}\text{O},\alpha\text{p}\gamma$), 2645.45 12 from ($^{14}\text{N},\alpha\text{pn}\gamma$), 2645.70 8 from (p,$\gamma$), and 2645.74 17 from (p,p'γ). Others: 2645.1 10 from ^{35}Ar ε decay, 2645.8 6 from ($^{28}\text{Si},\alpha\text{p}\gamma$), and 2645.8 8 from ($\alpha,\text{p}\gamma$). J^π: spin=7/2 from p$\gamma$($\theta$) in ($\alpha,\text{p}\gamma$) and γ(θ) in (p,γ); parity from 882γ, M1+E2 to 5/2⁺, 1763 level in ($\alpha,\text{p}\gamma$). T_{1/2}: lifetime τ=0.229 ps 30: weighted average of 0.30 ps 9 from ($\alpha,\text{p}\gamma$) (1969In04, DSAM), 0.255 ps 65 (1972Hu11, DSAM), 0.200 ps 30 (1976Me12, DSAM) in (p,γ), 0.27 ps 9 in (n,n'γ) (1989Ge09, DSAM), 0.185 ps 50 (1972Va06, DSAM), 0.26 ps 6 (1972Br45, DSAM), 0.35 ps +10-7 (1969Du08, DSAM) in (p,p'γ), and 0.26 ps 9 in Coulomb excitation (1977Sc36, DSAM). Other: τ<1.3 ps in ($^{28}\text{Si},\alpha\text{p}\gamma$) (2007Ks01, DSAM).
2694.00 20	3/2 ⁺	32 fs 6	B LM O QRSTU WXY	XREF: Others: AA XREF: L(2693?)X(2676)y(2650) E(level): weighted average of 2693.8 2 from ^{35}Ar ε decay, 2693.6 13 from ($\alpha,\text{p}\gamma$), 2693.9 3 from (p,γ), 2694.50 28 from (p,p'γ). J^π: spin=3/2 from γ(θ) in (p,p'γ); L=2 from 0⁺ in ($^3\text{He},\text{d}$) and (d,^3He). T_{1/2}: lifetime τ=46 fs 9: unweighted average of 70 fs 30 from ($\alpha,\text{p}\gamma$) (1972Br33, DSAM), 21 fs 3 (1972Hu11, DSAM), 19 fs 4 (1973Fa07, DSAM), 20 fs 4 (1976Me12, DSAM) from (p,γ) 65 fs 20 (1972Va06, DSAM), 62 fs 16 (1969Du08, DSAM) from (p,p'γ), and 62 fs 8 from (n,n'γ) (1989Ge09, DSAM). Others: 115 fs +95-59 using 931γ from (p,p'γ) (1969Du08), but is considered less reliable by authors, 11 fs 7 from (p,p'γ) (1971Ca40, DSAM) is considered an outlier.

Continued on next page (footnotes at end of table)

Adopted Levels, Gammas (continued) ^{35}Cl Levels (continued)

E(level) [†]	J ^π [#]	T _{1/2} [‡]	XREF	Comments
3002.64 11	5/2 ⁺	22 fs 6	B F HI LM O QRSTU WXY	E(level): weighted average of 3002.3 3 from ^{35}Ar ε decay, 3003.1 8 from ($^{28}\text{Si}, \alpha\gamma$), 3002.0 10 from ($^{16}\text{O}, \alpha\gamma$), 3001.98 22 from ($^{14}\text{N}, \alpha\text{pn}\gamma$), 3002.6 15 from ($\alpha, \text{p}\gamma$), 3002.75 8 from (p, γ), 3002.6 5 from ($\text{p}, \text{p}'\gamma$). J ^π : L=2 from 0 ⁺ in (pol d, ^3He) and L+1/2 transfer from analyzing power. Discrepancy: L=3 from 0 ⁺ in (d, n). T _{1/2} : lifetime $\tau=32$ fs 8: weighted average of 48 fs 9 from ($\alpha, \text{p}\gamma$) (1972Br33, DSAM), 22 fs 3 (1972Hu11, DSAM), 14 fs 2 (1973Fa07, DSAM), 16 fs 4 (1976Me12, DSAM) from (p, γ) 72 fs 12 from ($\text{n}, \text{n}'\gamma$) (1989Ge09, DSAM), 18 fs +26-15 (1972Va06, DSAM), and 31 fs 13 (1969Du08, DSAM) from ($\text{p}, \text{p}'\gamma$).
3162.87 9	7/2 ⁻	30.6 ps 9	FGHI KLM O QR TU WX	T=1/2 E(level): weighted average of 3163.1 6 from ($^{28}\text{Si}, \alpha\gamma$), 3163.43 19 from ($^{24}\text{Mg}, \alpha\gamma$), 3163.0 3 from ($^{16}\text{O}, \alpha\gamma$), 3162.64 9 from ($^{14}\text{N}, \alpha\text{pn}\gamma$), 3163.1 11 from ($\alpha, \text{p}\gamma$), 3162.99 7 from (p, γ), 3162.75 10 from ($\text{p}, \text{p}'\gamma$). J ^π : spin=7/2 from $\text{p}\gamma(\theta)$ in ($\alpha, \text{p}\gamma$) and $\gamma(\theta)$ in (p, γ) and ($\text{p}, \text{p}'\gamma$); L=3 from 0 ⁺ in (d, n), ($^3\text{He}, \text{d}$), ($\text{d}, ^3\text{He}$), and (pol d, ^3He). T _{1/2} : lifetime $\tau=30.6$ ps 9: weighted average of 42 ps 2 (1974Va13, RDM), 41.7 ps 17 (1976Ke02, RDM), 45 ps 6 (1991Ja11, RDM) from $^{24}\text{Mg}(^{16}\text{O}, \alpha\gamma)$, 41.8 ps 18 from $^{27}\text{Al}(^{14}\text{N}, \alpha\text{pn}\gamma)$ (1976Wa11, RDM), 60 ps 7 from ($\alpha, \text{p}\gamma$) (1969In04, DSAM), 37 ps 4 from (p, γ), (1971Ba23, delayed coincidence), and 46 ps 6 from ($\text{p}, \text{p}'\gamma$) (1974Hu09, RDM). Others: $\tau=140$ ps 40 (1968Az02, delayed coincidence; obtained using NaI(Tl) detectors with broad time resolution) from (p, γ).
3918.47 12	3/2 ⁺	4.9 fs 14	B M O RS u x	XREF: R(3908)u(3930)x(3948) E(level): weighted average of 3918.4 5 from ^{35}Ar ε decay, 3915 2 from ($\alpha, \text{p}\gamma$), 3918.48 10 from (p, γ). J ^π : spin=3/2 from feeding 5162 $\gamma(\theta)$ from 5/2 ⁺ , 9081 level in (p, γ); parity from 3918 γ , M1+E2 to 3/2 ⁺ ground state. L=2 from 0 ⁺ in (pol d, ^3He). T _{1/2} : lifetime $\tau=7$ fs 2 from (p, γ), (1972Hu11, DSAM). Others: <22 fs from ($\alpha, \text{p}\gamma$) (1972Br33, DSAM).
3942.9 2	9/2 ⁺	0.18 ps 3	F HI LM O	E(level): weighted average of 3942.0 7 from ($^{28}\text{Si}, \alpha\gamma$), 3943.4 10 from ($^{16}\text{O}, \alpha\gamma$), 3942.3 10 from ($^{14}\text{N}, \alpha\text{pn}\gamma$), 3943.4 12 from ($\alpha, \text{p}\gamma$), and 3942.9 2 from (p, γ). J ^π : spin=9/2 from $\text{p}\gamma(\theta)$ in ($\alpha, \text{p}\gamma$); 2180 γ E2 to 5/2 ⁺ . T _{1/2} : lifetime $\tau=0.26$ ps 5: weighted average of 0.33 ps 5 in ($\alpha, \text{p}\gamma$) (1972Br33, DSAM), 0.39 ps 11 (1972Hu11, DSAM), 0.145 ps 50 (1973Fa07, DSAM), and 0.29 ps 5 (1976Me12, DSAM) from (p, γ). Other: $\tau<0.5$ ps in ($^{28}\text{Si}, \alpha\gamma$) (2007Ks01, DSAM).
3967.6 4	1/2 ⁺	11.8 fs 28	B M O R	E(level): weighted average of 3967.7 4 from ^{35}Ar ε decay, 3968 2 from ($\alpha, \text{p}\gamma$), and 3967.3 6 from (p, γ). J ^π : L=0 from 0 ⁺ in ($^3\text{He}, \text{d}$). T _{1/2} : lifetime $\tau=17$ fs 4: weighted average of 13 fs 4 (1972Hu11, DSAM), 20 fs 5 (1973Fa07, DSAM), and 20 fs 4 (1976Me12, DSAM) in (p, γ). Other: $\tau<54$ fs from ($\alpha, \text{p}\gamma$) (1972Br33, DSAM).

Continued on next page (footnotes at end of table)

Adopted Levels, Gammas (continued) ^{35}Cl Levels (continued)

E(level) [†]	J ^π [#]	T _{1/2} [‡]	XREF	Comments
4059.1 4	(3/2) ⁻	14.3 fs 21	LM O QR	E(level): weighted average of 4056.9 27 from (α,p), 4058 3 from (α,pγ), 4059.2 4 from (p,γ). J ^π : L=1 from 0 ⁺ in (³ He,d); (3/2 ⁻) from FRESKO-DWBA analysis of measured σ(θ) of (α,p) (2019Se09). T _{1/2} : lifetime τ=20 fs 3: weighted average of 31 fs 13 from (α,pγ) (1972Br33, DSAM), 22 fs 3 (1972Hu11, DSAM), 20 fs 4 (1973Fa07, DSAM) and 18 fs 3 (1976Me12, DSAM) from (p,γ).
4112.4 16	7/2 ⁺	49 fs 11	LM O Q	E(level): weighted average of 4111.1 29 from (α,p), 4108 2 from (α,pγ), and 4113.7 10 from (p,γ). J ^π : 4113.4γ E2(+M3) to 3/2 ⁺ . T _{1/2} : lifetime τ=70 fs 16 from (α,pγ) (1972Br33, DSAM).
4173.43 23	(5/2 ⁺)	35 fs 6	M O q	T=1/2 XREF: q(4170) E(level): weighted average of 4171 2 from (α,pγ) and 4173.45 19 from (p,γ). J ^π : based on primary transitions from proton resonances in (p,γ): 3827.3γ from 8001.0, (7/2 ⁺), 4456.4γ from 8630, 7/2 ⁻ level, 4243.0γ from 8416.7, 1/2 ⁺ level. Note that primary 3099.1γ from 7272.7, 1/2 ⁻ resonance disfavors 5/2 ⁺ assignment, but this γ branch is relatively weak. T _{1/2} : lifetime τ=51 fs 8: weighted average of 68 fs 24 in (α,pγ) (1972Br33, DSAM), 105 fs +35-30 (1971Wi13, DSAM), 55 fs 6 (1972Hu11, DSAM), and 34 fs 9 (1976Me12, DSAM) in (p,γ).
4177.9 2	3/2 ⁻	27.0 fs 28	L O qR	XREF: q(4170)R(4167) J ^π : spin=3/2 from γ(θ) in (p,γ); L=1 from 0 ⁺ in (³ He,d). T _{1/2} : lifetime τ=39 fs 4: weighted average of 32 fs 5 (1972Hu11, DSAM), 42 fs 4 (1973Fa07, DSAM), and 47 fs 10 (1976Me12, DSAM) from (p,γ).
4347.74 8	9/2 ⁻	1.0 ps 6	FGHI LM O	E(level): weighted average of 4347.8 6 from (²⁸ Si,αpγ), 4348.29 24 from (²⁴ Mg,αpγ), 4347.4 3 from (¹⁶ O,αpγ), 4347.43 17 from (¹⁴ N,αpγ), 4347.8 12 from (α,pγ), and 4347.76 6 from (p,γ). J ^π : spin from γ(θ) in (p,γ); parity from 1184.6γ, M1+E2 to 7/2 ⁻ , 3163 level in (p,γ) (1974Lo17). T _{1/2} : lifetime τ=1.5 ps 8: unweighted average of 1.31 ps +30-23 from (²⁸ Si,αpγ) (2007Ks01, DSAM), 2.9 ps +14-7 (1972Br33, DSAM), and 0.33 ps 6 (1976Me12, DSAM) from (p,γ).
4624.3 3	3/2,5/2	40 fs 17	B LM O	XREF: B(?) E(level): other: 4625.3 28 from (α,p); 4618 2 from (α,pγ) is discrepant. J ^π : from pγ(θ) in (α,pγ). T _{1/2} : lifetime τ=58 fs 24 from (α,pγ) in 1972Br33 with DSAM.
4769.86 23	(5/2 ⁻)	76 fs 20	LM O	E(level): others: 4770.4 26 from (α,p), 4770 15 from (α,pγ). J ^π : from primary transitions in (p,γ): 4059.2γ from 8829.3, 1/2 ⁻ level, 2778.9γ from 7548.9, 7/2 ⁻ level, 3231.0γ from 8001.0, (7/2 ⁺) level. Other: 7/2,9/2 from

Continued on next page (footnotes at end of table)

Adopted Levels, Gammas (continued) ^{35}Cl Levels (continued)

<u>E(level)[†]</u>	<u>J^π</u>	<u>T_{1/2}[‡]</u>	<u>XREF</u>	<u>Comments</u>
4839.10 11	(1/2 ⁺ ,3/2)	9.7 fs 35	0 x	$p\gamma(\theta)$ for a group at E=4770 15 in ($\alpha,p\gamma$) is inconsistent and disfavored by γ rays to 1/2⁺ and 3/2⁺. T_{1/2}: lifetime τ=110 fs 29: weighted average of 270 fs 95 (1972Hu11, DSAM) and 105 fs 17 (1976Me12, DSAM) in (p,γ). XREF: x(4839) J^π: from primary γ transitions in (p,γ): 2339.4γ from 7178.6, 1/2⁺ level, 2355.4γ from 7194.6, 1/2⁻ level, 2867.2γ from 7706.7, 5/2⁺ level. T_{1/2}: lifetime τ=14 fs 5 from (p,γ) (1976Me12, DSAM).
4854.4 19	(1/2,3/2)	4.9 fs 14	L 0 x	XREF: x(4839) E(level): other: 4859.2 28 from (α,p). J^π: from primary γ transitions in (p,γ): 2440.1γ from 7194.6, 1/2⁻ level, 3183.9γ from 8038.5, 1/2⁺ level. T_{1/2}: lifetime τ=7 fs 2 from (p,γ) (1976Me12, DSAM).
4880.96 15	7/2 ⁽⁺⁾	5.6 fs 14	LM 0	E(level): others: 4881.8 29 from (α,p), 4880.8 21 from ($\alpha,p\gamma$). J^π: spin=7/2 from $p\gamma(\theta)$ in ($\alpha,p\gamma$); 3417.1 primary γ from 8298.2, 3/2⁺ resonance in (p,γ). T_{1/2}: lifetime τ=8 fs 2: weighted average of 7 fs 2 (1972Hu11, DSAM), 7 fs 3 (1973Fa07, DSAM), and 10 fs 3 (1976Me12, DSAM) from (p,γ). Other: τ=0.28 ps 6 from ($\alpha,p\gamma$) in 1972Br33 with DSAM.
5009.1 18	(1/2 ⁺ ,3/2)	7.6 fs 21	LM 0 R	E(level): weighted average of 5006.9 27 from (α,p) and 5010.1 18 from (p,γ). Others: 5015 20 from ($\alpha,p\gamma$) and 5010 15 from ($^3\text{He},d$). J^π: primary 2184.4γ from 7194.6, 1/2⁻ level and 3406.5γ from 8416.7, 1/2⁺ level; possible 3246.9γ to 5/2⁺. T_{1/2}: lifetime τ=11 fs 3 from (p,γ) (1976Me12, DSAM).
5158.6 29	3/2 ⁺ ,5/2 ⁺		Lm R WX	XREF: m(5175) E(level): weighted average of 5158.9 29 from (α,p), 5163 15 from ($^3\text{He},d$), 5150 20 from $^{36}\text{Ar}(d,^3\text{He})$, and 5155 11 from $^{36}\text{Ar}(\text{pol } d,^3\text{He})$. J^π: L=2 from 0⁺ in ($\text{pol } d,^3\text{He}$) and ($d,^3\text{He}$).
5163.34 22	7/2 ⁻		m 0	XREF: m(5175) J^π: spin=7/2 from feeding 3466$\gamma(\theta)$ from 8630, 7/2⁻ level in (p,γ); parity=- from 3466γ M1+E2 from 7/2⁻.
5214.5 20	(5/2 ⁺) [@]	<4.85 fs	LM 0	XREF: M(5230) E(level): weighted average of 5209.5 28 from (α,p), 5230 20 from ($\alpha,p\gamma$), and 5215.8 15 from (p,γ). J^π: other: 3/2,5/2 from $p\gamma(\theta)$ in ($\alpha,p\gamma$).
5403.3 10	3/2 ⁻	11.8 fs 28	L 0 R	T_{1/2}: lifetime τ<7 fs from (p,γ) (1976Me12, DSAM). E(level): weighted average of 5402.0 29 from (α,p) and 5403.5 10 from (p,γ). Other: 5409 12 from ($^3\text{He},d$). J^π: L=1 from 0⁺ in ($^3\text{He},d$); 2400.7γ to 5/2⁺ rules out 1/2⁻ based on RUL, since it would require a large B(M2)(W.u.) exceeding RUL.

Continued on next page (footnotes at end of table)

Adopted Levels, Gammas (continued) ^{35}Cl Levels (continued)

E(level) [†]	J ^π #	T _{1/2} [‡]	XREF			Comments
						T _{1/2} : lifetime $\tau=17$ fs 4 from (p, γ) (1976Me12 , DSAM). Additional 36% of I γ transitions to unknown levels in (p, γ). E(level): weighted average of 5407.4 8 from ($^{28}\text{Si},\alpha\text{p}\gamma$), 5407.64 27 from ($^{24}\text{Mg},\alpha\text{p}\gamma$), 5406.7 3 from ($^{16}\text{O},\alpha\text{p}\gamma$), 5406.68 19 from ($^{14}\text{N},\alpha\text{p}\gamma$), and 5407.1 15 from ($\alpha,\text{p}\gamma$). J ^π : spin=11/2 from p γ (θ) in (p, γ); parity from 1059 γ , M1+E2 to 9/2 ⁻ , 4347 level. T _{1/2} : lifetime $\tau=0.4$ ps 1 from ($\alpha,\text{p}\gamma$) (1974Lo17 , DSAM). Other: $\tau<1.1$ ps from ($^{28}\text{Si},\alpha\text{p}\gamma$) (2007Ks01 , DSAM).
5406.95 ^a 21	11/2 ⁻	0.28 ps 7	FGHI	LM		
5531 5			LM	O		E(level): weighted average of 5531 5 from (α,p) and 5535 20 from ($\alpha,\text{p}\gamma$).
5586.0 20	5/2 ⁺		Lm	O	wx	XREF: m(5600)w(5600)x(5583) E(level): weighted average of 5586.0 26 from (α,p) and 5586 2 from (p, γ). J ^π : L=2 from 0 ⁺ in (pol d, ^3He) and L+1/2 transfer from analyzing power.
5598.4 22	3/2 ⁺ ,5/2 ⁺	2.1 fs 7	Lm	O	wx	XREF: m(5600)w(5600)x(5583) E(level): weighted average of 5594.6 26 from (α,p) and 5599.7 15 from (p, γ). J ^π : L(d, ^3He)=2 from 0 ⁺ . T _{1/2} : lifetime $\tau=3$ fs 1 from (p, γ) (1976Me12 , DSAM).
5633.6 30			Lm	r	w y	XREF: m(5650)r(5660)w(5600)y(5650) E(level): from (α,p). XREF: m(5650)r(5660)y(5650)
5646 2	(5/2,7/2,9/2 ⁺)	2.8 fs 7	Lm	O	r y	E(level): weighted average of 5645 4 from (α,p) and 5646 2 from (p, γ). J ^π : from primary transitions in (p, γ): 1903 γ from 7548.9, 7/2 ⁻ level, 2355 γ from 8001.0, (7/2 ⁺) level, 3261 γ from 8906.8, 5/2 ⁺ level. T _{1/2} : lifetime $\tau=4$ fs 1 from (p, γ) in 1976Me12 with DSAM.
5655 2	(3/2) ⁺	13.9 fs 35	Lm	O	r w y	T=3/2 XREF: m(5650)r(5660)w(5600)y(5650) E(level): weighted average of 5653 4 from (α,p) and 5655 2 from (p, γ). J ^π : L=2 from 0 ⁺ in ($^3\text{He},\text{d}$); primary 2142 γ from 7797.2, 1/2 ⁻ level in (p, γ). Note that a weak primary 1903 γ from 7548.9, 7/2 ⁻ level in (p, γ) favors 5/2 ⁺ . T _{1/2} : lifetime $\tau=20$ fs 5 from (p, γ) (1976Me12 , DSAM).
5682.2 20	1/2 ⁻ ,3/2 ⁻		Lm	O	R	Additional 18% for decay to other levels. XREF: m(5650) E(level): weighted average of 5679.7 34 from (α,p), 5683 2 from (p, γ), and 5684 12 from ($^3\text{He},\text{d}$). J ^π : L=1 from 0 ⁺ in ($^3\text{He},\text{d}$). Additional 67% of I γ transitions to unknown levels (p, γ). E(level): others: 5731.2 25 from (α,p), 5720 20 from $^{36}\text{Ar}(\text{d},^3\text{He})$, and 5720 11 from $^{36}\text{Ar}(\text{pol}$
5723.52 18	5/2 ⁺		L	O	WX	

Continued on next page (footnotes at end of table)

Adopted Levels, Gammas (continued)

^{35}Cl Levels (continued)				
E(level) [†]	J ^π #	T _{1/2} [‡]	XREF	Comments
				d, ^3He).
				J ^π : L=2 from 0 ⁺ in (pol d, ^3He) and (d, ^3He) and L+1/2 transfer from analyzing power.
				Additional 100% of I _γ transitions to unknown levels in (p,γ).
5758 3	(1/2, 3/2 ⁻)		L O R	E(level): weighted average of 5757 4 from (α,p), 5758 3 from (p,γ), and 5760 12 from (^3He , d).
				J ^π : L(^3He , d)=(0,1) from 0 ⁺ .
				Additional 56% of I _γ transitions to unknown levels in (p,γ).
5806.6 20	(1/2, 3/2, 5/2 ⁺)	3.5 fs 7	L O	E(level): weighted average of 5808.1 31 from (α,p) and 5806 2 from (p,γ).
				J ^π : primary 2891γ from 8696.9, 3/2 ⁻ level; 5806γ to 3/2 ⁺ .
				T _{1/2} : lifetime τ=5 fs 1 from (p,γ) (1976Me12, DSAM).
5830 4	(5/2 ⁻) [@]		LM	XREF: M(5850)
				E(level): weighted average of 5829 4 from (α,p) and 5850 25 from (α,pγ).
				J ^π : other: spin=5/2, 9/2 from pγ(θ) in (α,pγ).
5926.57 29	(11/2) ⁻	<0.28 ps	F HI LM	E(level): weighted average of 5927.1 7 from (^{28}Si , αpγ), 5926.5 5 from (^{16}O , αpγ), 5926.47 29 from (^{14}N , αpnγ), 5928 3 from (α,p), and 5927.1 19 from (α,pγ).
				J ^π : 1579γ M1+E2, ΔJ=1 to 4348, 9/2 ⁻ in ^{12}C (^{28}Si , αpγ); spin=7/2, 11/2 also from pγ(θ) in (α,pγ).
				T _{1/2} : lifetime τ<0.4 ps from (^{28}Si , αpγ) (2007Ks01, DSAM).
6087.23 26	13/2 ⁻	6.2 ps 6	FGHI LM	E(level): weighted average of 6087.6 8 from (^{28}Si , αpγ), 6088.05 29 from (^{24}Mg , αpγ), 6086.7 4 from (^{16}O , αpγ), 6086.89 23 from (^{14}N , αpnγ), 6084.2 29 from (α,p), and 6087.0 18 from (α,pγ).
				J ^π : spin=13/2 from pγ(θ) in (α,pγ); parity from 680γ, M1+E2 to 11/2 ⁻ , 5406 level in (α,pγ).
				T _{1/2} : lifetime τ=8.9 ps 8: weighted average of 9.3 ps 8 from (α,pγ) in 1974Lo17 with RDM and 7.7 ps 14 from (^{14}N , αpnγ) in 1976Wa11.
6106.2 21	(3/2, 5/2 ⁺)	8.3 fs 21	1 O wx	XREF: l(6104)w(6140)x(6136)
				E(level): weighted average of 6104 4 from (α,p) and 6106.2 21 from (p,γ).
				J ^π : 4886.4γ to 1/2 ⁺ , 4342.9γ to 5/2 ⁺ ; primary 6106.2γ from 7781.7, (5/2 ⁻) level.
				T _{1/2} : lifetime τ=12 fs 3 from (p,γ) in 1976Me12 with DSAM.
6141 4	5/2 ⁺		L wx	XREF: w(6140)x(6136)
				J ^π : L=2 from 0 ⁺ in (d, ^3He) and (pol d, ^3He), and L+1/2 transfer from analyzing power.
6181 3	(1/2 ⁺ to 7/2 ⁺)		L n0	XREF: n(6200)
				E(level): other: 6180 3 from (α,p).
				J ^π : 6180γ to 3/2 ⁺ g.s.; primary 1438γ from

Continued on next page (footnotes at end of table)

Adopted Levels, Gammas (continued) ^{35}Cl Levels (continued)

E(level) [†]	J ^π #	XREF		Comments
6224.4 30		L	n	7618.7, 5/2 ⁺ level in (p,γ). additional 122% of Iγ transitions to unknown levels in (p,γ). XREF: n(6200) J ^π : L(α,d)=6 for a group at 6200 10.
6283 3	(5/2 ⁺)@	L	R	E(level): weighted average of 6283 3 from (α,p) and 6284 4 from (³ He,d). J ^π : other: L=(2) from 0 ⁺ in (³ He,d). J ^π : L=(0,1) from 0 ⁺ in (³ He,d).
6329 4	(1/2,3/2 ⁻)		R	XREF: R(6377)
6380.1 8	(9/2 ⁻)@	H	L	E(level): weighted average of 6380.2 8 from (¹⁶ O,αpγ) and 6379.3 25 from (α,p). J ^π : other: L=(2,3) from 0 ⁺ in (³ He,d) for a group at E=6377 2 is inconsistent, probably indicating a different level.
6401.2 22	(1/2 ⁻)@	LM		
6428 2	(1/2 ⁺)@	L	R	E(level): weighted average of 6429 3 from (α,p) and 6427 2 from (³ He,d). J ^π : other: L=(3) from 0 ⁺ in (³ He,d) is inconsistent.
6469 3	(1/2 ⁻ ,3/2 ⁻)	L	R	E(level): weighted average of 6475 4 from (α,p) and 6468 2 from (³ He,d). J ^π : L=(1) from 0 ⁺ in (³ He,d).
6492 2	(3/2 ⁺)@	L	O	E(level): weighted average of 6492 3 from (α,p), 6492 3 from (p,γ), and 6491 2 from (³ He,d). J ^π : other: L=(2) from 0 ⁺ in (³ He,d).
6546 2	(1/2 ⁺)@	L	R	E(level): weighted average of 6549 3 from (α,p) and 6545 2 from (³ He,d). J ^π : other: L=(0,1) from 0 ⁺ in (³ He,d), indicating (1/2,3/2 ⁻).
6643 2	(1/2 ⁻ ,3/2 ⁻)	D	R	E(level): from (³ He,d). J ^π : from L=(1) from 0 ⁺ in (³ He,d).
6659.1 30	(7/2 ⁺)	L	y	XREF: y(6677) J ^π : from (α,p).
6660.2 8	(11/2 ⁻)	F		J ^π : 2312γ to 9/2 ⁻ , 3497γ to 7/2 ⁻ in (²⁸ Si,αpγ), assuming spin increasing as excitation energy increases.
6675.6 25		D	L	XREF: y(6677) E(level): weighted average of 6679.4 31 from (α,p) and 6674 2 from (³ He,d).
6761 2	3/2 ⁺ ,5/2 ⁺	D	R	XREF: X(6745) E(level): weighted average of 6761 2 from (³ He,d), 6750 20 from ³⁶ Ar(d, ³ He), and 6745 12 from ³⁶ Ar(pol d, ³ He). J ^π : L(d, ³ He)=2 from 0 ⁺ .
6779.0 20		D	L	E(level): weighted average of 6781.3 30 from (α,p) and 6778 2 from (³ He,d).
6800 4		D	L	XREF: D(6802) E(level): from (α,p).
6823 2		D	R	E(level): from (³ He,d).
6842 2		D	L	E(level): weighted average of 6842 4 from (α,p) and 6842 2 from (³ He,d).
6863 3	(9/2 ⁺)@	L		
6866.7 6	(1/2 ⁺ ,3/2)	D	O	E(level): other: 6866 2 from (³ He,d). J ^π : L(³ He,d)=(0,1) from 0 ⁺ .
6891.6 30	(9/2 ⁺)@	L		
6948.1 34	5/2 ⁺	L	WX	XREF: W(6930) E(level): weighted average of 6948.6 34 from (α,p) and 6930 20 from

Continued on next page (footnotes at end of table)

Adopted Levels, Gammas (continued) ^{35}Cl Levels (continued)

E(level) [†]	J ^π [#]	T _{1/2} [‡]	XREF			Comments
						$^{36}\text{Ar}(\text{d}, ^3\text{He})$. Other: 6953 43 from $^{36}\text{Ar}(\text{pol d}, ^3\text{He})$. J ^π : L(pol d, ^3He)=2 and L+1/2 from analyzing power.
6986.3 34	(9/2 ⁺) [@]		L			
7066.2 7	(5/2 ⁺) [@]		L	O	R	E(level): weighted average of 7066.2 7 from (p,γ), and 7066 2 from (^3He ,d).
7103.4 7	3/2 ⁽⁻⁾		L	O	R	XREF: L(7105?) E(level): weighted average of 7103.4 7 from (p,γ), and 7103 2 from (^3He ,d). J ^π : spin=3/2 from γ(θ) in (p,γ); parity from L(^3He ,d)=(1,3) from 0 ⁺ .
7121.7 24			L			
7178.6 8	1/2 ⁽⁺⁾			O		T=3/2 J ^π : spin=1/2 from γ(θ) in (p,γ); parity from IAS in ^{35}S .
7179 4	(7/2 ⁺) [@]		L	N		E(level): weighted average of 7180 4 from $^{32}\text{S}(\alpha, \text{p})$ and 7170 10 from $^{33}\text{S}(\alpha, \text{d})$. J ^π : also L=6 from 3/2 ⁺ in $^{33}\text{S}(\alpha, \text{d})$, indicating (7/2;17/2) ⁺ .
7185	5/2 ⁺			O	R	X
						XREF: R(7178)X(7181) J ^π : L(pol d, ^3He)=2 and L+1/2 from analyzing power for a group at E=7181 60. Other: L(^3He ,d)=(2) from 0 ⁺ for a group at E=7178 2 considered the same level.
7194.6 7	1/2 ⁻	0.027 keV 8		OP	R	
						J ^π : from resonance analysis of measured σ(θ) (1971BrXT) in $^{34}\text{S}(\text{p}, \text{p}), (\text{p}, \text{p}'\gamma)$:resonances. Other: 1/2, 3/2 from γ(θ) in (p,γ).
7213.7 25			L			y
7225.5 8	5/2			O		y
						J ^π : from γ(θ) in (p,γ).
7228.2 20	(3/2 ⁺) [@]		L		R	y
						XREF: R(7227)y(7250) E(level): weighted average of 7230.9 30 from $^{32}\text{S}(\alpha, \text{p})$, 7227 2 from $^{34}\text{S}(\text{He}, \text{d})$. J ^π : other: L=(0,1) from 0 ⁺ in (^3He ,d) is inconsistent. J ^π : spin=5/2 from γ(θ) in (p,γ); primary 6014.1γ to 1/2 ⁺ .
7234.0 8	5/2 ⁽⁺⁾			O		
7269.2 1				O		
7272.7 9	1/2 ⁻	0.014 keV 4		OP	R	y
						XREF: P(7274)R(7273)y(7250) J ^π : from resonance analysis of measured σ(θ) (1971BrXT) in $^{34}\text{S}(\text{p}, \text{p}), (\text{p}, \text{p}'\gamma)$:resonances. Other: (1/2, 3/2) from γ(θ) in (p,γ).
7348 3	(7/2) [@]		L			
7361.9 8	3/2 ⁽⁺⁾		L	O	R	
						E(level): weighted average of 7362 3 from (α,p), 7362.0 8 from (p,γ), and 7361 2 from (^3He ,d). J ^π : spin=3/2 from γ(θ) in (p,γ); 6142.0γ (M1+E2) to 1/2 ⁺ .
7396.4 10	7/2 ⁽⁻⁾			O	R	
						E(level): weighted average of 7396.0 10 from (p,γ), and 7398 2 from (^3He ,d). J ^π : spin=7/2 from γ(θ) in (p,γ); 4232.7γ (M1+E2) to 7/2 ⁻ .
7413.8 30			L			
7450.9 8	3/2		L	O		
						E(level): weighted average of 7447 3 from (α,p) and 7451.1 6 from (p,γ).

Continued on next page (footnotes at end of table)

Adopted Levels, Gammas (continued) ^{35}Cl Levels (continued)

E(level) [†]	J ^π [#]	T _{1/2} [‡]	XREF	Comments
7496			0	J ^π : from $\gamma(\theta)$ in (p, γ).
7501.1 8	(5/2 ⁺ , 7/2, 9/2 ⁺)		1 0	XREF: l(7503)
7502.9 7			1 0	J ^π : primary 3558.0 γ to 9/2 ⁺ , 4498.0 γ to 5/2 ⁺ .
7518.8 7	7/2		0	XREF: l(7503)
7548.9 6	7/2 ⁻	<0.69 fs	0P	J ^π : from $\gamma(\theta)$ in (p, γ). T=3/2
				XREF: P(7550)
				J ^π : from resonances analysis of measured $\sigma(\theta)$ in $^{34}\text{S}(\text{p,p}), (\text{p,p}'\gamma)$: resonances. Spin=7/2 also from $\gamma(\theta)$ in S(p, γ).
				T _{1/2} : lifetime $\tau < 1$ fs from (p, γ) (1973Fa07, DSAM).
				$\Gamma = 5.8$ eV 13 from (p, γ), 11.0 eV 15 from $^{34}\text{S}(\text{p,p}), (\text{p,p}'\gamma)$: resonances.
7561.3 6	1/2, 3/2		L 0	E(level): other: 7566.5 35 from (α ,p).
7565 20	5/2 ⁺			J ^π : from $\gamma(\theta)$ in (p, γ). J ^π : L=2 from 0 ⁺ in (pol d, ^3He) and L+1/2 transfer from analyzing power.
7590 4			L	
7600.8 8	5/2 ⁺	<13.9 fs	0	J ^π : spin=5/2 from $\gamma(\theta)$ in (p, γ); 7599.9 γ M1+E2 to 3/2 ⁺ .
				T _{1/2} : lifetime $\tau < 20$ fs from (p, γ) (1971Wi13, DSAM), which is equivalent to $\Gamma > 0.033$ eV.
7618.7 5	5/2 ⁺		0	J ^π : spin=5/2 from $\gamma(\theta)$ in (p, γ); 7617.8 γ M1+E2 to 3/2 ⁺ .
7648.1 30			L	
7656.6 8	(1/2, 3/2, 5/2 ⁺)		0	J ^π : primary 3689.1 γ to 1/2 ⁺ .
7670 10	(7/2 to 17/2) ⁺		N	J ^π : L(α ,d)=6 from 3/2 ⁺ .
7671.9 8	5/2 ⁺		1 Op	XREF: p(7686)
				J ^π : spin=5/2 from $\gamma(\theta)$ in (p, γ); 4668.8 γ M1+E2 to 5/2 ⁺ . But 3324.0 γ to 9/2 ⁻ seems favors 5/2 ⁻ .
7685.5 8	3/2 ⁻		Op	J ^π : spin=3/2 from $\gamma(\theta)$ in (p, γ) and 3/2 ⁻ from R-Matrix analysis in (p,p): resonances.
				$\Gamma_p = 446$ eV 15 from (p,p): resonances.
7693.9 8	(1/2 ⁺ , 3/2, 5/2)		0	J ^π : primary 7693.0 γ to 3/2 ⁺ , 3634.5 γ to (3/2) ⁻ , 4690.8 γ to 5/2 ⁺ .
7706.4 8	5/2 ⁺		L Op	E(level): other: 7709.9 25 from (α ,p).
				J ^π : spin=5/2 from $\gamma(\theta)$ in (p, γ) and 3/2 ⁺ , 5/2 ⁺ from R-Matrix analysis in (p,p): resonances.
				$\Gamma_p = 4$ eV 1 in (p,p).
7744.8 9	7/2 ⁻		L 0	E(level): other: 7744.1 30 from (α ,p).
				J ^π : spin=7/2 from $\gamma(\theta)$ in (p, γ); 4581.5 γ M1+E2 to 7/2 ⁻ .
7748	(7/2 to 17/2) ⁺		NO	XREF: N(7750)
				E(level): from (p, γ).
				J ^π : L(α ,d)=6 from 3/2 ⁺ .
7777.1 9	(5/2 ⁺)		L 0	E(level): other: 7770.3 33 from (α ,p) is discrepant, probably indicating a different level.
				J ^π : based on primary transitions: 4613.8 γ to 7/2 ⁻ , 5131.0 γ to 7/2 ⁺ , 6557.1 γ to 1/2 ⁺ .
7781.7 11	(5/2 ⁻)		0	J ^π : primary 3433.8 γ to 9/2 ⁻ ; relatively weak 6561.7 γ to 1/2 ⁺ .

Continued on next page (footnotes at end of table)

Adopted Levels, Gammas (continued)

^{35}Cl Levels (continued)					
E(level) [†]	J ^π #	T _{1/2} [‡]	XREF		Comments
7797.2 9	1/2 ⁻	0.031 keV 10	L	OP	E(level): weighted average of 7799.8 30 from (α,p), 7797.1 9 from (p,γ), and 7796 3 from $^{34}\text{S}(\text{p,p}), (\text{p,p}'\gamma)$: resonances. J ^π : spin=1/2 from γ(θ) in (p,γ); (1/2) ⁻ from R-Matrix analysis of in (p,p):resonances.
7837.2 10	3/2 ⁻	3.7 keV 4		OP	T=3/2 E(level): weighted average of 7837.2 10 from (p,γ), and 7837 3 from $^{34}\text{S}(\text{p,p}), (\text{p,p}'\gamma)$:resonances. T _{1/2} : Lifetime τ<5 fs from (p,γ) (1971Wi13, DSAM). J ^π : spin=3/2 from γ(θ) in (p,γ) and 3/2 ⁻ from R-Matrix analysis in (p,p):resonances.
7839.0 9	(5/2 ⁺)			0	J ^π : primary 6618.9γ to 1/2 ⁺ , 3895.9γ to 9/2 ⁺ .
7868.7 10	(1/2 ⁺ , 3/2, 5/2 ⁺)		L	0	E(level): other: 7868.5 35 from (α,p). J ^π : primary 6648.6γ to 1/2 ⁺ , 2652.8γ to (5/2 ⁺).
7872.83 ^b 30	13/2 ⁺		F HI	NO	XREF: N(7870) E(level): weighted average of 7873.3 8 from ($^{28}\text{Si}, \alpha\text{p}\gamma$), 7872.8 4 from ($^{16}\text{O}, \alpha\text{p}\gamma$), and 7872.78 30 from ($^{14}\text{N}, \alpha\text{pn}\gamma$). J ^π : 1946.35γ E1+M2, ΔJ=1 to 11/2 ⁻ , 1786.2γ D(+Q) to 13/2 ⁻ ; band head. J ^π : from γ(θ) in (p,γ) and R-Matrix analysis in (p,p):resonances.
7880.8 9	3/2 ⁺ , 5/2 ⁺	0.008 keV 4		OP	J ^π : from γ(θ) in (p,γ) and R-Matrix analysis in (p,p):resonances.
7885				0	
7889.1 15				0	
7899.2 7	(5/2)		1	0	XREF: I(7901) J ^π : primary 2735.8γ to 7/2 ⁻ , 3018.1γ to 7/2 ⁽⁺⁾ , 3721.1γ to 3/2 ⁻ , 5204.9γ to 3/2 ⁺ .
7903			1	0	XREF: I(7901)
7923.3 9	(3/2 ⁺ , 5/2 ⁺)			0	J ^π : primary 5277.2γ to 7/2 ⁺ , 6703.2γ to 1/2 ⁺ .
7930				0	
7949				0	
7970.3 10	(5/2 ⁻)			0	J ^π : primary 3622.3γ to 9/2 ⁻ , weak 6750.2γ to 1/2 ⁺ .
7978.6 35			L		
7987.8 10	3/2 ⁺			0	J ^π : spin=3/2 from γ(θ) in (p,γ); 7986.8γ M1+E2 to 3/2 ⁺ .
7995.6 10	(5/2)			0	J ^π : primary 2592.0γ to 3/2 ⁻ , 7994.6γ to 3/2 ⁺ , 2832.1γ to 7/2 ⁻ , 5349.5γ to 7/2 ⁺ .
8001.0 9	(7/2 ⁺)		L	0	E(level): other: 8000.7 41 from (α,p). J ^π : from primary transitions in (p,γ): 3653.0γ to 9/2 ⁻ , 4057.9 to 9/2 ⁺ , 3827.3γ to 5/2 ⁻ , 6237.4γ to 5/2 ⁺ , 8000.0γ to 3/2 ⁺ .
8005.0 10	5/2 ⁺	0.011 keV 5		OP	XREF: P(8004)W(8010)X(8007) J ^π : L(d, ^3He)=2 from 0 ⁺ ; spin=5/2 from

Continued on next page (footnotes at end of table)

Adopted Levels, Gammas (continued)

^{35}Cl Levels (continued)					
E(level) [†]	J ^π [#]	T _{1/2} [‡]	XREF		Comments
8010 10	(7/2 to 17/2) ⁺			N	γ(θ) in (p,γ).
8019.2 35			L		J ^π : L(α,d)=6 from 3/2 ⁺ .
8035.7 10	1/2 ⁻ , 3/2 ⁻	0.026 keV 7		OP	E(level): other: 8034 3 from ³⁴ S(p,p),(p,p'γ):resonances.
8038.4 11	1/2 ⁺ , 3/2 ⁺	0.30 keV 2		OP	E(level): weighted average of 8038.5 11 from (p,γ), and 8038 3 from ³⁴ S(p,p),(p,p'γ):resonances.
					J ^π : spin=3/2 from γ(θ) in (p,γ), but 1/2 ⁺ from R-Matrix analysis in (p,p); 8037.5γ M1+E2 to 3/2 ⁺ .
8075.9 10	(5/2)		L	0	E(level): weighted average of 8075.1 46 from (α,p) and 8075.9 10 from (p,γ).
					J ^π : primary 3897.8γ to 3/2 ⁻ , 8074.9γ to 3/2 ⁺ ; 2912.4g to 7/2 ⁻ , 3962.0γ to 7/2 ⁺ .
8080				0	
8096.1 11	(5/2)			0	J ^π : primary 2932.6γ to 7/2 ⁻ , 5449.9γ to 7/2 ⁺ ; 3918.0γ to 3/2 ⁻ , 4177.4γ to 3/2 ⁺ .
8100 10	(7/2 to 17/2) ⁺			N	J ^π : L(α,d)=6 from 3/2 ⁺ .
8106.4 11	3/2 ⁺		L	0	E(level): other: 8103.8 42 from (α,p).
					J ^π : spin=3/2 from γ(θ) in (p,γ); 8105.4γ M1+E2 to 3/2 ⁺ .
8113.3 11	(1/2 ⁺ , 3/2, 5/2 ⁺)		L	0	XREF: L(8121?)
					J ^π : primary 2897.4γ (5/2 ⁺), 4145.7γ to 1/2 ⁺ .
8147.2 11	3/2 ⁻	0.56 keV 6		OP	XREF: P(8150)
					J ^π : from R-Matrix analysis in (p,p); also proposed by 1976Me12 in (p,γ) based on strengths of primary γ rays.
					T _{1/2} : from ³⁴ S(p,p),(p,p'γ):resonances in 1977Ou01.
8148 3	1/2 ⁻	2.66 keV 27		P	
8150 3	3/2 ⁻	0.56 keV 6		P	
8156.8 11	(5/2 ⁺)			0	J ^π : primary 3978.7γ to 3/2 ⁻ , 8155.8γ to 3/2 ⁺ , 4213.6γ to 9/2 ⁺ .
8171.7 42			L		
8179.4 12	(3/2 ⁻ , 5/2, 7/2 ⁺)			0	J ^π : primary 5016.0γ to 7/2 ⁻ , 5485.0γ to 3/2 ⁺ .
8208.2 10	5/2 ⁺	0.033 keV 10		OP	T=3/2
				WX	XREF: W(8210)X(8204)
					E(level): other: 8209 3 from ³⁴ S(p,p),(p,p'γ):resonances.
					J ^π : L(pol d, ³ He)=2 and L+1/2 from analyzing power.
8211 3	1/2 ⁺	0.094 keV 15	L	OP	XREF: O(8209)
					E(level): from (p,p):resonances. Other: 8210.2 46 from (α,p).
8216.3 10	5/2 ⁺	0.014 keV 3	1	OP	XREF: w(8210)x(8204)
				WX	J ^π : from γ(θ) in (p,γ) and R-Matrix analysis in (p,p):resonances.
8242.4 12	3/2 ⁻	0.140 keV 15	1	OP	XREF: l(8246.0)P(8243)
					J ^π : from γ(θ) in (p,γ) and R-Matrix analysis in (p,p):resonances.
8251 5	(3/2 ⁺ , 5/2 ⁺)		1	0	XREF: l(8246.0)
					J ^π : primary 4137γ to 7/2 ⁺ , 7031γ to 1/2 ⁺ .

Continued on next page (footnotes at end of table)

Adopted Levels, Gammas (continued)

^{35}Cl Levels (continued)				
E(level) [†]	J ^π [#]	T _{1/2} [‡]	XREF	Comments
8269.0 5	5/2 ⁺	0.005 keV 3	L OP	XREF: L(8270.6) E(level): other: 8271 3 from $^{34}\text{S}(\text{p,p}), (\text{p,p}'\gamma)$:resonances. J ^π : 3/2 ⁺ , 5/2 ⁺ from R-Matrix analysis in (p,p):resonances; primary 5105.5γ to 7/2 ⁻ rules out 3/2 ⁺ , since it would require Mult=M2 which is unlikely based on RUL and Γ=0.005 keV.
8277.2 5	(5/2) ⁺	0.006 keV 3	OP	XREF: P(8279) J ^π : 3/2 ⁺ , 5/2 ⁺ from R-Matrix analysis in (p,p):resonances; primary 3939.2γ to 9/2 ⁻ disfavors both 3/2 ⁺ and 5/2 ⁺ , but 3/2 ⁺ is much less likely.
8282.4 5	(3/2 ⁻ , 5/2)		0	J ^π : primary 3118.9γ to 7/2 ⁻ , 4104.2γ to 3/2 ⁻ , 8281.4γ to 3/2 ⁺ .
8287.82 30	1/2 ⁻	0.04 keV 1	L OP	XREF: L(8292)P(8290) J ^π : from γ(θ) in (p,γ) and R-Matrix analysis in (p,p):resonances.
8297.2 9	3/2 ⁻	0.073 keV 15	OP	XREF: P(8300)
8298.2 5	3/2 ⁺		1 0	XREF: l(8292) J ^π : spin=3/2 from γ(θ) in (p,γ); 8297.1γ M1+E2 to 3/2 ⁺ .
8318.1 3	(5/2 ⁺)		L 0	XREF: L(8321.0) J ^π : primary 4350.5γ to 1/2 ⁺ , 5154.7γ to 7/2 ⁻ , 5671.9γ to 7/2 ⁺ .
8319.3 ^a 5	15/2 ⁻	12.5 fs 7	F H	E(level): weighted average of 8319.4 9 from ($^{28}\text{Si}, \alpha\text{p}\gamma$) and 8319.2 5 from ($^{16}\text{O}, \alpha\text{p}\gamma$). J ^π : 2911.9γ E2, ΔJ=2 to 11/2 ⁻ , 2232.7γ to 13/2 ⁻ ; band assignment. T _{1/2} : lifetime τ=18 fs 1 from ($^{28}\text{Si}, \alpha\text{p}\gamma$) (2013Bi10, DSAM). T _{1/2} : from (HI,xnγ).
8323.1 13	(3/2 ⁺ , 5/2 ⁺)		L 0	XREF: L(8321.0) J ^π : primary 4209.1γ to 7/2 ⁺ , 7102.9γ to 1/2 ⁺ .
8346 5			0	
8381.8 8	5/2 ⁺	0.023 keV 7	OP	XREF: P(8383) J ^π : from γ(θ) in (p,γ) and R-Matrix analysis in (p,p):resonances.
8389.4 16			0	
8402.9 5	5/2 ⁻	0.002 keV 1	L OP	XREF: L(8398.8)P(8404) E(level): others: 8398.8 38 from (α,p), and 8404 3 from $^{34}\text{S}(\text{p,p}), (\text{p,p}'\gamma)$:resonances. J ^π : 5/2 ⁻ , 7/2 ⁻ from R-Matrix analysis in (p,p):resonances; primary 5708.5γ to 3/2 ⁺ rules out based on RUL and Γ=0.002 keV.
8405.8 14	(5/2 ⁻)	0.001 keV 1	OP	E(level): weighted average of 8405.7 14 from (p,γ), and 8406 3 from $^{34}\text{S}(\text{p,p}), (\text{p,p}'\gamma)$:resonances. J ^π : (5/2 ⁻ , 7/2 ⁻) from R-Matrix analysis in (p,p):resonances; primary 8404.6γ to 3/2 ⁺ .
8408.4 18	1/2 ⁻	0.125 keV 15	OP	E(level): weighted average of 8409.6 18 from $^{34}\text{S}(\text{p},\gamma)$ and 8409 3 from $^{34}\text{S}(\text{p,p}), (\text{p,p}'\gamma)$:resonances.
8416.7 5	1/2 ⁺	0.026 keV 7	OP	E(level): weighted average of 8416.7 5 from

Continued on next page (footnotes at end of table)

Adopted Levels, Gammas (continued)

^{35}Cl Levels (continued)					
E(level) [†]	J ^π [#]	T _{1/2} [‡]	XREF		Comments
8430.3 5			L	O	(p,γ), and 8418 3 from $^{34}\text{S}(\text{p,p}),(\text{p,p}'\gamma)$:resonances.
8434.8 5	(3/2) ⁺	0.090 keV 15		OP	J ^π : 1/2 ⁺ from R-Matrix analysis in (p,p):resonances; but primary 4473.5γ to 9/2 ⁺ and 7196.5γ to 1/2 ⁺ in (p,γ) gives (5/2 ⁺), which may indicate two different resonances with close energies.
8464.3 5	(1/2 ⁺ ,3/2,5/2 ⁺)			O	E(level): other: 8429.2 40 from (α,p). XREF: P(8435)
8484.4 5	(3/2) ⁺	0.012 keV 5	1	OP	J ^π : primary 4496.7γ to 1/2 ⁺ , 3248.3γ to (5/2 ⁺). E(level): weighted average of 8484.4 5 from (p,γ), and 8486 3 from $^{34}\text{S}(\text{p,p}),(\text{p,p}'\gamma)$:resonances.
8486.2 5	3/2 ⁻	0.150 keV 15	1	OP	J ^π : 3/2 ⁺ ,5/2 ⁺ from R-Matrix analysis in (p,p):resonances; 3/2 ⁺ proposed by 2001Vo24 in (p,γ) based on proposed isobaric analog to 2940, 3/2 ⁺ level in ^{35}S . E(level): weighted average of 8486.1 5 from (p,γ), and 8488 3 from $^{34}\text{S}(\text{p,p}),(\text{p,p}'\gamma)$:resonances.
8487.0 5	15/2 ⁻	<70 fs	F	H	E(level): weighted average of 8487.8 10 from ($^{28}\text{Si},\alpha\text{p}\gamma$) and 8486.8 5 from ($^{16}\text{O},\alpha\text{p}\gamma$). J ^π : 3080.0γ E2, ΔJ=2 to 11/2 ⁻ , 2399.8γ D(+Q) to 13/2 ⁻ . T _{1/2} : lifetime τ<0.1 ps from ($^{28}\text{Si},\alpha\text{p}\gamma$) (2007Ks01, DSAM).
8506.5 5			L	O	E(level): other: 8505.1 47 from (α,p).
8514.3 5	1/2 ⁻	0.150 keV 15		OP	XREF: P(8515)
8533.9 5			L	O	E(level): weighted average of 8532.4 43 from (α,p) and 8533.9 5 from (p,γ).
8558.2? 13	(13/2 ⁻)	<0.14 ps	F		XREF: F(?) J ^π : 2631γ D(+Q), ΔJ=1 to 11/2 ⁻ in ($^{28}\text{Si},\alpha\text{p}\gamma$). T _{1/2} : lifetime τ<0.2 ps from ($^{28}\text{Si},\alpha\text{p}\gamma$) (2007Ks01, DSAM).
8571.9 5	(5/2) ⁺	0.08 keV 1	L	OP	T=(3/2) XREF: P(8573)X(8580) E(level): weighted average of 8571.9 5 from (p,γ), and 8573 3 from $^{34}\text{S}(\text{p,p}),(\text{p,p}'\gamma)$:resonances. Other: 8567.9 37 from (α,p).
8580.5 5	1/2 ⁺	0.75 keV 8		OP	XREF: P(8582)
8585.8 5				O	
8590.1 5	5/2 ⁺	0.003 keV 2		OP	XREF: P(8592) E(level): weighted average of 8590.0 5 from (p,γ), and 8592 3 from $^{34}\text{S}(\text{p,p}),(\text{p,p}'\gamma)$:resonances.
8612.0 5	3/2 ⁺ ,5/2 ⁺		1	O	J ^π : 3/2 ⁺ ,5/2 ⁺ from R-Matrix analysis in (p,p):resonances; primary 5426.6γ to 7/2 ⁻ ruled out 3/2 ⁺ based on RUL and Γ=0.003 keV. XREF: W(8610)
8613.7 5	5/2 ⁺	0.175 keV 20	1	OP	J ^π : L(d, ^3He)=2 from 0 ⁺ ; XREF: P(8616)

Continued on next page (footnotes at end of table)

Adopted Levels, Gammas (continued) ^{35}Cl Levels (continued)

E(level) [†]	J ^π [#]	T _{1/2} [‡]	XREF	Comments
8618.2 5	3/2 ⁺ , 5/2 ⁺	0.002 keV 1	1 OP	XREF: P(8620) E(level): weighted average of 8618.1 5 from (p,γ), and 8620 3 from $^{34}\text{S}(\text{p,p}), (\text{p,p}'\gamma)$:resonances.
8630.1 3	7/2 ⁻	0.001 keV 1	0	J ^π : 4282.1γ M1+E2 to 9/2 ⁻ , 6866.4γ D+Q to 5/2 ⁺ .
8633 3	(3/2, 5/2 ⁺)	0.001 keV 1	P	
8640	(5/2 ⁺)		0	J ^π : spin=(5/2) from γ(θ) in (p,γ); primary 4697γ to 9/2 ⁺ .
8641.4 5	3/2 ⁺ , 5/2 ⁺	0.003 keV 2	OP	XREF: P(8643)
8654 4			L	
8686.2 5	5/2 ⁻	0.001 keV 1	OP	E(level): weighted average of 8686.2 5 from (p,γ), and 8687 3 from $^{34}\text{S}(\text{p,p}), (\text{p,p}'\gamma)$:resonances.
8688.3 5	1/2 ⁺	0.20 keV 2	OP	J ^π : 5/2 ⁻ , 7/2 ⁻ from R-Matrix analysis in (p,p):resonances; primary 7446.0γ to 1/2 ⁺ disfavors both 5/2 ⁻ and 7/2 ⁻ , but 7/2 ⁻ is much less likely to 3/2 and decay schemes in (p,γ).
8691 3	1/2 ⁻	6.4 keV 7	P	XREF: P(8689)
8696.9 5	3/2 ⁻	0.8 keV 1	L OP	XREF: L(8698)P(8698)
8700 10	(7/2 to 17/2) ⁺		N	J ^π : L(α,d)=6 from 3/2 ⁺ .
8706.3 5			0	
8717.8 5			L 0	E(level): weighted average of 8720.9 37 from (α,p) and 8717.7 5 from (p,γ).
8750.9 5	3/2 ⁻	0.30 keV 3	OP	E(level): weighted average of 8750.9 5 from (p,γ), and 8750 3 from $^{34}\text{S}(\text{p,p}), (\text{p,p}'\gamma)$:resonances.
8767.0 5			0	
8772.9 5	1/2 ⁻	0.57 keV 6	OP	XREF: P(8775)
8779.8 5	3/2 ⁻	0.214 keV 25	OP	XREF: P(8781)
				E(level): weighted average of 8779.8 5 from (p,γ), and 8781 3 from $^{34}\text{S}(\text{p,p}), (\text{p,p}'\gamma)$:resonances.
8787.2 5	(3/2 ⁺)	0.001 keV 1	L OP	E(level): others: 8786.3 38 from (α,p), and 8788 3 from $^{34}\text{S}(\text{p,p}), (\text{p,p}'\gamma)$:resonances.
				J ^π : (3/2 ⁺ , 5/2 ⁺) from R-Matrix analysis in (p,p):resonances; primary 3623.7γ to 7/2 ⁻ .
8788.3 ^b 6	(15/2 ⁺)	<0.28 ps	F H	XREF: F(?) J ^π : 915.4γ D, ΔJ=1 to 13/2 ⁺ ; band assignment.
				T _{1/2} : lifetime τ<0.4 ps from ($^{28}\text{Si}, \alpha\text{p}\gamma$) (2007Ks01, DSAM).
8798.4 5	(3/2 ⁺ , 5/2 ⁺)	0.001 keV 1	OP	E(level): weighted average of 8798.4 5 from (p,γ), and 8799 3 from $^{34}\text{S}(\text{p,p}), (\text{p,p}'\gamma)$:resonances.
8820.9 5	(1/2 ⁺ to 7/2 ⁺)		0	J ^π : primary 5817.6γ to 5/2 ⁺ , 5817.6γ to 3/2 ⁺ .
8824.2 5	1/2 ⁺	1.70 keV 17	OP	XREF: P(8825)
				E(level): weighted average of 8824.2 5 from (p,γ), and 8825 3 from $^{34}\text{S}(\text{p,p}), (\text{p,p}'\gamma)$:resonances.
8829.3 5	1/2 ⁻	12.2 keV 12	OP	E(level): weighted average of 8829.3 5 from (p,γ), and 8829 3 from

Continued on next page (footnotes at end of table)

Adopted Levels, Gammas (continued)

^{35}Cl Levels (continued)					
E(level) [†]	J ^π [#]	T _{1/2} [‡]	XREF		Comments
					$^{34}\text{S}(\text{p,p}), (\text{p,p}'\gamma)$: resonances. J ^π : 1/2 ⁻ from R-Matrix analysis in (p,p): resonances; but primary 6183.0γ to 7/2 ⁺ gives J≥3/2, which may indicate a different level.
8830.3	1/2 ⁺	0.080 keV 15		P	
8833.1 5	(5/2)			O	J ^π : primary 3617.1γ to (5/2 ⁺), 4063.0γ to (5/2 ⁻); 4719.1γ to 7/2 ⁺ , 5669.6γ to 7/2 ⁻ . XREF: P(8839)
8838.1 5	5/2 ⁻ , 7/2 ⁻	0.001 keV 1	1	OP	XREF: N(8840)
8844.3 ^b 4	17/2 ⁺	5.9 ps 11	F HI	N	E(level): weighted average of 8845.1 9 from ($^{28}\text{Si}, \alpha\text{p}\gamma$), 8844.1 5 from ($^{16}\text{O}, \alpha\text{p}\gamma$), and 8844.2 4 from ($^{14}\text{N}, \alpha\text{p}\gamma$). J ^π : 971.38γ E2, ΔJ=2 to 13/2 ⁺ , 526γ to 15/2 ⁻ ; band assignment. T _{1/2} : weighted average of 6.1 ps 11 from ($^{16}\text{O}, \alpha\text{p}\gamma$) (1991Ja11, DSAM) and 5.5 ps 14 from ($^{14}\text{N}, \alpha\text{p}\gamma$) (1976Wa11, DSAM). E(level): weighted average of 8856.1 5 from (p,γ), and 8858 3 from $^{34}\text{S}(\text{p,p}), (\text{p,p}'\gamma)$: resonances. J ^π : 3/2 ⁺ , 5/2 ⁺ from R-Matrix analysis in (p,p): resonances; primary 5692.6γ to 7/2 ⁻ rules out 3/2 ⁺ based on RUL and Γ=0.010 keV. E(level): weighted average of 8868.6 5 from (p,γ), and 8870 3 from $^{34}\text{S}(\text{p,p}), (\text{p,p}'\gamma)$: resonances.
8856.2 5	5/2 ⁺	0.010 keV 5		OP	
8868.6 5	3/2 ⁺ , 5/2 ⁺	0.027 keV 10		OP	
8884.1 5				O	
8886.1 5	(5/2)		L	O	E(level): other: 8887.5 41 from (α,p). J ^π : primary 3722.6γ to 7/2 ⁻ , 6239.8γ to 7/2 ⁺ ; 4707.9γ to 3/2 ⁻ , 4967.2γ to 3/2 ⁺ . J ^π : primary 4950.0γ to 9/2 ⁺ , 6198.8γ to 3/2 ⁺ . J ^π : primary 4726.6γ to 3/2 ⁻ , 6210.3γ to 3/2 ⁺ , 7684.5γ to 1/2 ⁺ . XREF: P(8909)x(8921)
8893.3 5	(5/2 ⁺ , 7/2 ⁺)			O	J ^π : 3/2 ⁺ , 5/2 ⁺ from R-Matrix analysis in (p,p): resonances; primary 4963.5γ to 9/2 ⁺ rules out 3/2 ⁺ based on RUL and Γ=0.002 keV.
8904.8 5	(1/2, 3/2, 5/2 ⁺)			O	XREF: x(8921)
8906.8 5	5/2 ⁺	0.002 keV 1		OP	J ^π : primary 4571.7γ to 9/2 ⁻ , 7156.0γ to 5/2 ⁺ .
8919.8 5	(5/2 ⁻ , 7/2, 9/2 ⁺)			O	XREF: x(8921)
8933.3 5				O	
8953.1 5	3/2 ⁺	0.075 keV 15		OP	J ^π : spin=3/2 from γ(θ) in (p,γ) and (3/2) ⁺ from R-Matrix analysis in (p,p): resonances. XREF: P(8959)
8957.9 5	(1/2 ⁻ , 3/2 ⁻)	0.04 keV 1	L	OP	E(level): others: 8957.7 44 from (α,p) and 8959 3 from (p,p): resonances. XREF: P(8983)
8981.9 5	5/2 ⁻ , 7/2 ⁻	0.003 keV 2		OP	

Continued on next page (footnotes at end of table)

Adopted Levels, Gammas (continued)

^{35}Cl Levels (continued)				
E(level) [†]	J ^π #	T _{1/2} [‡]	XREF	Comments
8984.2 5	5/2 ⁺	0.025 keV 10	OP	E(level): weighted average of 8981.9 5 from (p,γ), and 8983 3 from $^{34}\text{S}(\text{p,p}),(\text{p,p}'\gamma)$:resonances. XREF: P(8985) J ^π : 3/2 ⁺ , 5/2 ⁺ from R-Matrix analysis in (p,p):resonances; primary 5820.7γ to 7/2 ⁻ rules out 3/2 ⁺ based on RUL and Γ=0.025 keV.
8988.4 5			0	
8992.4 5			0	
8996.7 5	5/2 ⁻ , 7/2 ⁻	0.002 keV 1	L OP	XREF: L(8996.9)P(8998)
9001.1 5			0	
9019.3 5	3/2 ⁻	3.27 keV 30	OP	XREF: P(9021) T _{1/2} : weighted average of 3.1 keV 3 from (p,γ) and 3.50 keV 35 from (p,p):resonances.
9024.5 5			1 0	
9029.9 5	(5/2) ⁺	0.04 keV 1	1 OP	XREF: P(9031)
9033.1 5			0	
9038.4 5	1/2 ⁻	0.292 keV 30	OP	XREF: P(9040) E(level): weighted average of 9038.3 5 from (p,γ), and 9040 3 from $^{34}\text{S}(\text{p,p}),(\text{p,p}'\gamma)$:resonances.
9048.4 5	5/2 ⁻ , 7/2 ⁻	0.001 keV 1	OP	XREF: P(9048)
9050 3	(5/2) ⁺	0.095 keV 15	P	
9081.5 4	5/2 ⁺	59 eV 10	L OP	XREF: L(9085.7)P(9084) E(level): weighted average of 9081.4 4 from (p,γ), and 9084 3 from $^{34}\text{S}(\text{p,p}),(\text{p,p}'\gamma)$:resonances. J ^π : spin=5/2 from γ(θ) in (p,γ) and (5/2) ⁺ from R-Matrix analysis in (p,p):resonances. T _{1/2} : weighted average of 65 eV 20 from (p,γ) and 57 eV 10 from (p,p):resonances.
9088.5 5			L OP	XREF: L(9085.7)P(9089) E(level): weighted average of 9088.5 5 from (p,γ), and 9089 4 from $^{34}\text{S}(\text{p,p}),(\text{p,p}'\gamma)$:resonances.
9099.3 5			0	
9100.4 5	3/2 ⁻	0.20 keV 2	OP	XREF: P(9101)
9107.4 5			L OP	XREF: L(9109.3)P(9108) E(level): weighted average of 9107.4 5 from (p,γ), and 9108 4 from $^{34}\text{S}(\text{p,p}),(\text{p,p}'\gamma)$:resonances.
9110.1 5			L OP	XREF: L(9109.3)P(9114) E(level): weighted average of 9110.0 5 from (p,γ), and 9114 4 from $^{34}\text{S}(\text{p,p}),(\text{p,p}'\gamma)$:resonances.
9124.0 5			OP	XREF: P(9120)
9127 4	5/2 ^{&}		J P	E(level): from (p,p):resonances. Other: 9127 9 from $^{31}\text{P}(\alpha,\text{p})$:resonances.
9135.1 5		2.0 keV 4	Op	XREF: p(9136) T _{1/2} : from (p,γ).
9138.3 5			Op	XREF: p(9136)
9147 4	(7/2 to 17/2) ⁺		N P	XREF: N(9150)P(9147) J ^π : L(α,d)=6 from 3/2 ⁺ .

Continued on next page (footnotes at end of table)

Adopted Levels, Gammas (continued) ^{35}Cl Levels (continued)

E(level) [†]	J ^π #	XREF	Comments
9155.7 5		O	
9157.1 5	5/2 ⁻	OP	XREF: P(9159) J ^π : spin=5/2 from $\gamma(\theta)$ in (p, γ); 5993.6 γ M1+E2 to 7/2 ⁻ .
9161 4	1/2 ^{&}	J 1	XREF: l(9160.0)
9163.3 5	(3/2 ⁻)	1 OP	XREF: l(9160.0)P(9168) E(level): other: 9168 4 from (p,p):resonances.
9184.2 5		Op	XREF: p(9185)
9188.8 5		Op	XREF: p(9185)
9194.1 5		L OP	E(level): others: 9194.0 39 from (α ,p), and 9196 4 from (p,p):resonances.
9207 4	9/2 ⁺	P	
9222 4	5/2 ⁺	P	X XREF: X(9220) J ^π : L(pol d, ^3He)=2 and L+1/2 from analyzing power for a group at E=9220 120.
9227 4	5/2 ⁻	P	
9241 4	7/2 ⁺	P	
9255 4	1/2 ^{&}	J P	E(level): from (p,p):resonances. Other: 9256 4 from $^{31}\text{P}(\alpha,p)$:resonances.
9264.9 39		L	
9271 4		P	
9276 4		P	
9284 4	5/2 ⁻	P	
9294 4		P	
9299 4		P	
9316.2 39		L	
9325 4	5/2 ⁺	P	
9335 4	7/2 ⁺	L P	E(level): weighted average of 9336 4 from $^{34}\text{S}(\text{p,p}),(\text{p,p}'\gamma)$:resonances and 9334 5 from $^{32}\text{S}(\alpha,p)$.
9345 4		P	
9349 4		P	
9355 4		P	
9376 4		L	
9400 9	1/2 ^{&}	J	
9450 10	(7/2 to 17/2) ⁺	N	J ^π : L(α ,d)=6 from 3/2 ⁺ .
9456 4	3/2 ^{&}	J	
9476 4		L	
9481.0 18	3/2 ^{&}	J	
9508 4	(1/2 ⁻ , 3/2, 5/2 ⁺)	L	X XREF: X(9514) J ^π : L(pol d, ^3He)=(1,2) from 0 ⁺ .
9551 6	5/2 ^{&}	J	
9673 6	1/2 ^{&}	J	
9712.7 27	1/2 ^{&}	J L	E(level): weighted average of 9713.0 27 from $^{31}\text{P}(\alpha,p),(\alpha,n)$:resonances and 9711.8 44 from (α ,p).
9740 4		L	
9751.1 27	7/2 ^{&}	J	
9799 5		L	
9814.0 27	5/2 ⁽⁺⁾	J	X XREF: X(9820) J ^π : spin=5/2 from angular in (α ,p):resonance; L(d, ^3He)=(1,2) from 0 ⁺ .
9836 5		L	
9869.8 27	1/2 ^{&}	J L	XREF: L(9870)
9900.8 27	3/2 ^{&}	J	

Continued on next page (footnotes at end of table)

Adopted Levels, Gammas (continued)

^{35}Cl Levels (continued)				
E(level) [†]	J ^π #	T _{1/2} [‡]	XREF	Comments
9923 4	3/2 ⁻ &		J	
9958 6	3/2&		J	
9969 5	(1/2,3/2)&		J L	XREF: J(9968) E(level): from (α,p) and (α,n):resonances.
10031 5	(1/2 ⁺ ,3/2 ⁻)&		J	
10058 5	3/2&		J	
10075 5	(1/2)&		J L	E(level): from (α,p) and (α,n):resonances.
10090 5	5/2 ⁺ &		J	
10132 5	3/2 ⁻ &		J	
10163 5	&		L	
10172 5	3/2 ⁻ ,5/2 ⁻ &		J	
10179 5			L	
10180.6 ^a 5	19/2 ⁻	42.3 fs 28	F H	E(level): weighted average of 10181.1 10 from (²⁸ Si,αpγ) and 10180.5 5 from (¹⁶ O,αpγ). J ^π : 1693.5γ E2, ΔJ=2 to 15/2 ⁻ ; 1336.3γ D, ΔJ=1 to 17/2 ⁺ ; band assignment. T _{1/2} : lifetime τ=61 fs 4 from (²⁸ Si,αpγ) (2013Bi10, DSAM).
10215 5	3/2&		J L	XREF: L(10218.0)
10221.9 11	(17/2 ⁻)	<0.28 ps	F H	XREF: F(?) E(level): from (¹⁶ O,αpγ). J ^π : 1377.8γ to 17/2 ⁺ consistent with ΔJ=0. T _{1/2} : lifetime τ<0.4 ps from (²⁸ Si,αpγ) (2007Ks01, DSAM).
10236 5	3/2 ⁻ &		J	
10278 5	1/2&		J	
10295 5			J	
10322 5	5/2 ⁺ &		J	
10340 5	(3/2,5/2)&		J	
10359 5	3/2&		J	
10385 5			J	
10395 5			L	
10399 5	3/2 ⁺ ,7/2 ⁺ &		J	
10431 5	5/2 ⁺ &		J	
10465 5	5/2 ⁺ &		J L	E(level): weighted average of 10467 5 from ³¹ P(α,p),(α,n):resonances and 10463.2 49 from (α,p).
10486 5	5/2 ⁺ &		J	
10517 6			L	
10534 5	1/2 ⁺ ,5/2 ⁺ &		J	
10548 6			L	
10584 5	3/2 ⁺ &		J L	E(level): weighted average of 10587 5 from ³¹ P(α,p),(α,n):resonances and 10579.5 54 from (α,p).
10642 5	5/2 ⁺ &		J L	E(level): weighted average of 10641 5 from ³¹ P(α,p),(α,n):resonances and 10643.0 50 from (α,p).
10678 5	3/2 ⁺ ,5/2 ⁺ &		J	

Continued on next page (footnotes at end of table)

Adopted Levels, Gammas (continued) ^{35}Cl Levels (continued)

E(level) [†]	J ^π [#]	T _{1/2} [‡]	XREF	Comments
10696 5	7/2 ⁻ &		J	
10733 5	3/2 ⁺ &		J L	E(level): weighted average of 10734 5 from $^{31}\text{P}(\alpha, \text{p}), (\alpha, \text{n})$: resonances and 10732.1 50 from (α, p) .
10760 5	3/2 ⁺ , 7/2 ⁻ &		J L	E(level): weighted average of 10760 5 from $^{31}\text{P}(\alpha, \text{p}), (\alpha, \text{n})$: resonances and 10759.1 55 from (α, p) .
10800 5	5/2 ⁺ &		J	
10816 5	1/2 ⁺ , 3/2 ⁻ &		J	
10844 5	3/2 ⁻ &		J	
10858.5 ^b 8	19/2 ⁺	<0.28 ps	F H	XREF: F(?) J ^π : 2069.8γ E2, ΔJ=2 to 15/2 ⁺ , 2014.7γ D, ΔJ=1 to 17/2 ⁺ ; band assignment. T _{1/2} : lifetime τ<0.4 ps from ($^{28}\text{Si}, \alpha \text{p} \gamma$) (2007Ks01, DSAM).
10870 5	(5/2, 7/2) &		J	
10886 5			J	
10909 5	3/2 ⁺ , 5/2 ⁻ &		J	
10928 5	7/2 &		J	
10942 5			J	
10964 5	5/2 ⁺ &		J	
10973 5			J	
10998 5	3/2 ⁺ , 7/2 ⁻ &		J	
11034 5	3/2 ⁺ &		J	
11063 5	1/2 ⁺ , 9/2 ⁺ &		J	
11082 5	5/2 ⁺ , 9/2 ⁺ &		J	
11105 5			J	
11123 5	3/2 &		J	
11142 5	5/2 ⁺ &		J	
11154 5			J	
11169 5			J	
11179 5	3/2 ⁻ , 5/2 ⁺ &		J	
11185 5			J	
11198 5	3/2 &		J	
11230 5			J	
11246 5	3/2 &		J	
11265 5	3/2 ⁻ &		J	
11287 5			J	
11307 5			J	
11314 5	1/2 ⁻ , 5/2 ⁺ &		J	
11327 5	1/2 ⁻ , 5/2 ⁺ &		J	
11357 5			J	
11375 5			J	
11390 5			J	
11410 5			J	
11421 5	3/2 ⁻ &		J	
11434 5			J	
11458.7 ^b 7	21/2 ⁺	43.0 fs 2I	F H J	E(level): weighted average of 11459.2 I2 from

Continued on next page (footnotes at end of table)

Adopted Levels, Gammas (continued)

^{35}Cl Levels (continued)				
E(level) [†]	J^π [#]	$T_{1/2}$ [‡]	XREF	Comments
				($^{28}\text{Si}, \alpha\gamma$) and 11458.5 7 from ($^{16}\text{O}, \alpha\gamma$). Other: 11460 5 from $^{31}\text{P}(\alpha, p), (\alpha, n)$:resonances. J^π : 2614.5 γ E2, $\Delta J=2$ to $17/2^+$; band assignment. $T_{1/2}$: lifetime $\tau=62$ fs 3 from ($^{28}\text{Si}, \alpha\gamma$) (2013Bi10, DSAM).
11473 5	$1/2^-, 5/2^- \&$		J	
11492 5			J	
11505 5	$3/2^-, 5/2^- \&$		J	
11525 5			J	
11540 5			J	
11551 5	$3/2^-, 5/2^+ \&$		J	
11565 5			J	
11589 5			J	
11609 5	$5/2^+, 7/2^- \&$		J	
11627 5	$1/2^+, 5/2^+ \&$		J	
11638 5			J	
11651 5			J	
11684 5	$1/2^+, 5/2^+ \&$		J	
11773 5	$3/2^-, 7/2^- \&$		J	
11783 5	$3/2^- \&$		J	
1.25×10^4 5			J	X T=3/2 XREF: J(12900) E(level): from (pol d, ^3He). E(level): weighted average of 12572.2 12 from ($^{28}\text{Si}, \alpha\gamma$) and 12571.6 6 from ($^{16}\text{O}, \alpha\gamma$). J^π : 2390.8 γ E2, $\Delta J=2$ to $19/2^-$, 1113.3 γ D, $\Delta J=1$ to $21/2^+$; band assignment. $T_{1/2}$: lifetime $\tau=48$ fs 6 from ($^{28}\text{Si}, \alpha\gamma$) (2013Bi10, DSAM).
12571.7 ^a 6	$23/2^-$	33 fs 4	F H	T=3/2 J^π : 3734 γ to $23/2^-$; band assignment. $T_{1/2}$: lifetime $\tau < 13$ fs from ($^{28}\text{Si}, \alpha\gamma$) (2013Bi10, DSAM).
13900			J	
16306.4 ^a 16	($27/2^-$)	<9.0 fs	F	

[†] Most of Adopted Levels are reported in (p, γ) dataset with precise E(level) values, deduced based on measured γ -ray energies which however are not reported in references in (p, γ) dataset, while E(level) values from E_γ values with uncertainties in other datasets are less precise and the numbers of γ rays are much less than that of (p, γ) datasets. Therefore, a least-squares fit to available E_γ values with uncertainties would not produce E(level) values more precise and complete than those already reported in (p, γ) and the evaluators have adopted E(level) values from (p, γ) or averages of values based on E_γ data from all available datasets where applicable, as noted.

[‡] Sources of $T_{1/2}$ values are listed under comments; Γ from $^{34}\text{S}(p, p), (p, p'\gamma)$:resonances, unless otherwise noted.

[#] For proton-resonance levels populated in (p,p):resonances, J^π assignments are from R-Matrix analysis, unless otherwise noted.

[@] From FRESKO-DWBA analysis of measured $\sigma(\theta)$ of (α, p) (2019Se09).

[&] From angular distributions in (α, p):resonance.

^a Band(A): Band based on $f_{7/2}$ orbital.

^b Band(B): Band based on $13/2^+$.

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult.#	$\delta^\#$	Comments
1219.39	1/2 ⁺	1219.4 1	100	0.0	3/2 ⁺	M1+E2	0.093 10	B(M1)(W.u.)=0.101 +14-11; B(E2)(W.u.)=2.2 +6-5 E _γ : weighted average of 1219.3 2 from ³⁵ Ar ε decay and 1219.4 1 from (p,p'γ). Other: 1219.5 10 from (α,pγ). Mult.,δ: ΔJ=1 from γ(θ) in (p,p'γ), E1+M2 ruled out by RUL. δ from adopted B(E2)†=0.00076 7 measured in Coulomb excitation and adopted lifetime τ=0.172 ps 20.
1763.18	5/2 ⁺	543.79 1763.16 10	<0.2 100	1219.39 0.0	1/2 ⁺ 3/2 ⁺	[E2] M1+E2	-2.85 10	I _γ : from (p,γ). Other: <0.5 from (α,pγ). B(M1)(W.u.)=0.00122 +14-12; B(E2)(W.u.)=12.1 +11-9 E _γ : weighted average of 1763.0 2 from ³⁵ Ar ε decay, 1763.1 2 from (²⁴ Mg,αpγ), 1763.3 2 from (¹⁶ O,αpγ), 1763.15 10 from (¹⁴ N,αpnγ), and 1763.2 1 from (p,p'γ). Other: 1763.2 6 from (α,pγ). Mult.: from γγ(ADO) in (¹⁶ O,αpγ) with ΔJ=1 and γ(lin pol) in (²⁸ Si,αpγ). δ: weighted average of -3.0 1 from (α,pγ), -2.66 12 from (p,p'γ), -2.95 45 from Coulomb excitation, -2.6 4 from (¹⁶ O,αpγ), all from γ(θ) or pγ(θ). Other: -0.42 17 from γγ(θ) in (p,γ); 1.8 +10-4; deduced by the evaluators from adopted B(E2)†=0.0106 7 and T _{1/2} =0.36 ps 3.
2645.66	7/2 ⁺	882.80 11	11.4 10	1763.18	5/2 ⁺	M1+E2	+0.21 5	B(M1)(W.u.)=0.0196 +35-27; B(E2)(W.u.)=4.2 +23-18 E _γ : weighted average of 882.9 1 from (²⁴ Mg,αpγ), 882.7 5 from (¹⁶ O,αpγ), 882.32 35 from (¹⁴ N,αpnγ), 882.4 10 from (α,pγ), and 882.3 3 from (p,p'γ). I _γ : unweighted average of 8.7 6 from (²⁸ Si,αpγ), 15.2 13 from (²⁴ Mg,αpγ), 11.08 28 from (¹⁶ O,αpγ), 11.1 22 from (α,pγ), 9.5 11 from (p,γ), and 13 5 from (p,p'γ). Mult.: D+Q from pγ(θ) in (α,pγ); E1+M2 ruled out by RUL. δ: weighted average of +0.25 5 from (α,pγ) and +0.17 5 from (p,p'γ). I _γ : from (α,pγ). Other: <2.2 from (p,γ). B(E2)(W.u.)=3.6 +6-4 E _γ : weighted average of 2645.5 4 from (²⁴ Mg,αpγ), 2645.8 5 from (¹⁶ O,αpγ), 2645.79 30 from (¹⁴ N,αpnγ), and 2645.7 2 from (p,p'γ). Other: 2646.2 15 from (α,pγ). I _γ : others: 100.0 31 from (²⁴ Mg,αpγ), 100.0 22 from (¹⁶ O,αpγ), 100.0 22 from (α,pγ), and 100.0 20 from (p,p'γ). Other: 100 6 from (²⁸ Si,αpγ). Mult.: from γγ(θ)(DCO) and γγ(lin pol) in (²⁸ Si,αpγ); γ(θ) and γ(lin pol) in (p,p'γ).
		1426.24 2645.70 20	<1.2 100.0 11	1219.39 0.0	1/2 ⁺ 3/2 ⁺	[M3] E2		B(M1)(W.u.)=0.110 +26-19; B(E2)(W.u.)=3.9 +33-22 E _γ : weighted average of 930.7 2 from ³⁵ Ar ε decay, 930.5 15 from (α,pγ), and 931.3 3 from (p,p'γ). I _γ : weighted average of 14.3 18 from ³⁵ Ar ε decay, 18.2 26 from (α,pγ), 16.3 13 from (p,γ), and 18.6 25 from (p,p'γ). Mult.,δ: from γ(θ) and γ(lin pol) in (p,p'γ).
2694.00	3/2 ⁺	930.9 2	16.3 13	1763.18	5/2 ⁺	M1+E2	+0.09 3	

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult.#	$\delta^\#$	Comments
2694.00	3/2 ⁺	1474.8 3	9.5 10	1219.39	1/2 ⁺	M1+E2	-0.63 21	B(M1)(W.u.)=0.0116 +37-29; B(E2)(W.u.)=8.0 +45-39 E _γ : weighted average of 1474.8 3 from ³⁵ Ar ε decay and 1475 2 from (p,p'γ). I _γ : weighted average of 9.3 10 from ³⁵ Ar ε decay, 11.7 26 from (α,pγ), 9.5 13 from (p,γ), and 8.3 25 from (p,p'γ). Mult.,δ: from γ(θ) and γ(lin pol) in (p,p'γ). B(M1)(W.u.)=0.026 +6-4; B(E2)(W.u.)=0.92 +26-19 E _γ : weighted average of 2693.7 2 from ³⁵ Ar ε decay and 2694.1 6 from (p,p'γ). Other: 2693.2 20 from (α,pγ). I _γ : weighted average of 100.0 17 from ³⁵ Ar ε decay, 100 4 from (α,pγ), 100.0 25 from (p,γ), and 100.0 30 from (p,p'γ). Mult.: from γ(θ) and γ(lin pol) in (p,p'γ). δ: weighted average of +0.17 8 from (α,pγ) and +0.26 2 from (p,p'γ).
		2693.7 2	100.0 17	0.0	3/2 ⁺	M1+E2	+0.26 2	
3002.64	5/2 ⁺	308.64	<5	2694.00	3/2 ⁺	[M1,E2]		I _γ : from (α,pγ).
		356.98	<1	2645.66	7/2 ⁺	[M1,E2]		I _γ : from (α,pγ).
		1239.44	2.2 11	1763.18	5/2 ⁺	[M1,E2]		I _γ : from (p,p'γ).
		1783.20	3.3 11	1219.39	1/2 ⁺	[E2]		B(M1)(W.u.)=0.011 +8-5 if M1, B(E2)(W.u.)=26 +19-12 if E2. B(E2)(W.u.)=6.4 +35-22 I _γ : from (p,p'γ).
		3002.3 4	100.0 10	0.0	3/2 ⁺	M1+E2	+0.08 2	B(M1)(W.u.)=0.034 +14-7; B(E2)(W.u.)=0.09 +7-4 E _γ : weighted average of 3002.1 4 from ³⁵ Ar ε decay, 3002.5 15 from (α,pγ), and 3002.5 5 from (p,p'γ). I _γ : weighted average of 100.0 19 from ³⁵ Ar ε decay and 100.0 10 from (p,p'γ). Mult.: from γ(θ) and γ(lin pol) in (p,p'γ). δ: weighted average of +0.09 3 from (α,pγ) and +0.07 2 from (p,p'γ). Other: -0.22 1 from (p,γ).
3162.87	7/2 ⁻	160.67 20	1.5 4	3002.64	5/2 ⁺	[E1]		B(E1)(W.u.)=6.6×10 ⁻⁵ 17 E _γ : weighted average of 161 1 from (¹⁶ O,αpγ) and 160.66 20 from (¹⁴ N,αpnγ). I _γ : unweighted average of 1.10 30 from (²⁸ Si,αpγ) and 1.89 22 from (p,γ). B(E1)(W.u.)=1.61×10 ⁻⁵ 20 E _γ : weighted average of 517.7 1 from (²⁴ Mg,αpγ), 517.3 5 from (¹⁶ O,αpγ), 517.26 10 from (¹⁴ N,αpnγ), 517.4 15 from (α,pγ), and 517.0 8 from (p,p'γ). I _γ : unweighted average of 9.2 5 from (²⁸ Si,αpγ), 14.3 8 from (²⁴ Mg,αpγ), 11.1 13 from (¹⁶ O,αpγ), 11.1 33 from (α,pγ), 8.9 11 from (p,γ), and 19 8 from (p,p'γ). Mult.: D from γγ(θ)(DCO) in (²⁸ Si,αpγ); Δπ=yes from level scheme.
		517.47 11	12.3 16	2645.66	7/2 ⁺	(E1)		B(E1)(W.u.)=1.88×10 ⁻⁸ +31-34; B(M2)(W.u.)=0.0085 +43-38 I _γ : weighted average of 0.50 20 from (¹⁶ O,αpγ) and 0.33 5 from (p,γ). Mult.,δ: D+Q from γ(θ) in (p,γ) (1971Pr11); Δπ=yes from level scheme.
		1399.9 7	0.34 5	1763.18	5/2 ⁺	(E1+M2)	+0.44 12	
		1943.42	<0.22	1219.39	1/2 ⁺	[E3]		

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult.#	$\delta^\#$	Comments
3162.87	7/2 ⁻	3162.53 10	100.0 11	0.0	3/2 ⁺	M2+E3	+0.17 2	B(M2)(W.u.)=0.255 9; B(E3)(W.u.)=4.2 +11-9 E _γ : weighted average of 3162.7 4 from (²⁴ Mg,αpγ), 3162.6 3 from (¹⁶ O,αpγ), 3162.43 10 from (¹⁴ N,αpnγ), and 3162.6 1 from (p,p'γ). Other: 3162.6 15 from (α,pγ). I _γ : weighted average of 100 5 from (²⁸ Si,αpγ), 100.0 30 from (²⁴ Mg,αpγ), 100.0 20 from (¹⁶ O,αpγ), 100.0 33 from (α,pγ), 100.0 11 from (p,γ), and 100.0 20 from (p,p'γ). Mult.: from γ(θ) and γ(lin pol) in (p,γ). δ: weighted average of +0.16 1 from (²⁸ Si,αpγ), +0.18 7 from (¹⁶ O,αpγ), and +0.25 4 from (p,γ). Others: -0.14 2 from (α,pγ) and -0.22 5 from (p,p'γ).
3918.47	3/2 ⁺	1272.79 2155.1 15	<3.7 24.2 28	2645.66 1763.18	7/2 ⁺ 5/2 ⁺	[E2] [M1,E2]		E _γ : from ³⁵ Ar ε decay (1.7755 s). I _γ : weighted average of 29 12 from ³⁵ Ar ε decay and 23.9 28 from (p,γ). B(M1)(W.u.)=0.075 +32-18 if M1, B(E2)(W.u.)=61 +26-14 if E2. I _γ : from (α,pγ). B(M1)(W.u.)=0.030 +15-9 if M1, B(E2)(W.u.)=16 +8-5 if E2.
		2698.97	19 5	1219.39	1/2 ⁺	[M1,E2]		I _γ : weighted average of 100 6 from ³⁵ Ar ε decay, 100 5 from (α,pγ), and 100.0 19 from (p,γ).
		3918.24	100.0 19	0.0	3/2 ⁺	M1+E2		δ: -0.21 2 or +20 +10-5 from γ(θ) in (p,γ). B(M1)(W.u.)=0.052 +22-11 if M1, B(E2)(W.u.)=13 +6-3 if E2.
3942.9	9/2 ⁺	1297.4 15	8.0 21	2645.66	7/2 ⁺	[M1,E2]		E _γ : from (α,pγ). I _γ : weighted average of 6.2 31 from (²⁸ Si,αpγ), 8.1 21 from (α,pγ), and 8.7 22 from (p,γ).
		2180.1 10	100.0 21	1763.18	5/2 ⁺	E2		B(M1)(W.u.)=0.0042 +13-12 if M1, B(E2)(W.u.)=9.3 +29-27 if E2. B(E2)(W.u.)=8.7 +17-13 E _γ : weighted average of 2180 1 from (¹⁶ O,αpγ) and 2180.2 15 from (α,pγ). I _γ : weighted average of 100.0 21 from (α,pγ) and 100.0 22 from (p,γ). Other: 100 11 from (²⁸ Si,αpγ).
3967.6	1/2 ⁺	1273.6 2204.4 2748.5 6	2.7 6 ≤1.3 100 6	2694.00 1763.18 1219.39	3/2 ⁺ 5/2 ⁺ 1/2 ⁺	[M1,E2] [E2] [M1]		B(M1)(W.u.)=0.019 +8-6 if M1, B(E2)(W.u.)=45 +18-13 if E2. B(M1)(W.u.)=0.071 +23-14 E _γ : from ³⁵ Ar ε decay (1.7755 s). I _γ : other: 100 7 from ³⁵ Ar ε decay.
		3967.4	23 4	0.0	3/2 ⁺	[M1,E2]		I _γ : weighted average of 22 5 from ³⁵ Ar ε decay and 24 4 from (p,γ). B(M1)(W.u.)=0.0054 +19-13 if M1, B(E2)(W.u.)=1.31 +46-30 if E2.
4059.1	(3/2) ⁻	1365.1 2295.8	1.3 4 5.5 16	2694.00 1763.18	3/2 ⁺ 5/2 ⁺	[E1] [E1]		B(E1)(W.u.)=2.1×10 ⁻⁴ +8-7 B(E1)(W.u.)=1.9×10 ⁻⁴ +7-6 I _γ : weighted average of 8 4 from (α,pγ) and 5.1 16 from (p,γ). B(E1)(W.u.)=0.00179 +34-22
		2839.6	100.0 21	1219.39	1/2 ⁺	[E1]		I _γ : weighted average of 100 4 from (α,pγ) and 100.0 21 from (p,γ).

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult.#	$\delta^\#$	Comments
4059.1	$(3/2)^-$	4058.9	<2.1	0.0	$3/2^+$	[E1]		
4112.4	$7/2^+$	2349.1	89 6	1763.18	$5/2^+$	M1+E2	-0.16 2	B(M1)(W.u.)=0.0159 +48-31; B(E2)(W.u.)=0.28 +12-8
		4112.1	100 6	0.0	$3/2^+$	E2(+M3)	0.00 10	I_γ : other: 16 5 from ($\alpha,\text{p}\gamma$). B(E2)(W.u.)=0.76 +22-15 I_γ : other: 100 5 from ($\alpha,\text{p}\gamma$). B(M1)(W.u.)=0.048 +20-12 if M1, B(E2)(W.u.)=82 +34-21 if E2. B(E2)(W.u.)=82 +34-21 upper bound exceeds RUL=100 if E2.
4173.43	$(5/2^+)$	1479.40	45 14	2694.00	$3/2^+$	[M1,E2]		B(M1)(W.u.)=0.0068 +33-19 if M1, B(E2)(W.u.)=4.5 +21-13 if E2.
		1527.73	<17	2645.66	$7/2^+$	[M1,E2]		
		2410.16	28 9	1763.18	$5/2^+$	[M1,E2]		
		2953.91	≤ 5.2	1219.39	$1/2^+$	[E2]		
		4173.16	100 17	0.0	$3/2^+$	[M1,E2]		B(M1)(W.u.)=0.0047 +15-7 if M1, B(E2)(W.u.)=1.02 +33-14 if E2.
4177.9	$3/2^-$	1175.2	<0.82	3002.64	$5/2^+$	[E1]		
		1483.9	13.1 33	2694.00	$3/2^+$	[E1]		B(E1)(W.u.)=5.7 $\times 10^{-4}$ +16-14
		1532.2	<0.82	2645.66	$7/2^+$	[M2]		
		2414.6	<1.6	1763.18	$5/2^+$	[E1]		
		2958.4	51 8	1219.39	$1/2^+$	(E1+M2)	+0.11 4	B(E1)(W.u.)=2.8 $\times 10^{-4}$ +5-4; B(M2)(W.u.)=1.7 +16-11 Mult., δ : D+Q from $\gamma(\theta)$ in (p,γ); $\Delta\pi$ =yes from level scheme. B(M2)(W.u.)=1.7 +16-11 upper bound exceeds RUL=3.
		4177.6	100 8	0.0	$3/2^+$	(E1+M2)	+0.06 3	B(E1)(W.u.)=1.93 $\times 10^{-4}$ +28-20; B(M2)(W.u.)=0.18 +24-14 Mult., δ : D+Q from $\gamma(\theta)$ in (p,γ); $\Delta\pi$ =yes from level scheme.
4347.74	$9/2^-$	1184.90 20	100.0 29	3162.87	$7/2^-$	M1+E2	-0.36 3	B(M1)(W.u.)=0.007 +7-3; B(E2)(W.u.)=2.5 +27-10 E_γ : weighted average of 1185.0 2 from ($^{24}\text{Mg},\alpha\text{p}\gamma$), 1184.9 3 from ($^{16}\text{O},\alpha\text{p}\gamma$), and 1184.80 20 from ($^{14}\text{N},\alpha\text{p}\gamma$). Other: 1184.6 10 from ($\alpha,\text{p}\gamma$). I_γ : weighted average of 100 4 from ($^{28}\text{Si},\alpha\text{p}\gamma$), 100.0 31 from ($^{24}\text{Mg},\alpha\text{p}\gamma$), 100 4 from ($^{16}\text{O},\alpha\text{p}\gamma$), 100.0 31 from ($\alpha,\text{p}\gamma$), and 100.0 29 from (p,γ). Mult.: from $\gamma\gamma(\text{DCO})$ and $\gamma\gamma(\text{lin pol})$ in ($^{28}\text{Si},\alpha\text{p}\gamma$), $\Delta J=1$. δ : weighted average of -0.36 3 from ($\alpha,\text{p}\gamma$), -0.3 1 from ($^{28}\text{Si},\alpha\text{p}\gamma$), and -0.65 30 from ($^{16}\text{O},\alpha\text{p}\gamma$).
		1701.88 25	46.8 24	2645.66	$7/2^+$	E1+M2	-0.018 12	B(E1)(W.u.)=3.7 $\times 10^{-5}$ +39-14; B(M2)(W.u.)=0.019 +47-17 E_γ : weighted average of 1702.0 3 from ($^{24}\text{Mg},\alpha\text{p}\gamma$), 1701.7 5 from ($^{16}\text{O},\alpha\text{p}\gamma$), and 1701.85 25 from ($^{14}\text{N},\alpha\text{p}\gamma$). Other: 1702.1 15 from ($\alpha,\text{p}\gamma$). I_γ : unweighted average of 45.6 24 from ($^{28}\text{Si},\alpha\text{p}\gamma$), 39.5 24 from ($^{24}\text{Mg},\alpha\text{p}\gamma$), 52.0 14 from ($^{16}\text{O},\alpha\text{p}\gamma$), 52 7 from ($\alpha,\text{p}\gamma$), and 44.9 29 from (p,γ). Mult.: from $\gamma\gamma(\text{DCO})$ and $\gamma\gamma(\text{lin pol})$ in ($^{28}\text{Si},\alpha\text{p}\gamma$) with $\Delta J=1$ and $\gamma(\theta)$ and $\gamma(\text{lin pol})$ in ($\alpha,\text{p}\gamma$). δ : from (p,γ). Others: +0.5 +5-3 from ($^{28}\text{Si},\alpha\text{p}\gamma$), +0.5 4 from ($^{16}\text{O},\alpha\text{p}\gamma$).
		2584.46	≤ 4.4	1763.18	$5/2^+$	[M2]		

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult.#	$\delta^\#$	Comments
4347.74	9/2 ⁻	4347.8 8	12.4 11	0.0	3/2 ⁺	[E3]		B(E3)(W.u.)=44 +46-17 E _γ , I _γ : from (¹⁶ O, αpγ). E _γ : other: 4618 2 from (α,pγ). Mult., δ: from pγ(θ) with -0.08 14 for J=3/2; +0.36 15 for J=5/2 in (α,pγ). If J(4624=5/2), Mult would be M1+E2 based on RUL. B(M1)(W.u.)=0.040 +23-5 if M1, B(E2)(W.u.)=58 +34-7 if E2. B(E1)(W.u.)=4.6×10 ⁻⁴ +30-9
4624.3	3/2, 5/2	4624.0	100	0.0	3/2 ⁺	D+Q		
4769.86	(5/2 ⁻)	1606.95	100 16	3162.87	7/2 ⁻	[M1,E2]		
		1767.17	54 16	3002.64	5/2 ⁺	[E1]		
		2075.79	<15	2694.00	3/2 ⁺	[E1]		
		3550.28	<15	1219.39	1/2 ⁺	[M2]		
		4769.51	<15	0.0	3/2 ⁺	[E1]		
4839.10	(1/2 ⁺ , 3/2)	1836.41	<12	3002.64	5/2 ⁺			
		2145.03	<6.8	2694.00	3/2 ⁺			
		2193.37 ^a	<5.1	2645.66	7/2 ⁺			
		3075.78	100 9	1763.18	5/2 ⁺			
		3619.51	<12	1219.39	1/2 ⁺			
		4838.74	70 9	0.0	3/2 ⁺			
4854.4	(1/2, 3/2)	3634.8	100 7	1219.39	1/2 ⁺			
		4854.0	33 7	0.0	3/2 ⁺			
4880.96	7/2 ⁽⁺⁾	1878.27	15 5	3002.64	5/2 ⁺	[M1,E2]		B(M1)(W.u.)=0.051 +30-16 if M1, B(E2)(W.u.)=55 +32-17 if E2.
		2186.89	<16	2694.00	3/2 ⁺	[E2]		
		2235.22	47 8	2645.66	7/2 ⁺	[M1,E2]		I _γ : other: 52 9 from (α,pγ) for a doublet. B(M1)(W.u.)=0.095 +43-18 if M1, B(E2)(W.u.)=72 +33-14 if E2. B(E2)(W.u.)=72 +33-14 upper bound exceeds RUL=100 if E2. E _γ : from (α,pγ). I _γ : weighted average of 100 9 from (α,pγ) and 100 8 from (p,γ). B(M1)(W.u.)=0.075 +33-12 if M1, B(E2)(W.u.)=29 +13-5 if E2.
		3117.4 20	100 8	1763.18	5/2 ⁺	[M1,E2]		
5009.1	(1/2 ⁺ , 3/2)	4880.60	<8.1	0.0	3/2 ⁺	[E2]		
		835.7	<25	4173.43	(5/2 ⁺)			
		950.0	<10	4059.1	(3/2) ⁻			
		1846.2 ^a	<50	3162.87	7/2 ⁻			
		2006.4	<20	3002.64	5/2 ⁺			
		2315.0	<25	2694.00	3/2 ⁺			
		2363.4 ^a	<5	2645.66	7/2 ⁺			
		3245.8	<10	1763.18	5/2 ⁺			
		3789.5	<7	1219.39	1/2 ⁺			
		5008.7	100	0.0	3/2 ⁺			
5163.34	7/2 ⁻	2000.41	93 12	3162.87	7/2 ⁻	D+Q	+0.44 20	
		2160.63	23 9	3002.64	5/2 ⁺			
		3399.98	100 12	1763.18	5/2 ⁺	D(+Q)	0.00 3	
		3943.71 ^a	<23	1219.39	1/2 ⁺			
		5162.93	<23	0.0	3/2 ⁺			

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. #	$\delta^\#$	Comments
5214.5	(5/2 ⁺)	205.4	<23	5009.1	(1/2 ⁺ , 3/2)			
		1041.1	<60	4173.43	(5/2 ⁺)			
		1155.4	<10	4059.1	(3/2) ⁻			
		2051.6	<15	3162.87	7/2 ⁻			
		2211.8	<25	3002.64	5/2 ⁺			
		2520.4	<15	2694.00	3/2 ⁺			
		2568.7	<15	2645.66	7/2 ⁺			
		3451.1	<6	1763.18	5/2 ⁺			
		3994.9	<4	1219.39	1/2 ⁺			
		5214.1	100	0.0	3/2 ⁺	[M1,E2]		
5403.3	3/2 ⁻	2400.6	46 9	3002.64	5/2 ⁺	[E1]		B(E1)(W.u.)=0.0011 +6-2
		3639.9	<18	1763.18	5/2 ⁺	[E1]		
		4183.6	100 18	1219.39	1/2 ⁺	[E1]		B(E1)(W.u.)=4.5×10 ⁻⁴ +20-6
		5402.9	<13	0.0	3/2 ⁺	[E1]		
5406.95	11/2 ⁻	1059.32 20	14.5 22	4347.74	9/2 ⁻	M1+E2	+0.14 3	B(M1)(W.u.)=0.0082 +30-20; B(E2)(W.u.)=0.54 +35-23 E _γ : weighted average of 1059.3 2 from (²⁴ Mg,αpγ), 1059.5 2 from (¹⁶ O,αpγ), and 1059.17 20 from (¹⁴ N,αpnγ). Other: 1059.3 10 from (α,pγ). I _γ : unweighted average of 12.0 12 from (²⁸ Si,αpγ), 16.9 17 from (²⁴ Mg,αpγ), 19.2 5 from (¹⁶ O,αpγ), and 9.9 22 from (α,pγ). Mult.: from γγ(DCO) and γγ(lin pol) in (²⁸ Si,αpγ), ΔJ=1. δ: weighted average of +0.12 3 from (¹⁶ O,αpγ), +0.2 1 from (²⁸ Si,αpγ), and +0.25 8 from (α,pγ). B(E2)(W.u.)=4.6 +15-9 E _γ : weighted average of 2244.2 3 from (²⁴ Mg,αpγ), 2243.3 2 from (¹⁶ O,αpγ), 2244.00 25 from (¹⁴ N,αpnγ), and 2244 3 from (α,pγ). I _γ : weighted average of 100.0 33 from (²⁸ Si,αpγ), 100.0 32 from (²⁴ Mg,αpγ), 100.0 23 from (¹⁶ O,αpγ), and 100.0 22 from (α,pγ). Mult.: from γγ(DCO) and γγ(lin pol) in (²⁸ Si,αpγ), ΔJ=2.
		2243.71 23	100.0 22	3162.87	7/2 ⁻	E2		
5586.0	5/2 ⁺	2940.2	67 17	2645.66	7/2 ⁺			
		3822.6	100 17	1763.18	5/2 ⁺			
		4366.3	<17	1219.39	1/2 ⁺			
		5585.5	<17	0.0	3/2 ⁺			
5598.4	3/2 ⁺ , 5/2 ⁺	3835.0	37 14	1763.18	5/2 ⁺	[M1,E2]		B(M1)(W.u.)=0.050 +29-19 if M1, B(E2)(W.u.)=13 +8-5 if E2.
		5597.9	100 14	0.0	3/2 ⁺	[M1,E2]		B(M1)(W.u.)=0.044 +23-12 if M1, B(E2)(W.u.)=5.3 +28-14 if E2.
5646	(5/2, 7/2, 9/2 ⁺)	2483	100	3162.87	7/2 ⁻			
		2643		3002.64	5/2 ⁺			
		3883		1763.18	5/2 ⁺			

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. #	$\delta^\#$	Comments
5646	(5/2,7/2,9/2 ⁺)	5646	<8	0.0	3/2 ⁺			
5655	(3/2) ⁺	2652	100 6	3002.64	5/2 ⁺	[M1,E2]		B(M1)(W.u.)=0.076 +30-14 if M1, B(E2)(W.u.)=41 +16-7 if E2.
		3892	7.5 25	1763.18	5/2 ⁺	[M1,E2]		B(M1)(W.u.)=0.0018 +10-6 if M1, B(E2)(W.u.)=0.45 +25-15 if E2.
		4435	<7.5	1219.39	1/2 ⁺	[M1,E2]		
5682.2	1/2 ⁻ ,3/2 ⁻	1504.3	100	4177.9	3/2 ⁻			
5723.52	5/2 ⁺	3029.38	63 8	2694.00	3/2 ⁺			
		3960.10	100 11	1763.18	5/2 ⁺			
5758	(1/2,3/2 ⁻)	1580	67 22	4177.9	3/2 ⁻			
		4538	<13	1219.39	1/2 ⁺			
		5758	100 22	0.0	3/2 ⁺			
5806.6	(1/2,3/2,5/2 ⁺)	5806.1	100	0.0	3/2 ⁺			
5830	(5/2 ⁻)	2667	100	3162.87	7/2 ⁻	D+Q		I _γ : from (α,pγ). δ: -2.2 5 or -0.6 1 for J=5/2 from γ(θ) in (α,pγ).
5926.57	(11/2) ⁻	1579.09 30	100 11	4347.74	9/2 ⁻	M1+E2	-0.2 1	E _γ : weighted average of 1578.9 5 from (¹⁶ O,αpγ), 1579.15 30 from (¹⁴ N,αpnγ), and 1579.3 15 from (α,pγ). Mult.,δ: D+Q and δ from γγ(θ)(DCO) in (²⁸ Si,αpγ) with ΔJ=1; E1+M2 ruled out by RUL. Other: -0.8 4 from γ(θ) in (α,pγ). I _γ : from (²⁸ Si,αpγ).
		1983.6	11.1 22	3942.9	9/2 ⁺	[E1]		
		2763.58		3162.87	7/2 ⁻			
6087.23	13/2 ⁻	680.35 10	100.0 21	5406.95	11/2 ⁻	M1+E2	+0.020 15	B(M1)(W.u.)=0.0107 10 E _γ : weighted average of 680.4 1 from (²⁴ Mg,αpγ), 680.5 5 from (¹⁶ O,αpγ), and 680.22 15 from (¹⁴ N,αpnγ). Other: 679.9 10 from (α,pγ). I _γ : weighted average of 100.0 31 from (²⁸ Si,αpγ), 100.0 21 from (¹⁶ O,αpγ), and 100 6 from (α,pγ). Mult.,δ: from γ(θ) and γ(lin pol) in (α,pγ). Other: δ=+0.1 1 from γγ(DCO) in (²⁸ Si,αpγ), ΔJ=1.
		1739.4 4	6.9 9	4347.74	9/2 ⁻	E2		B(E2)(W.u.)=0.054 +9-8 E _γ : weighted average of 1739.6 5 from (¹⁶ O,αpγ) and 1739.3 4 from (¹⁴ N,αpnγ). I _γ : weighted average of 6.3 6 from (²⁸ Si,αpγ), 7.9 9 from (¹⁶ O,αpγ), and 18 6 from (α,pγ). Mult.: from γγ(θ)(DCO) and γγ(lin pol) in (²⁸ Si,αpγ), ΔJ=2.
6106.2	(3/2,5/2 ⁺)	4342.7	10.3 35	1763.18	5/2 ⁺			
		4886.4	62 17	1219.39	1/2 ⁺			
		6105.6	100 17	0.0	3/2 ⁺			
6181	(1/2 ⁺ to 7/2 ⁺)	6180	100	0.0	3/2 ⁺			
6380.1	(9/2 ⁻)	296 1		6087.23	13/2 ⁻			I _γ : from (¹⁶ O,αpγ).
		971 1		5406.95	11/2 ⁻			I _γ : from (¹⁶ O,αpγ).
6492	(3/2 ⁺)	6491	100	0.0	3/2 ⁺			

Adopted Levels, Gammas (continued)

<u>$\gamma(^{35}\text{Cl})$ (continued)</u>							
<u>$E_i(\text{level})$</u>	<u>J_i^π</u>	<u>$E_\gamma^\dagger$</u>	<u>$I_\gamma^\ddagger$</u>	<u>E_f</u>	<u>J_f^π</u>	<u>Mult.[#]</u>	<u>$\delta^\#$</u>
6660.2	(11/2 ⁻)	2312.4		4347.74	9/2 ⁻		
		3497.1		3162.87	7/2 ⁻		
7066.2	(5/2 ⁺)	1902.8	1.8	5163.34	7/2 ⁻		
		2892.6	1.0	4173.43	(5/2 ⁺)		
		2953.7	1.2	4112.4	7/2 ⁺		
		3007.0	2.9	4059.1	(3/2 ⁻)		
		3903.1	12	3162.87	7/2 ⁻		
		4063.3	6.1	3002.64	5/2 ⁺		
		4371.9	10	2694.00	3/2 ⁺		
		4420.2	3.3	2645.66	7/2 ⁺		
		5302.6	29	1763.18	5/2 ⁺		
		5846.3	37	1219.39	1/2 ⁺		
		7065.4	100	0.0	3/2 ⁺		
7103.4	3/2 ⁽⁻⁾	2925.4	2.3	4177.9	3/2 ⁻		
		3184.8	2.3	3918.47	3/2 ⁺		
		4100.5	4.7	3002.64	5/2 ⁺		
		4409.1	17	2694.00	3/2 ⁺	D(+Q)	0.00 3
		5339.8	6.3	1763.18	5/2 ⁺		
		5883.5	100	1219.39	1/2 ⁺	D(+Q)	0.00 3
		7102.6	23	0.0	3/2 ⁺		
7178.6	1/2 ⁽⁺⁾	1072.4	7.9	6106.2	(3/2,5/2 ⁺)		
		1775.3	4.5	5403.3	3/2 ⁻		
		2339.4	4	4839.10	(1/2 ⁺ ,3/2)		
		3005.0	2.1	4173.43	(5/2 ⁺)		
		3119.4	58	4059.1	(3/2 ⁻)		
		3210.8	24	3967.6	1/2 ⁺		
		3260.0	7.9	3918.47	3/2 ⁺		
		4484.3	11	2694.00	3/2 ⁺		
		5958.7	45	1219.39	1/2 ⁺		
		7177.8	100	0.0	3/2 ⁺		
7185	5/2 ⁺	1970	<1.1	5214.5	(5/2 ⁺)		
		2176	<1.1	5009.1	(1/2 ⁺ ,3/2)		
		3011	<11	4173.43	(5/2 ⁺)		
		3126	57 13	4059.1	(3/2 ⁻)		
		4022	<6.5	3162.87	7/2 ⁻		
		4182	<6.5	3002.64	5/2 ⁺		
		4491	4.4 22	2694.00	3/2 ⁺		
		4539	13 5	2645.66	7/2 ⁺		
		5421	9 7	1763.18	5/2 ⁺		
		5965	35 15	1219.39	1/2 ⁺		
		7184	100 22	0.0	3/2 ⁺		

Adopted Levels, Gammas (continued)

<u>$\gamma(^{35}\text{Cl})$ (continued)</u>							
<u>$E_i(\text{level})$</u>	<u>J_i^π</u>	<u>E_γ</u>	<u>I_γ</u>	<u>E_f</u>	<u>J_f^π</u>	<u>Mult. #</u>	<u>$\delta^\#$</u>
7194.6	$1/2^-$	2185.4	1.2	5009.1	$(1/2^+, 3/2)$		
		2340.1	3.0	4854.4	$(1/2, 3/2)$		
		2355.4	2.7	4839.10	$(1/2^+, 3/2)$		
		3016.6	24	4177.9	$3/2^-$		
		3135.4	4.5	4059.1	$(3/2)^-$		
		3226.8	2.7	3967.6	$1/2^+$		
		3276.0	6	3918.47	$3/2^+$		
		4500.3	1.6	2694.00	$3/2^+$		
		5431.0	0.75	1763.18	$5/2^+$		
		5974.7	100	1219.39	$1/2^+$		
		7193.8	3.0	0.0	$3/2^+$		
7225.5	$5/2$	1119.3	3.1	6106.2	$(3/2, 5/2^+)$		
		1418.9		5806.6	$(1/2, 3/2, 5/2^+)$		
		1543.3	3.4	5682.2	$1/2^-, 3/2^-$		
		1571	1.4	5655	$(3/2)^+$		
		1822.2	1.5	5403.3	$3/2^-$		
		2010.9	12	5214.5	$(5/2^+)$		
		3047.5	14	4177.9	$3/2^-$		
		3166.3	3.4	4059.1	$(3/2)^-$		
		3306.9	2.5	3918.47	$3/2^+$		
		4222.6		3002.64	$5/2^+$		
		4531.2	17	2694.00	$3/2^+$	D+Q	+0.36 4
		5461.9	12	1763.18	$5/2^+$	D+Q	-0.36 7
		7224.7	100	0.0	$3/2^+$	D+Q	+0.36 4
7234.0	$5/2^{(+)}$	1579	0.22	5655	$(3/2)^+$		
		3056.0		4177.9	$3/2^-$		
		4539.7	1.9	2694.00	$3/2^+$		
		4588.0	3.2	2645.66	$7/2^+$		
		5470.4	0.22	1763.18	$5/2^+$		
		6014.1	1.9	1219.39	$1/2^+$		
		7233.2	100	0.0	$3/2^+$	D+Q	-0.10
7272.7	$1/2^-$	1515	1.7	5758	$(1/2, 3/2^-)$		
		1869.4	0.29	5403.3	$3/2^-$		
		2263.5	2.0	5009.1	$(1/2^+, 3/2)$		
		2433.5	1.5	4839.10	$(1/2^+, 3/2)$		
		3099.1	0.58	4173.43	$(5/2^+)$		
		3213.4	1.3	4059.1	$(3/2)^-$		
		3304.9	1.2	3967.6	$1/2^+$		
		4578.4	1.6	2694.00	$3/2^+$		
		6052.8	35	1219.39	$1/2^+$		
		7271.9	100	0.0	$3/2^+$		
7361.9	$3/2^{(+)}$	1707	0.41	5655	$(3/2)^+$		

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)								
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. #	$\delta^\#$	Comments
7361.9	$3/2^{(+)}$	2352.7	0.82	5009.1	$(1/2^+, 3/2)$			
		2522.7	1.4	4839.10	$(1/2^+, 3/2)$			
		3183.8	0.69	4177.9	$3/2^-$			
		3302.6	2.7	4059.1	$(3/2)^-$			
		3394.1	4.1	3967.6	$1/2^+$			
		3443.3	1.1	3918.47	$3/2^+$			
		4359.0	0.41	3002.64	$5/2^+$			
		4667.6	0.69	2694.00	$3/2^+$			
		5598.2	11	1763.18	$5/2^+$	D+Q		δ : $>+2.5$ or -0.01 10 from $\gamma(\theta)$ in (p, γ).
		6141.9	100	1219.39	$1/2^+$	(M1+E2)		Mult.: D+Q from $\gamma(\theta)$ in (p, γ); E1+M2 is less likely for this strong primary transition.
7396.4	$7/2^{(-)}$	7361.1	14	0.0	$3/2^+$	D+Q		δ : -0.05 3 or $+2.5$ 3 in (p, γ).
		3048.5	23	4347.74	$9/2^-$			δ : $-0.16 < \delta(Q/D) < -0.12$ or $+7.5 < \delta(Q/D) < +10$ in (p, γ).
		3283.8	17	4112.4	$7/2^+$			
		3453.3	11	3942.9	$9/2^+$			
		4233.3	100	3162.87	$7/2^-$	(M1+E2)	+0.28 25	Mult.: D+Q from $\gamma(\theta)$ in (p, γ); E1+M2 is less likely for this strong primary transition.
		4393.5	30	3002.64	$5/2^+$	D(+Q)	+0.011 18	
7450.9	$3/2$	4750.4	28	2645.66	$7/2^+$			
		5632.7	4.3	1763.18	$5/2^+$			
		3532.2	6.9 34	3918.47	$3/2^+$			
		4448.0	13.7 14	3002.64	$5/2^+$			
		4756.6	11.0 11	2694.00	$3/2^+$			
		5687.2	5.5 28	1763.18	$5/2^+$			
7501.1	$(5/2^+, 7/2, 9/2^+)$	6230.9	100 10	1219.39	$1/2^+$			
		3558.0	46 6	3942.9	$9/2^+$			
		4498.2	23.6 24	3002.64	$5/2^+$			
		4855.1	100 11	2645.66	$7/2^+$			
		5737.4	12.7 13	1763.18	$5/2^+$			
7502.9		1396.7 @	14 @	6106.2	$(3/2, 5/2^+)$			
		1904.4 @	4 @	5598.4	$3/2^+, 5/2^+$			
		2493.7 @	4 @	5009.1	$(1/2^+, 3/2)$			
		3324.8 @	6 @	4177.9	$3/2^-$			
		3443.6 @	100 @	4059.1	$(3/2)^-$			
		3559.8 @	4 @	3942.9	$9/2^+$			
		4339.7 @	8 @	3162.87	$7/2^-$			
		4500.0 @	8 @	3002.64	$5/2^+$			
		4856.9 @	14 @	2645.66	$7/2^+$			

Adopted Levels, Gammas (continued) $\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult.#	$\delta^\#$	Comments
7502.9		5739.2 @	1.4 @	1763.18	5/2 ⁺			
		6282.9 @	36 @	1219.39	1/2 ⁺			
		7502.0 @	0.6 @	0.0	3/2 ⁺			
7518.8	7/2	1920.3	2.9 15	5598.4	3/2 ⁺ , 5/2 ⁺			
		1932.7	4.4 22	5586.0	5/2 ⁺			
		3170.9	2.9 15	4347.74	9/2 ⁻			
		3575.7	2.9 15	3942.9	9/2 ⁺			
		4355.6	100 10	3162.87	7/2 ⁻	D(+Q)	+0.1 2	
		4872.8	4.4 22	2645.66	7/2 ⁺			
		5755.1	29.4 30	1763.18	5/2 ⁺	D+Q	+0.098 22	
7548.9	7/2 ⁻	1903	0.59 7	5646	(5/2, 7/2, 9/2 ⁺)			
		1962.8	0.11	5586.0	5/2 ⁺	[E1]		
		2778.9	1.49 11	4769.86	(5/2 ⁻)	[M1, E2]		
		4385.7	100.0 11	3162.87	7/2 ⁻	M1+E2	-0.07 2	
		4545.9	2.13 11	3002.64	5/2 ⁺	(E1+M2)	+0.6 4	Mult., δ : D+Q and δ from $\gamma(\theta)$ in (p, γ); $\Delta\pi$ =yes from level scheme.
		4902.9	0.72 8	2645.66	7/2 ⁺	[E1]		
		5785.2	0.35 4	1763.18	5/2 ⁺	[E1]		
		7548.0	0.287 32	0.0	3/2 ⁺	(M2)		Mult.: Q(+O) with $\delta(O/Q)=0.1$ 2 from $\gamma(\theta)$ in (p, γ); $\Delta\pi$ =yes from level scheme; E3 component is less likely based on RUL.
7561.3	1/2, 3/2	3383.2	2.6	4177.9	3/2 ⁻			
		3502.0	17	4059.1	(3/2 ⁻)			
		3593.5	1.4	3967.6	1/2 ⁺			
		3642.6	4.6	3918.47	3/2 ⁺			
		4866.9	63	2694.00	3/2 ⁺			
		6341.3	100	1219.39	1/2 ⁺			
		7560.4	97	0.0	3/2 ⁺			
7600.8	5/2 ⁺	2437.4	5.8	5163.34	7/2 ⁻	[E1]		
		2719.7	6.1	4880.96	7/2 ⁽⁺⁾			
		2761.6	6.1	4839.10	(1/2 ⁺ , 3/2)			
		2830.8	3.9	4769.86	(5/2 ⁻)			
		2976.4	5.5	4624.3	3/2, 5/2			
		3422.7	9.1	4177.9	3/2 ⁻	[E1]		
		3427.2		4173.43	(5/2 ⁺)			
		3541.5	6.1	4059.1	(3/2 ⁻)	[E1]		
		4437.6	15	3162.87	7/2 ⁻	[E1]		
		4597.8	24	3002.64	5/2 ⁺	M1+E2	+0.50 22	Mult.: D+Q from $\gamma(\theta)$ in (p, γ); E1+M2 is less likely for this strong primary transition.
		4906.4	55	2694.00	3/2 ⁺	(M1+E2)		Mult.: D+Q from $\gamma(\theta)$ in (p, γ); $\Delta\pi$ =no from level scheme.
		4954.8	5.8	2645.66	7/2 ⁺	[M1, E2]		
		5837.1	58	1763.18	5/2 ⁺	M1+E2	+0.31 7	

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)								
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult.#	$\delta^\#$	Comments
7600.8	5/2 ⁺	6380.8	3.3	1219.39	1/2 ⁺	[E2]		
		7599.9	100	0.0	3/2 ⁺	M1+E2	-0.19 4	
7618.7	5/2 ⁺	1438	0.77	6181	(1/2 ⁺ to 7/2 ⁺)			
		2404.1	2.6	5214.5	(5/2 ⁺)			
		2455.3	0.9	5163.34	7/2 ⁻			
		2779.5	0.9	4839.10	(1/2 ⁺ , 3/2)			
		3440.6	3.9	4177.9	3/2 ⁻			
		3559.4	2.6	4059.1	(3/2) ⁻			
		4455.5	3.9	3162.87	7/2 ⁻			
		5855.0	13	1763.18	5/2 ⁺	M1+E2	+0.48 11	
		7617.8	100	0.0	3/2 ⁺	M1+E2	+2.2 3	
7656.6	(1/2, 3/2, 5/2 ⁺)	1550.4	5.3 26	6106.2	(3/2, 5/2 ⁺)			
		1974.3	1.8 9	5682.2	1/2 ⁻ , 3/2 ⁻			
		2002	1.8 9	5655	(3/2) ⁺			
		2817.4	5.3 26	4839.10	(1/2 ⁺ , 3/2)			
		3478.5	1.8 9	4177.9	3/2 ⁻			
		3597.3	56 6	4059.1	(3/2) ⁻			
		3688.8	3.5 18	3967.6	1/2 ⁺			
		7655.7	100 11	0.0	3/2 ⁺			
7671.9	5/2 ⁺	2073.4	2.1	5598.4	3/2 ⁺ , 5/2 ⁺			
		2508.5	7	5163.34	7/2 ⁻			
		2790.8	2.1	4880.96	7/2 ⁽⁺⁾			
		2901.9	1.8	4769.86	(5/2) ⁻			
		3324.0	11	4347.74	9/2 ⁻			
		3498.3	8.8	4173.43	(5/2 ⁺)			
		4508.7	3.2	3162.87	7/2 ⁻			
		4668.9	16	3002.64	5/2 ⁺	M1+E2	-1.4 5	
		5025.9	23	2645.66	7/2 ⁺	D(+Q)	+0.3 22	
		5908.2	100	1763.18	5/2 ⁺	M1+E2	+0.86 14	
		7671.0	1.4	0.0	3/2 ⁺			
7685.5	3/2 ⁻	1927		5758	(1/2, 3/2 ⁻)			
		2087.0	2.9	5598.4	3/2 ⁺ , 5/2 ⁺			
		2282.1	2.9	5403.3	3/2 ⁻			
		2470.9	1	5214.5	(5/2 ⁺)			
		2676.3	0.86	5009.1	(1/2 ⁺ , 3/2)			
		3507.4	4.3	4177.9	3/2 ⁻			
		3511.9	5.7	4173.43	(5/2 ⁺)			
		3626.2	8.6	4059.1	(3/2) ⁻			
		4682.5	14	3002.64	5/2 ⁺	D(+Q)	-0.01 5	
		5921.8	1	1763.18	5/2 ⁺			
		6465.5	1.4	1219.39	1/2 ⁺			
		7684.6	100	0.0	3/2 ⁺	D+Q	+6.0 7	Mult.: D+Q with $\delta=+6.0$ 7 from $\gamma(\theta)$ in (p, γ); $\Delta\pi=\text{yes}$ from

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	E_γ †	I_γ ‡	E_f	J_f^π	Mult. #	δ #	Comments
								level scheme; but E1+M2 with such a large $\delta(\text{M2/E1})$ is unlikely for this strong primary transition based on RUL, which might suggest a different level in (p, γ).
7693.9	(1/2 ⁺ , 3/2, 5/2)	2095.4	1.0 5	5598.4	3/2 ⁺ , 5/2 ⁺			
		3634.6	1.0 5	4059.1	(3/2) ⁻			
		4690.9	2.1 11	3002.64	5/2 ⁺			
		7693.0	100 11	0.0	3/2 ⁺			
7706.4	5/2 ⁺	2051	4.8	5655	(3/2) ⁺	M1+E2		δ : <-2 or ≥ 13 from $\gamma(\theta)$ in (p, γ).
		2825.3	0.48	4880.96	7/2 ⁽⁺⁾			
		2867.2	1.6	4839.10	(1/2 ⁺ , 3/2)			
		3082.0	0.96	4624.3	3/2, 5/2			
		3528.3	2.4	4177.9	3/2 ⁻			
		3593.8	2.4	4112.4	7/2 ⁺			
		3787.7	0.72	3918.47	3/2 ⁺			
		4703.4	1.2	3002.64	5/2 ⁺			
		5060.4	4.8	2645.66	7/2 ⁺	M1+E2		δ : -1.01 10 or ≥ 13 from (p, γ).
		5942.7	0.48	1763.18	5/2 ⁺			
		6486.4	0.6	1219.39	1/2 ⁺			
		7705.5	100	0.0	3/2 ⁺	D(+Q)	+0.1 1	
7744.8	7/2 ⁻	2863.7	47	4880.96	7/2 ⁽⁺⁾			
		2974.8	4.7	4769.86	(5/2 ⁻)			
		3632.2	14	4112.4	7/2 ⁺			
		3801.7	23	3942.9	9/2 ⁺	(E1)		Mult.: D(+Q) with $\delta=-0.16$ 17 from $\gamma(\theta)$ in (p, γ); $\Delta\pi$ =yes from level scheme; M2 component is less likely for the primary transition.
		4581.6	37	3162.87	7/2 ⁻	M1+E2	+0.18 8	
		5098.7	100	2645.66	7/2 ⁺	(E1)		Mult.: D(+Q) with $\delta=+0.1$ 30 from $\gamma(\theta)$ in (p, γ); $\Delta\pi$ =yes from level scheme; M2 component is less likely for the primary transition.
		5981.1	5.6	1763.18	5/2 ⁺			
		6524.8 ^a	1.4	1219.39	1/2 ⁺	[E3]		This transition is considered questionable by the evaluators since it would require Mult=E3 which is unlikely for this primary transition based on RUL.
7777.1	(5/2 ⁺)	2122	1.3	5655	(3/2) ⁺			
		2613.7	2.8	5163.34	7/2 ⁻			
		2937.9	5	4839.10	(1/2 ⁺ , 3/2)			
		3007.1	7.5	4769.86	(5/2 ⁻)			
		3599.0	18	4177.9	3/2 ⁻			
		3717.8	2.8	4059.1	(3/2) ⁻			
		4613.9	5	3162.87	7/2 ⁻			
		4774.1	10	3002.64	5/2 ⁺			
		5082.7	23	2694.00	3/2 ⁺			
		5131.0	3.3	2645.66	7/2 ⁺			
		6013.4	100	1763.18	5/2 ⁺			

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)								
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. [#]	$\delta^\#$	Comments
7777.1	(5/2 ⁺)	6557.1	28	1219.39	1/2 ⁺			
		7776.2	45	0.0	3/2 ⁺			
7781.7	(5/2 ⁻)	1675.5	3.9	6106.2	(3/2, 5/2 ⁺)			
		2183.2		5598.4	3/2 ⁺ , 5/2 ⁺			
		2378.3		5403.3	3/2 ⁻			
		2567.1	4.2	5214.5	(5/2 ⁺)			
		3011.7	19	4769.86	(5/2 ⁻)			
		3433.8	5.6	4347.74	9/2 ⁻			
		3608.1	100	4173.43	(5/2 ⁺)			
		3669.1	17	4112.4	7/2 ⁺			
		3722.4	3.3	4059.1	(3/2) ⁻			
		3863.0	31	3918.47	3/2 ⁺			
		4618.5	14	3162.87	7/2 ⁻			
		5135.6	22	2645.66	7/2 ⁺			
		6018.0	50	1763.18	5/2 ⁺			
		6561.7	3.3	1219.39	1/2 ⁺			
		7780.8	4.7	0.0	3/2 ⁺			
7797.2	1/2 ⁻	2039		5758	(1/2, 3/2 ⁻)			
		2142	1.6	5655	(3/2) ⁺			
		2393.8	2.4	5403.3	3/2 ⁻			
		2788.0	2.4	5009.1	(1/2 ⁺ , 3/2)			
		2942.7	1.6	4854.4	(1/2, 3/2)			
		2958.0	1.3	4839.10	(1/2 ⁺ , 3/2)			
		3737.9	0.36	4059.1	(3/2) ⁻			
		6577.2	11	1219.39	1/2 ⁺			
		7796.3	100	0.0	3/2 ⁺			
7837.2	3/2 ⁻	1656 ^{&}	1.4 ^{&}	6181	(1/2 ⁺ to 7/2 ⁺)			
		2238.7 ^{&}	1.4 ^{&}	5598.4	3/2 ⁺ , 5/2 ⁺			
		2622.6 ^{&}	3.8 ^{&}	5214.5	(5/2 ⁺)			
		3659.1 ^{&}	76 ^{&}	4177.9	3/2 ⁻	(M1+E2)	-0.05 3	Mult.: D+Q from (p, γ); $\Delta\pi$ =no from level scheme. B(M1)(W.u.)=1.03 $\times$ 10 ³ 19 exceeds RUL=10. B(E2)(W.u.)=7 $\times$ 10 ² +11-6 exceeds RUL=100.
		3663.6 ^{&}	8.1 ^{&}	4173.43	(5/2 ⁺)			
		3777.9 ^{&}	5.4 ^{&}	4059.1	(3/2) ⁻			
		3894.1 ^{&a}	1.4 ^{&}	3942.9	9/2 ⁺			This γ goes to 9/2 ⁺ , which is unlikely from a 3/2 ⁻ resonance. It is likely from 7839.0 level.
		3918.5 ^{&}	0.54 ^{&}	3918.47	3/2 ⁺			
		4834.2 ^{&}	11 ^{&}	3002.64	5/2 ⁺			
		6073.5 ^{&}	5.1 ^{&}	1763.18	5/2 ⁺			

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)								
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. [#]	$\delta^\#$	Comments
7837.2	3/2 ⁻	6617.1&	100&	1219.39	1/2 ⁺	(E1)		Mult.: D(+Q) with $\delta=+0.06$ 3 from (p, γ); $\Delta\pi=\text{yes}$ from level scheme; M2 component is less likely for this strong primary transition. B(E1)(W.u.)=6.5 +10-8 exceeds RUL=0.1.
		7836.3&	57&	0.0	3/2 ⁺	(E1)		Mult.: D(+Q) with $\delta=+0.02$ 3 from (p, γ); $\Delta\pi=\text{yes}$ from level scheme; M2 component is less likely for this strong primary transition. B(E1)(W.u.)=2.24 +47-44 exceeds RUL=0.1.
7839.0	(5/2 ⁺)	1658&	1.4&	6181	(1/2 ⁺ to 7/2 ⁺)			
		2240.5&	1.4&	5598.4	3/2 ⁺ , 5/2 ⁺			
		2624.4&	3.8&	5214.5	(5/2 ⁺)			
		3660.9&	76&	4177.9	3/2 ⁻			
		3665.4&	8.1&	4173.43	(5/2 ⁺)			
		3779.7&	5.4&	4059.1	(3/2) ⁻			
		3895.9&	1.4&	3942.9	9/2 ⁺			
		3920.3&	0.54&	3918.47	3/2 ⁺			
		4836.0&	11&	3002.64	5/2 ⁺			
		6075.3&	5.1&	1763.18	5/2 ⁺			
		6618.9&	100&	1219.39	1/2 ⁺			
		7838.1&	57&	0.0	3/2 ⁺			
7868.7	(1/2 ⁺ , 3/2, 5/2 ⁺)	2654.1	1.6	5214.5	(5/2 ⁺)			
		2859.5	4.9	5009.1	(1/2 ⁺ , 3/2)			
		3695.1	8.2	4173.43	(5/2 ⁺)			
		6105.0	28	1763.18	5/2 ⁺			
		6648.6	21	1219.39	1/2 ⁺			
		7867.8	100	0.0	3/2 ⁺			
7872.83	13/2 ⁺	1212.6		6660.2	(11/2 ⁻)			E_γ : from ($^{28}\text{Si}, \alpha\text{p}\gamma$) only.
		1786.2 5	9.6 4	6087.23	13/2 ⁻	D(+Q)	-0.6 6	I_γ : weighted average of 8.2 18 from ($^{28}\text{Si}, \alpha\text{p}\gamma$) and 9.7 4 from ($^{16}\text{O}, \alpha\text{p}\gamma$). Mult., δ : from $\gamma\gamma(\text{DCO})$ in ($^{28}\text{Si}, \alpha\text{p}\gamma$).
		1946.35 30	48.0 26	5926.57	(11/2) ⁻	E1+M2	+0.2 1	E_γ : weighted average of 1946.2 5 from ($^{16}\text{O}, \alpha\text{p}\gamma$) and 1946.40 30 from ($^{14}\text{N}, \alpha\text{p}\text{n}\gamma$). I_γ : weighted average of 46 5 from ($^{28}\text{Si}, \alpha\text{p}\gamma$) and 48.5 26 from ($^{16}\text{O}, \alpha\text{p}\gamma$). Mult., δ : from $\gamma\gamma(\text{DCO})$ and $\gamma\gamma(\text{lin pol})$ in ($^{28}\text{Si}, \alpha\text{p}\gamma$), $\Delta J=1$.
		2465.94 30	100.0 20	5406.95	11/2 ⁻	E1+M2	-0.25 10	E_γ : weighted average of 2466.2 5 from ($^{16}\text{O}, \alpha\text{p}\gamma$) and

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)

<u>E_i(level)</u>	<u>J^{π}_i</u>	<u>E_{γ}[†]</u>	<u>I_{γ}[‡]</u>	<u>E_f</u>	<u>J^{π}_f</u>	<u>Comments</u>
2465.85 30 from (¹⁴ N, α p γ).						
I _{γ} : weighted average of 100 9 from (²⁸ Si, α p γ) and 100.0 20 from (¹⁶ O, α p γ).						
Mult., δ : from $\gamma\gamma$ (DCO) and $\gamma\gamma$ (lin pol) in (²⁸ Si, α p γ), $\Delta J=1$.						
7880.8	3/2 ⁺ ,5/2 ⁺	2226	40	5655	(3/2) ⁺	
		3041.6	5.7	4839.10	(1/2 ⁺ ,3/2)	
		3256.3	6.7	4624.3	3/2,5/2	
		3702.7	10	4177.9	3/2 ⁻	
		3768.2	4.3	4112.4	7/2 ⁺	
		3821.5	6.7	4059.1	(3/2) ⁻	
		3962.1	6.7	3918.47	3/2 ⁺	
		4877.8	37	3002.64	5/2 ⁺	
		5186.4	83	2694.00	3/2 ⁺	
		6117.1	100	1763.18	5/2 ⁺	
		7879.9	33	0.0	3/2 ⁺	
7899.2	(5/2)	2253	2.2	5646	(5/2,7/2,9/2 ⁺)	
		2735.8	1.3	5163.34	7/2 ⁻	
		3018.1	1.2	4880.96	7/2 ⁽⁺⁾	
		3274.7	0.74	4624.3	3/2,5/2	
		3721.1	12	4177.9	3/2 ⁻	
		3725.6	12	4173.43	(5/2 ⁺)	
		3839.9	4.4	4059.1	(3/2) ⁻	
		4736.0	5.9	3162.87	7/2 ⁻	
		5204.8	1.9	2694.00	3/2 ⁺	
		6135.4	100	1763.18	5/2 ⁺	
		7898.2	5.9	0.0	3/2 ⁺	
7923.3	(3/2 ⁺ ,5/2 ⁺)	3084.1	3.4	4839.10	(1/2 ⁺ ,3/2)	
		3153.3	4.9	4769.86	(5/2 ⁻)	
		3298.8	4.6	4624.3	3/2,5/2	
		3745.2	7.3	4177.9	3/2 ⁻	
		3749.7	12	4173.43	(5/2 ⁺)	
		3864.0	4.2	4059.1	(3/2) ⁻	
		3955.5	3.7	3967.6	1/2 ⁺	
		4004.6	2.9	3918.47	3/2 ⁺	
		4920.3	2.2	3002.64	5/2 ⁺	
		5228.9	27	2694.00	3/2 ⁺	
		5277.2	2.2	2645.66	7/2 ⁺	
		6159.5	3.7	1763.18	5/2 ⁺	
		6703.2	66	1219.39	1/2 ⁺	
		7922.3	100	0.0	3/2 ⁺	
7970.3	(5/2 ⁻)	2324	15	5646	(5/2,7/2,9/2 ⁺)	
		2384.2	15	5586.0	5/2 ⁺	

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)						
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult.#
7970.3	$(5/2^-)$	2755.7	15	5214.5	$(5/2^+)$	
		3089.2	50	4880.96	$7/2^{(+)}$	
		3200.3	25	4769.86	$(5/2^-)$	
		3622.4	40	4347.74	$9/2^-$	
		3792.2	30	4177.9	$3/2^-$	
		3857.7	30	4112.4	$7/2^+$	
		4807.1	50	3162.87	$7/2^-$	
		4967.3	25	3002.64	$5/2^+$	
		5324.2	100	2645.66	$7/2^+$	
		6206.5	100	1763.18	$5/2^+$	
		6750.2	5	1219.39	$1/2^+$	
7987.8	$3/2^+$	1881.6	0.83	6106.2	$(3/2, 5/2^+)$	
		2333	2.8	5655	$(3/2)^+$	
		3809.7	<1.3	4177.9	$3/2^-$	
		3814.2	<1.3	4173.43	$(5/2^+)$	
		3928.5	1.7	4059.1	$(3/2)^-$	
		6767.7	100	1219.39	$1/2^+$	M1+E2 δ : +0.21 4 or -3.1 4 from (p, γ).
		7986.8	60	0.0	$3/2^+$	M1+E2 δ : -0.36 2 or -11 3 from (p, γ).
7995.6	$(5/2)$	2188.9	0.77	5806.6	$(1/2, 3/2, 5/2^+)$	
		2397.1	1.4	5598.4	$3/2^+, 5/2^+$	
		2592.2	3.9	5403.3	$3/2^-$	
		2832.1	1.9	5163.34	$7/2^-$	
		3817.5	5.1	4177.9	$3/2^-$	
		3936.3	1.4	4059.1	$(3/2)^-$	
		4832.4	9	3162.87	$7/2^-$	
		4992.6	2.6	3002.64	$5/2^+$	
		5301.2	0.26	2694.00	$3/2^+$	
		5349.5	0.77	2645.66	$7/2^+$	
		6231.8	1.2	1763.18	$5/2^+$	
		7994.6	100	0.0	$3/2^+$	
8001.0	$(7/2^+)$	2355	2.6	5646	$(5/2, 7/2, 9/2^+)$	
		2786.4	1.1	5214.5	$(5/2^+)$	
		3119.9	1	4880.96	$7/2^{(+)}$	
		3231.0	11	4769.86	$(5/2^-)$	
		3653.1	2.7	4347.74	$9/2^-$	
		3827.4	14	4173.43	$(5/2^+)$	
		4057.9	4.3	3942.9	$9/2^+$	
		4837.8	1.1	3162.87	$7/2^-$	
		5354.9	2.3	2645.66	$7/2^+$	
		6237.2	100	1763.18	$5/2^+$	
		8000.0	2	0.0	$3/2^+$	

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)								
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult.#	$\delta^\#$	Comments
8005.0	$5/2^+$	2350	3.3	5655	$(3/2)^+$			
		3123.9	1.1	4880.96	$7/2^{(+)}$			
		3165.8	5.5	4839.10	$(1/2^+, 3/2)$			
		3826.9	<2.4	4177.9	$3/2^-$			I_γ : for decays to 4173 and/or 4178 levels.
		3831.3	<2.4	4173.43	$(5/2^+)$			I_γ : for decays to 4173 and/or 4178 levels.
		4841.8	27	3162.87	$7/2^-$			
		5002.0	3.6	3002.64	$5/2^+$			
		5310.6	3.1	2694.00	$3/2^+$			
		5358.9	7.3	2645.66	$7/2^+$			
		6241.2	27	1763.18	$5/2^+$			
		6784.9	1.1	1219.39	$1/2^+$			
		8004.0	100	0.0	$3/2^+$			
8035.7	$1/2^-, 3/2^-$	2229.0	10.7 11	5806.6	$(1/2, 3/2, 5/2^+)$			
		3857.6	26.8 27	4177.9	$3/2^-$			
		3976.4	19.6 20	4059.1	$(3/2)^-$			
		5341.3	9 5	2694.00	$3/2^+$			
		6271.9	9 5	1763.18	$5/2^+$			
		6815.6	100 11	1219.39	$1/2^+$			
		8034.7	3.6 18	0.0	$3/2^+$			
8038.4	$1/2^+, 3/2^+$	3183.8	1.1	4854.4	$(1/2, 3/2)$			
		3199.1	2.8	4839.10	$(1/2^+, 3/2)$			
		3413.9	5.3	4624.3	$3/2, 5/2$			
		3860.3	2.8	4177.9	$3/2^-$			
		3979.1	5.3	4059.1	$(3/2)^-$			
		4070.6	1.4	3967.6	$1/2^+$			
		4119.7	2.5	3918.47	$3/2^+$			
		5035.4	7	3002.64	$5/2^+$			
		5344.0	100	2694.00	$3/2^+$	M1+E2	-0.20 2	Mult., δ : D+Q with $\delta=-0.20$ 2 or +19 8 from (p, γ); E1+M2 and the larger $\delta(E2/M1)$ is ruled out by RUL. B(M1)(W.u.)=52.1 +43-37 exceeds RUL=10. B(E2)(W.u.)= 2.8×10^2 +6-5 exceeds RUL=100.
		6274.6	26	1763.18	$5/2^+$			
		6818.3	7	1219.39	$1/2^+$			
		8037.4	14	0.0	$3/2^+$	M1+E2	-0.13 4	B(M1)(W.u.)=2.20 44; B(E2)(W.u.)=2.2 +16-12 Mult., δ : D+Q with $\delta=-0.13$ 4 or +8.1 26 from (p, γ); E1+M2 and the larger $\delta(E2/M1)$ is ruled out by RUL.
8075.9	$(5/2)$	2430	20	5646	$(5/2, 7/2, 9/2^+)$			
		2489.8	20	5586.0	$5/2^+$			
		2861.3	14	5214.5	$(5/2^+)$			
		2912.4		5163.34	$7/2^-$			
		3194.8		4880.96	$7/2^{(+)}$			

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)							
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. #	Comments
8075.9	(5/2)	3305.9	7.1	4769.86	(5/2 ⁻)		
		3897.8	14	4177.9	3/2 ⁻		
		3963.3	3.6	4112.4	7/2 ⁺		
		5072.9	3.6	3002.64	5/2 ⁺		
		6312.1	100	1763.18	5/2 ⁺		
		8074.9		0.0	3/2 ⁺		
8096.1	(5/2)	2289.4	5.1	5806.6	(1/2,3/2,5/2 ⁺)		
		2450	13	5646	(5/2,7/2,9/2 ⁺)		
		2497.6	13	5598.4	3/2 ⁺ ,5/2 ⁺		
		2881.5	3.3	5214.5	(5/2 ⁺)		
		2932.6	5.1	5163.34	7/2 ⁻		
		3256.8	7.7	4839.10	(1/2 ⁺ ,3/2)		
		3326.1	5.1	4769.86	(5/2 ⁻)		
		3918.0	4.9	4177.9	3/2 ⁻		
		4036.8	5.1	4059.1	(3/2) ⁻		
		4177.4	4.6	3918.47	3/2 ⁺		
		4932.9	13	3162.87	7/2 ⁻		
		5401.7	13	2694.00	3/2 ⁺		
		5450.0	44	2645.66	7/2 ⁺		
		6332.3	21	1763.18	5/2 ⁺		
		8095.1	100	0.0	3/2 ⁺		
8106.4	3/2 ⁺	2507.9	1.4	5598.4	3/2 ⁺ ,5/2 ⁺		
		3251.8	1.6	4854.4	(1/2,3/2)		
		3267.1	1.4	4839.10	(1/2 ⁺ ,3/2)		
		3928.3	4.3	4177.9	3/2 ⁻		
		4047.1	3.5	4059.1	(3/2) ⁻		
		4138.5	8.1	3967.6	1/2 ⁺		
		5103.4	24	3002.64	5/2 ⁺		
		5412.0	24	2694.00	3/2 ⁺	M1+E2	δ : +4.9 10 or -0.05 4.
		6342.6	8.1	1763.18	5/2 ⁺		
		6886.3	95	1219.39	1/2 ⁺		
		8105.4	100	0.0	3/2 ⁺	M1+E2	δ : +1.2 1 or -0.46 4.
		1621	19	6492	(3/2 ⁺)		
8113.3	(1/2 ⁺ ,3/2,5/2 ⁺)	2007.0	19	6106.2	(3/2,5/2 ⁺)		
		2306.6	3.4	5806.6	(1/2,3/2,5/2 ⁺)		
		2458	2.8	5655	(3/2) ⁺		
		2514.8		5598.4	3/2 ⁺ ,5/2 ⁺		
		2898.7	5.6	5214.5	(5/2 ⁺)		
		3939.6	22	4173.43	(5/2 ⁺)		
		4054.0	4.1	4059.1	(3/2) ⁻		
		4145.4	16	3967.6	1/2 ⁺		
		5110.3	5.9	3002.64	5/2 ⁺		

Adopted Levels, Gammas (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	$\gamma(^{35}\text{Cl})$ (continued)		Comments
						Mult. #	$\delta^\#$	
8113.3	$(1/2^+, 3/2, 5/2^+)$	5418.9	41	2694.00	$3/2^+$			
		6349.5	16	1763.18	$5/2^+$			
		6893.2	100	1219.39	$1/2^+$			
		8112.3	78	0.0	$3/2^+$			
8147.2	$3/2^-$	2464.9	1.6	5682.2	$1/2^-, 3/2^-$			
		3138.0	4.8	5009.1	$(1/2^+, 3/2)$			
		3969.1	13	4177.9	$3/2^-$			
		4087.8	4.8	4059.1	$(3/2)^-$			
		4179.3	3.2	3967.6	$1/2^+$			
		4228.5	4.8	3918.47	$3/2^+$			
		5452.7	4.8	2694.00	$3/2^+$			
		6927.1	24	1219.39	$1/2^+$			
		8146.2	100	0.0	$3/2^+$			
8156.8	$(5/2^+)$	2511	7.8	5646	$(5/2, 7/2, 9/2^+)$			
		2570.7	16	5586.0	$5/2^+$			
		2942.2	12	5214.5	$(5/2^+)$			
		3386.8	2.0	4769.86	$(5/2^-)$			
		3978.7	9.8	4177.9	$3/2^-$			
		4044.2	14	4112.4	$7/2^+$			
		4097.4	2.0	4059.1	$(3/2)^-$			
		4213.6	3.9	3942.9	$9/2^+$			
		4993.6	9.8	3162.87	$7/2^-$			
		5153.8	12	3002.64	$5/2^+$			
		5510.7	100	2645.66	$7/2^+$			
		6393.0	2.0	1763.18	$5/2^+$			
		8155.8	5.9	0.0	$3/2^+$			
8179.4	$(3/2^-, 5/2, 7/2^+)$	3554.9	13.2 <i>I3</i>	4624.3	$3/2, 5/2$			
		5016.1	11.8 <i>I2</i>	3162.87	$7/2^-$			
		5176.4	5.9 <i>30</i>	3002.64	$5/2^+$			
		5484.9	4.4 <i>22</i>	2694.00	$3/2^+$			
		6415.6	100 <i>I0</i>	1763.18	$5/2^+$			
		8178.4	11.8 <i>I2</i>	0.0	$3/2^+$			
8208.2	$5/2^+$	2553	0.9	5655	$(3/2)^+$			
		3327.1	1.0	4880.96	$7/2^{(+)}$			
		3368.9	0.51	4839.10	$(1/2^+, 3/2)$			
		4148.8	1.9	4059.1	$(3/2)^-$			
		4289.5		3918.47	$3/2^+$			
		5205.1	1.4	3002.64	$5/2^+$			
		5513.7	0.64	2694.00	$3/2^+$			
		6444.4	18	1763.18	$5/2^+$	M1+E2	+0.33 4	B(M1)(W.u.)=0.75 +35-22; B(E2)(W.u.)=7.5 +41-25
		6988.1	3.9	1219.39	$1/2^+$			
		8207.2	100	0.0	$3/2^+$	M1+E2	-0.36 4	B(M1)(W.u.)=2.0 +9-5; B(E2)(W.u.)=15 +7-4

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)								
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult.#	$\delta^\#$	Comments
8216.3	$5/2^+$	2561	1.8	5655	$(3/2)^+$			
		3446.3	1.8	4769.86	$(5/2^-)$			
		4038.2	<6.7	4177.9	$3/2^-$			
		4042.6	<6.7	4173.43	$(5/2^+)$			
		5053.0	91	3162.87	$7/2^-$	(E1)		B(E1)(W.u.)=0.062 +21-12 Mult.: D+Q with $\delta=+0.11$ 2 from $\gamma(\theta)$ in (p, γ); $\Delta\pi$ =yes from level scheme; but M2 component is unlikely for this strong primary transition.
8242.4	$3/2^-$	5213.2	11	3002.64	$5/2^+$			
		5521.8	6.7	2694.00	$3/2^+$			
		6452.5	3.1	1763.18	$5/2^+$			
		8215.3	100	0.0	$3/2^+$	M1+E2	+0.06 3	B(M1)(W.u.)=0.55 +18-9; B(E2)(W.u.)=0.11 +16-8
		4068.7	7.7	4173.43	$(5/2^+)$			
8251	$(3/2^+, 5/2^+)$	5239.3	2.2	3002.64	$5/2^+$			
		7022.3	100	1219.39	$1/2^+$			
		4138	4.9	4112.4	$7/2^+$			
		5248	7.3	3002.64	$5/2^+$			
		7031	100	1219.39	$1/2^+$			
8269.0	$5/2^+$	8250	9.8	0.0	$3/2^+$			
		3499.0	4.4	4769.86	$(5/2^-)$			
		4090.8	27	4177.9	$3/2^-$			
		4350.2	6.7	3918.47	$3/2^+$			
		5105.7	16	3162.87	$7/2^-$			
		5265.9	29	3002.64	$5/2^+$			
		5574.5	13	2694.00	$3/2^+$			
		5622.9	8.9	2645.66	$7/2^+$			
		6505.2	18	1763.18	$5/2^+$			
		8268.0	100	0.0	$3/2^+$			
8277.2	$(5/2)^+$	2622	5.2	5655	$(3/2)^+$			
		3396.1	17	4880.96	$7/2^{(+)}$			
		3437.9	38	4839.10	$(1/2^+, 3/2)$			
		3507.2	4.1	4769.86	$(5/2^-)$			
		3652.7	10	4624.3	$3/2, 5/2$			
		3929.2	6.2	4347.74	$9/2^-$			
		4103.5	6.9	4173.43	$(5/2^+)$			
		4217.8	5.2	4059.1	$(3/2)^-$			
		4358.4	17	3918.47	$3/2^+$			
		5274.1	35	3002.64	$5/2^+$			
		5582.7	10	2694.00	$3/2^+$			
		5631.1	10	2645.66	$7/2^+$			
		6513.4	79	1763.18	$5/2^+$			
		8276.2	100	0.0	$3/2^+$			
8282.4	$(3/2^-, 5/2)$	2696.3	29	5586.0	$5/2^+$			

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. [#]	Comments
43	(3/2 ⁻ , 5/2)	8282.4	29	5214.5	(5/2 ⁺)		
			11	5163.34	7/2 ⁻		
			50	4177.9	3/2 ⁻		
			21	4059.1	(3/2) ⁻		
			93	3162.87	7/2 ⁻		
			25	1763.18	5/2 ⁺		
			100	0.0	3/2 ⁺		
		8287.82	4.6	5655	(3/2) ⁺		
			6.8	5009.1	(1/2 ⁺ , 3/2)		
			21	4177.9	3/2 ⁻		
	1/2 ⁻		4.6	4059.1	(3/2) ⁻		
			6.8	3967.6	1/2 ⁺		
			2.3	3918.47	3/2 ⁺		
			52	2694.00	3/2 ⁺		
			100	1219.39	1/2 ⁺	(E1)	B(E1)(W.u.)=0.069 +24-14 Mult.: D+Q with $\delta(Q/D)=+0.21$ 2 or -3.2 2 in (p, γ); but M2 component is unlikely for this primary transition based on resonance $\Gamma=0.04$ keV.
			30	0.0	3/2 ⁺		
		8298.2	13	5009.1	(1/2 ⁺ , 3/2)		
			17	4880.96	7/2 ⁽⁺⁾		
			21	4854.4	(1/2, 3/2)		
			25	4173.43	(5/2 ⁺)		
	3/2 ⁺		42	4059.1	(3/2) ⁻		
			4.2	3162.87	7/2 ⁻		This γ is questionable if $J^\pi(8298)=3/2^+$, since it would require Mult=M2 which is probably unlikely; however $J^\pi(8298)=3/2^-$ would make the strong 7078.0 γ and 8297.1 γ less likely.
			4.2	3002.64	5/2 ⁺		
			42	2694.00	3/2 ⁺		
			25	2645.66	7/2 ⁺		
			33	1763.18	5/2 ⁺		
			100	1219.39	1/2 ⁺	M1+E2	δ : -1.7 1 or -0.02 2 from $\gamma(\theta)$ in (p, γ).
			54	0.0	3/2 ⁺	M1+E2	δ : -0.15 4 or +9.5 36 from $\gamma(\theta)$ in (p, γ).
		8318.1	0.9	5806.6	(1/2, 3/2, 5/2 ⁺)		
			0.9	5646	(5/2, 7/2, 9/2 ⁺)		
	(5/2 ⁺)		1.9	5598.4	3/2 ⁺ , 5/2 ⁺		
			0.39	4769.86	(5/2) ⁻		
			2.3	4624.3	3/2 ⁺ , 5/2		
			<3.9	4177.9	3/2 ⁻		
			<3.9	4173.43	(5/2 ⁺)		
			1.5	4059.1	(3/2) ⁻		
			3.9	3967.6	1/2 ⁺		
			0.64	3162.87	7/2 ⁻		

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)								
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. [#]	$\delta^\#$	Comments
8318.1	(5/2 ⁺)	5315.0	3.9	3002.64	5/2 ⁺			
		5623.6	2.1	2694.00	3/2 ⁺			
		5672.0	1.2	2645.66	7/2 ⁺			
		6554.3	2.3	1763.18	5/2 ⁺			
		7097.9	2.6	1219.39	1/2 ⁺			
		8317.0	100	0.0	3/2 ⁺			
8319.3	15/2 ⁻	2232.7 6	57.2 28	6087.23	13/2 ⁻	[M1,E2]		E_γ, I_γ : from (¹⁶ O, $\alpha\gamma$). B(M1)(W.u.)=0.0577 4I if M1, B(E2)(W.u.)=43.8 3I if E2. B(E2)(W.u.)=20.3 12
		2911.9 8	100 5	5406.95	11/2 ⁻	E2		$E_\gamma, I_\gamma, \text{Mult.}$: from (¹⁶ O, $\alpha\gamma$). Mult=Q, $\Delta J=2$ from $\gamma\gamma(\text{DCO})$; M2 ruled out by RUL.
8323.1	(3/2 ⁺ , 5/2 ⁺)	4210.4	1.2	4112.4	7/2 ⁺			
		5320.0	3.7	3002.64	5/2 ⁺			
		5677.0	4.9	2645.66	7/2 ⁺			
		6559.3	<2.4	1763.18	5/2 ⁺			
		7102.9	9.8	1219.39	1/2 ⁺			
		8322.0	100	0.0	3/2 ⁺			
8381.8	5/2 ⁺	3611.7		4769.86	(5/2 ⁻)			
		3757.3		4624.3	3/2, 5/2			
		4269.1	23.5 24	4112.4	7/2 ⁺			
		4463.0	71 7	3918.47	3/2 ⁺	M1+E2	+0.15 3	B(M1)(W.u.)=2.9 +13-7; B(E2)(W.u.)=13 +8-5
		5378.7	74 8	3002.64	5/2 ⁺	M1+E2	+0.02 3	B(M1)(W.u.)=1.8 +14-7
		5687.3	2.9 15	2694.00	3/2 ⁺	M1+E2	-0.72 3I	B(M1)(W.u.)=0.039 +32-2I; B(E2)(W.u.)=2.4 +23-18
		5735.6	15 8	2645.66	7/2 ⁺			
		6618.0	100 10	1763.18	5/2 ⁺	M1+E2	+0.06 3	B(M1)(W.u.)=1.3 +6-3; B(E2)(W.u.)=0.4 +6-3
		8380.7	9 5	0.0	3/2 ⁺	D+Q	1.65	
8402.9	5/2 ⁻	3632.8	16 8	4769.86	(5/2 ⁻)			
		4224.7	29 16	4177.9	3/2 ⁻			
		4343.5	74 20	4059.1	(3/2 ⁻)			
		5708.4	32 16	2694.00	3/2 ⁺			
		5756.7	23 13	2645.66	7/2 ⁺			
		6639.0	48 13	1763.18	5/2 ⁺			
		8401.8	100 26	0.0	3/2 ⁺			
8405.8	(5/2 ⁻)	3524.7	35 9	4880.96	7/2 ⁽⁺⁾			
		4232.1	3.7 19	4173.43	(5/2 ⁺)			
		4293.1	3.3 16	4112.4	7/2 ⁺			
		5402.7	35 9	3002.64	5/2 ⁺			
		6641.9	100 26	1763.18	5/2 ⁺			
		8404.7	56 14	0.0	3/2 ⁺			
8416.7	1/2 ⁺	3407.4	14	5009.1	(1/2 ⁺ , 3/2)			
		4243.0	31	4173.43	(5/2 ⁺)			

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)										
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult.#	$\delta^\#$	Comments		
8416.7	$1/2^+$	4473.5	8.6	3942.9	$9/2^+$					
		5413.6	20	3002.64	$5/2^+$					
		5770.5	23	2645.66	$7/2^+$					
		6652.8	63	1763.18	$5/2^+$					
		7196.5	26	1219.39	$1/2^+$					
		8415.6	100	0.0	$3/2^+$					
8464.3	$(1/2^+, 3/2, 5/2^+)$	2809	47 24	5655	$(3/2)^+$					
		3249.6	34 17	5214.5	$(5/2^+)$					
		3455.0	47 24	5009.1	$(1/2^+, 3/2)$					
		3609.7	19 10	4854.4	$(1/2, 3/2)$					
		3839.8	25 13	4624.3	$3/2, 5/2$					
		4286.1	31 15	4177.9	$3/2^-$					
		4290.6	32 17	4173.43	$(5/2^+)$					
		4496.4	31 15	3967.6	$1/2^+$					
		4545.5	15 8	3918.47	$3/2^+$					
		5461.2	100 24	3002.64	$5/2^+$					
		5769.8	94 24	2694.00	$3/2^+$					
		7244.1	53 24	1219.39	$1/2^+$					
		8463.2	59 18	0.0	$3/2^+$					
		8484.4	$(3/2)^+$	2829	12 6	5655	$(3/2)^+$			
				3603.2	15 8	4880.96	$7/2^{(+)}$			
3859.9	2.2 11			4624.3	$3/2, 5/2$					
4516.5	2.6 13			3967.6	$1/2^+$					
4565.6	11 6			3918.47	$3/2^+$					
5481.3	15 9			3002.64	$5/2^+$					
5789.9	44 11			2694.00	$3/2^+$					
5838.2	6.1 31			2645.66	$7/2^+$					
6720.5	100 26			1763.18	$5/2^+$					
8483.3	9 5			0.0	$3/2^+$					
8486.2	$3/2^-$			2728	2.4 13	5758	$(1/2, 3/2^-)$			
		3861.7	3.9 20	4624.3	$3/2, 5/2$					
		4308.0	10 5	4177.9	$3/2^-$					
		4312.5	10 5	4173.43	$(5/2^+)$					
		4426.8	4.1 22	4059.1	$(3/2)^-$					
		4518.3	15 9	3967.6	$1/2^+$					
		5483.1	17 9	3002.64	$5/2^+$					
		5791.7	41 11	2694.00	$3/2^+$					
		6722.3	8 4	1763.18	$5/2^+$					
		7266.0	14 7	1219.39	$1/2^+$					
		8485.1	100 26	0.0	$3/2^+$					
8487.0	$15/2^-$	2399.8 8	100.0 34	6087.23	$13/2^-$	(M1(+E2))	+0.2 +3-2	Mult., δ : D+Q and δ from $\gamma\gamma(\text{DCO})$ in ($^{28}\text{Si}, \alpha p \gamma$); $\Delta\pi$ =no from level scheme.		

I_γ : final level is uncertain, either 4173 or 4178 (1976Sp08).
 I_γ : final level is uncertain, either 4173 or 4178 (1976Sp08).

Adopted Levels, Gammas (continued) $\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. #	$\delta^\#$	Comments
8487.0	$15/2^-$	3080.0 7	31.8 20	5406.95	$11/2^-$	E2		$E_\gamma, I_\gamma, \text{Mult.}$: from ($^{16}\text{O}, \alpha p \gamma$), with Mult=Q, $\Delta J=2$ from $\gamma\gamma(\text{ADO})$ and M2 ruled out by RUL.
8514.3	$1/2^-$	3659.7	16 8	4854.4	$(1/2, 3/2)$			
		4336.1	7 4	4177.9	$3/2^-$			
		4454.9	100 26	4059.1	$(3/2)^-$			
		4546.4	85 22	3967.6	$1/2^+$			
		4595.5	24 12	3918.47	$3/2^+$			
		7294.1	37 11	1219.39	$1/2^+$			
		8513.2	100 26	0.0	$3/2^+$			
8558.2?	$(13/2^-)$	2631.5 ^a	100	5926.57	$(11/2)^-$	D(+Q)	-0.2 +2-3	$E_\gamma, \text{Mult.}, \delta$: from ($^{28}\text{Si}, \alpha p \gamma$); $\Delta\pi=\text{no}$ from level scheme; Mult=Q, $\Delta J=1$.
8571.9	$(5/2)^+$	2848.3	3.1 16	5723.52	$5/2^+$			
		2917	1.9 10	5655	$(3/2)^+$			
		2973.4	1.0 6	5598.4	$3/2^+, 5/2^+$			
		3408.4	2.6 13	5163.34	$7/2^-$			
		3690.7	1.3 7	4880.96	$7/2^{(+)}$			
		4398.2	1.0 6	4173.43	$(5/2^+)$			
		4459.2	1.6 9	4112.4	$7/2^+$			
		4653.1	1.6 9	3918.47	$3/2^+$			
		5408.6	2.3 12	3162.87	$7/2^-$			
		5568.8	6.4 33	3002.64	$5/2^+$			
		5877.4	0.9 4	2694.00	$3/2^+$			
		5925.7	11 6	2645.66	$7/2^+$			
		7351.7	8 4	1219.39	$1/2^+$			
		8570.8	100 10	0.0	$3/2^+$			
8580.5	$1/2^+$	2088	8 4	6492	$(3/2^+)$			
		3741.2	2.4 13	4839.10	$(1/2^+, 3/2)$			
		4402.3	8 4	4177.9	$3/2^-$			
		4521.1	3.7 19	4059.1	$(3/2)^-$			
		4661.7	4.3 22	3918.47	$3/2^+$			
		5886.0	10 5	2694.00	$3/2^+$			
		7360.3	100 11	1219.39	$1/2^+$			
		8579.4	48 13	0.0	$3/2^+$			
8590.1	$5/2^+$	2832	22 11	5758	$(1/2, 3/2^-)$			
		2935	17 9	5655	$(3/2)^+$			
		3820.0	73 18	4769.86	$(5/2^-)$			
		4416.4	59 14	4173.43	$(5/2^+)$			
		4477.4	16 8	4112.4	$7/2^+$			
		5426.8	36 18	3162.87	$7/2^-$			
		5587.0	100 27	3002.64	$5/2^+$			
		5943.9	25 13	2645.66	$7/2^+$			
		6826.2	64 18	1763.18	$5/2^+$			

Adopted Levels, Gammas (continued) $\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. [#]	$\delta^\#$	Comments
8590.1	5/2 ⁺	8589.0	46 14	0.0	3/2 ⁺			
8612.0	3/2 ⁺ , 5/2 ⁺	6848.1	100 10	1763.18	5/2 ⁺			
		7391.8	34 8	1219.39	1/2 ⁺			
		8610.9	66 16	0.0	3/2 ⁺			
8613.7	5/2 ⁺	3732.5	2.9 14	4880.96	7/2 ⁽⁺⁾			
		4554.3	10 5	4059.1	(3/2) ⁻			
		4670.5	7.1	3942.9	9/2 ⁺			
		4694.9	13 7	3918.47	3/2 ⁺			
		5610.6	4.5 23	3002.64	5/2 ⁺			
		5919.2	8 4	2694.00	3/2 ⁺			
		5967.5	41 11	2645.66	7/2 ⁺			
		6849.8	16	1763.18	5/2 ⁺			
		7393.5	3.6	1219.39	1/2 ⁺			
		8612.6	100 11	0.0	3/2 ⁺			
8618.2	3/2 ⁺ , 5/2 ⁺	2894.6	6.4 33	5723.52	5/2 ⁺			
		2963	5.2 27	5655	(3/2) ⁺			
		3763.6	4.2 21	4854.4	(1/2, 3/2)			
		3993.7	4.6 24	4624.3	3/2, 5/2			
		4650.3	9 5	3967.6	1/2 ⁺			
		4699.4	7 4	3918.47	3/2 ⁺			
		5615.1	49 12	3002.64	5/2 ⁺			
		5923.7	42 12	2694.00	3/2 ⁺			
		5972.0	5.2 27	2645.66	7/2 ⁺			
		6854.3	70 18	1763.18	5/2 ⁺			
		8617.1	100 24	0.0	3/2 ⁺			
8630.1	7/2 ⁻	2906.5	0.6 4	5723.52	5/2 ⁺			
		3466.6	17 9	5163.34	7/2 ⁻	M1+E2	+0.60 10	Mult., δ : D+Q and δ from $\gamma(\theta)$ in (p, γ); E1+M2 ruled out by RUL.
		3860.0	3.0 15	4769.86	(5/2) ⁻			
		4282.1	26 7	4347.74	9/2 ⁻	M1+E2	-0.184 14	Mult., δ : D+Q and δ from $\gamma(\theta)$ in (p, γ); E1+M2 ruled out by RUL.
		4456.4	8 4	4173.43	(5/2) ⁺			
		4517.4	21 11	4112.4	7/2 ⁺	(E1+M2)	-0.06 2	Mult., δ : D+Q and δ from $\gamma(\theta)$ in (p, γ); $\Delta\pi$ =yes from level scheme.
		4686.9	21 11	3942.9	9/2 ⁺			
		5466.8	7 4	3162.87	7/2 ⁻			
		5627.0	3.0 15	3002.64	5/2 ⁺			
		5983.9	5.5 28	2645.66	7/2 ⁺			
		6866.2	100 26	1763.18	5/2 ⁺	(E1+M2)	-0.011 3	Mult., δ : D+Q and δ from $\gamma(\theta)$ in (p, γ); $\Delta\pi$ =yes from level scheme.
		8629.0	1.1 7	0.0	3/2 ⁺			
8640	(5/2 ⁺)	4697	61	3942.9	9/2 ⁺			
		5477	53	3162.87	7/2 ⁻			
		5637	16	3002.64	5/2 ⁺			
		5946	34	2694.00	3/2 ⁺			
		6876	100	1763.18	5/2 ⁺			

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)								
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Comments		
8641.4	$3/2^+, 5/2^+$	2986	10 6	5655	$(3/2)^+$	I_γ : final level is uncertain, either 4173 or 4178 (1976Sp08). I_γ : final level is uncertain, either 4173 or 4178 (1976Sp08).		
		3055.3	9 5	5586.0	$5/2^+$			
		3237.9	17 9	5403.3	$3/2^-$			
		3802.1	9 5	4839.10	$(1/2^+, 3/2)$			
		3871.3	86 22	4769.86	$(5/2^-)$			
		4016.9	7 4	4624.3	$3/2, 5/2$			
		4463.2	50 14	4177.9	$3/2^-$			
		4467.7	50 14	4173.43	$(5/2^+)$			
		4582.0	32 18	4059.1	$(3/2)^-$			
		4722.6	21 11	3918.47	$3/2^+$			
		5946.9	19 9	2694.00	$3/2^+$			
		8640.3	100 25	0.0	$3/2^+$			
		8686.2	$5/2^-$	3100.1	2.8 15		5586.0	$5/2^+$
				3282.7	4.1 21		5403.3	$3/2^-$
				4508.0	2.6 13		4177.9	$3/2^-$
				4626.8	4.6 24		4059.1	$(3/2)^-$
5522.9	4.9 25			3162.87	$7/2^-$			
5991.7	3.7 19			2694.00	$3/2^+$			
6922.3	6.3 32			1763.18	$5/2^+$			
7466.0	2.5 13			1219.39	$1/2^+$			
8685.0	100 11			0.0	$3/2^+$			
8696.9	$3/2^-$			2890.2	3.6 19	5806.6	$(1/2, 3/2, 5/2^+)$	
		3098.4	6.0 30	5598.4	$3/2^+, 5/2^+$			
		7476.7	15 4	1219.39	$1/2^+$			
		8695.7	100 10	0.0	$3/2^+$			
		8750.9	$3/2^-$	2944.2	4.2 21	5806.6	$(1/2, 3/2, 5/2^+)$	
3347.4	9 5			5403.3	$3/2^-$			
3536.2	4.5 23			5214.5	$(5/2^+)$			
4572.7	4.0 21			4177.9	$3/2^-$			
4577.2	4.3 23			4173.43	$(5/2^+)$			
6056.3	5.7 28			2694.00	$3/2^+$			
6987.0	57 15			1763.18	$5/2^+$			
7530.6	3.8			1219.39	$1/2^+$			
8749.7	100 10			0.0	$3/2^+$			
8779.8	$3/2^-$			3181.2	16 8	5598.4	$3/2^+, 5/2^+$	
		3565.1	24 12	5214.5	$(5/2^+)$			
		4601.6	56 15	4177.9	$3/2^-$			
		4606.0	56 15	4173.43	$(5/2^+)$			
		4861.0	16 8	3918.47	$3/2^+$			
		5776.7	100 26	3002.64	$5/2^+$			
		6085.2	44 11	2694.00	$3/2^+$			
		6133.6	48 11	2645.66	$7/2^+$			

Adopted Levels, Gammas (continued)

<u>$\gamma(^{35}\text{Cl})$ (continued)</u>							
<u>$E_i(\text{level})$</u>	<u>J_i^π</u>	<u>$E_\gamma^\dagger$</u>	<u>$I_\gamma^\ddagger$</u>	<u>E_f</u>	<u>J_f^π</u>	<u>Mult.[#]</u>	<u>Comments</u>
8779.8	3/2 ⁻	7015.9	25 13	1763.18	5/2 ⁺		
		8778.6	41 11	0.0	3/2 ⁺		
8787.2	(3/2 ⁺)	3063.5	27 14	5723.52	5/2 ⁺		
		3623.7	76 20	5163.34	7/2 ⁻		
		4017.1	27 14	4769.86	(5/2 ⁻)		
		4609.0	36 20	4177.9	3/2 ⁻		
		5623.8	23 12	3162.87	7/2 ⁻		
		5784.1	80 20	3002.64	5/2 ⁺		
		7023.3	100 24	1763.18	5/2 ⁺		
		8786.0	32 16	0.0	3/2 ⁺		
8788.3	(15/2 ⁺)	915.4 4	100	7872.83	13/2 ⁺	(M1)	E_γ , Mult.: from ($^{16}\text{O}, \alpha p \gamma$), with Mult=D, $\Delta J=1$ from $\gamma\gamma(\text{ADO})$; $\Delta\pi=(\text{no})$ from level scheme.
8798.4	(3/2 ⁺ , 5/2 ⁺)	4739.0	13 7	4059.1	(3/2 ⁻)		
		5795.2	26 7	3002.64	5/2 ⁺		
		7034.5	11 6	1763.18	5/2 ⁺		
		7578.1	11 6	1219.39	1/2 ⁺		
		8797.2	100 10	0.0	3/2 ⁺		
8820.9	(1/2 ⁺ to 7/2 ⁺)	5817.7	17 10	3002.64	5/2 ⁺		
		6126.3	35 10	2694.00	3/2 ⁺		
		7057.0	40 10	1763.18	5/2 ⁺		
		8819.7	100 10	0.0	3/2 ⁺		
8824.2	1/2 ⁺	5821.0	100 24	3002.64	5/2 ⁺		
		6129.6	16.2 8	2694.00	3/2 ⁺		
		7060.3	43 11	1763.18	5/2 ⁺		
		7603.9	60 16	1219.39	1/2 ⁺		
		8823.0	51 14	0.0	3/2 ⁺		
8829.3	1/2 ⁻	3174	30 15	5655	(3/2 ⁺)		
		3990.0	3.9 21	4839.10	(1/2 ⁺ , 3/2)		
		4059.2	4.6 24	4769.86	(5/2 ⁻)		
		4910.5	14 7	3918.47	3/2 ⁺		
		5826.1	67 18	3002.64	5/2 ⁺		
		6134.7	39 9	2694.00	3/2 ⁺		
		6183.1	100 24	2645.66	7/2 ⁺		
		7065.4	27 15	1763.18	5/2 ⁺		
		7609.0	8 4	1219.39	1/2 ⁺		
		8828.1	12 6	0.0	3/2 ⁺		
8833.1	(5/2)	3618.4	30 15	5214.5	(5/2 ⁺)		
		4063.0	11 6	4769.86	(5/2 ⁻)		
		4659.3	61 18	4173.43	(5/2 ⁺)		
		4720.4	20 11	4112.4	7/2 ⁺		
		4773.7	48 13	4059.1	(3/2 ⁻)		
		4914.3	48 13	3918.47	3/2 ⁺		

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)							Comments
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. [#]	
8833.1	(5/2)	5669.7	16 8	3162.87	7/2 ⁻		
		5829.9	29 15	3002.64	5/2 ⁺		
		6138.5	20 10	2694.00	3/2 ⁺		
		6186.9	100 26	2645.66	7/2 ⁺		
		7069.2	39 22	1763.18	5/2 ⁺		
		8831.9	13 7	0.0	3/2 ⁺		
8838.1	5/2 ⁻ , 7/2 ⁻	3674.6		5163.34	7/2 ⁻		
		3956.9		4880.96	7/2 ⁽⁺⁾		
		4068.0	13 7	4769.86	(5/2 ⁻)		
		4490.1		4347.74	9/2 ⁻		
		4778.7		4059.1	(3/2 ⁻)		
		5674.7	46 11	3162.87	7/2 ⁻		
		7074.2	100 10	1763.18	5/2 ⁺		
		971.38 20	100.0 30	7872.83	13/2 ⁺	E2	B(E2)(W.u.)=16.3 +37-26 E _γ : weighted average of 971.4 5 from (¹⁶ O,αpγ) and 971.38 20 from (¹⁴ N,αpnγ).
8856.2	5/2 ⁺	2757.0		6087.23	13/2 ⁻		
		3975.0	6.7 33	4880.96	7/2 ⁽⁺⁾		
		4231.6	10 5	4624.3	3/2, 5/2		
		4678.0	7 4	4177.9	3/2 ⁻		
		4937.4	22 6	3918.47	3/2 ⁺		
		5692.8	100 10	3162.87	7/2 ⁻		
		6161.6	5.5 28	2694.00	3/2 ⁺		
		6210.0		2645.66	7/2 ⁺		
		7092.3	14 8	1763.18	5/2 ⁺		
		8855.0	31 8	0.0	3/2 ⁺		
8868.6	3/2 ⁺ , 5/2 ⁺	4690.4	5.3 27	4177.9	3/2 ⁻		
		5865.4	33 8	3002.64	5/2 ⁺		
		6174.0	25 6	2694.00	3/2 ⁺		
		7104.7	9 5	1763.18	5/2 ⁺		
		7648.3	33 8	1219.39	1/2 ⁺		
		8867.4	100 25	0.0	3/2 ⁺		
		3231	23 12	5655	(3/2 ⁺)		
		3287.5	25 13	5598.4	3/2 ⁺ , 5/2 ⁺		
8886.1	(5/2)	3671.4	9 4	5214.5	(5/2 ⁺)		
		3722.6	35 16	5163.34	7/2 ⁻		
		4116.0	17 9	4769.86	(5/2 ⁻)		
		4707.9	31 16	4177.9	3/2 ⁻		
		4967.3	16 8	3918.47	3/2 ⁺		
		5882.9	31 16	3002.64	5/2 ⁺		
		6239.8	100 27	2645.66	7/2 ⁺		

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)							
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. #	Comments
8886.1	(5/2)	7122.1	46 12	1763.18	5/2 ⁺		
		8884.9	54 16	0.0	3/2 ⁺		
8893.3	(5/2 ⁺ , 7/2 ⁺)	3307.1	4.3 22	5586.0	5/2 ⁺		
		4780.6	24 11	4112.4	7/2 ⁺		
		4950.0	11 6	3942.9	9/2 ⁺		
		5729.9	78 19	3162.87	7/2 ⁻		
		6198.7	100 24	2694.00	3/2 ⁺		
		7129.3	51 14	1763.18	5/2 ⁺		
		8892.1	2.7 14	0.0	3/2 ⁺		
8904.8	(1/2, 3/2, 5/2 ⁺)	2724	15 8	6181	(1/2 ⁺ to 7/2 ⁺)		
		3250	12 6	5655	(3/2) ⁺		
		3895.5	13 7	5009.1	(1/2 ⁺ , 3/2)		
		4050.2	21 11	4854.4	(1/2, 3/2)		
		4065.5	16 8	4839.10	(1/2 ⁺ , 3/2)		
		4726.6	62 17	4177.9	3/2 ⁻		
		4845.3	20 10	4059.1	(3/2) ⁻		
		6210.2	52 14	2694.00	3/2 ⁺		
		7684.5	35 10	1219.39	1/2 ⁺		
		8903.6	100 24	0.0	3/2 ⁺		
8906.8	5/2 ⁺	3252		5655	(3/2) ⁺		
		3261	5.4 28	5646	(5/2, 7/2, 9/2 ⁺)		
		3308.2	3.2 16	5598.4	3/2 ⁺ , 5/2 ⁺		
		3320.6	5.8	5586.0	5/2 ⁺		
		3376	3.6 19	5531			
		3692.1	6.4 32	5214.5	(5/2 ⁺)		
		4136.7	8 4	4769.86	(5/2) ⁻		
		4963.5	19 5	3942.9	9/2 ⁺		
		7142.8	100 10	1763.18	5/2 ⁺		
8919.8	(5/2 ⁻ , 7/2, 9/2 ⁺)	3274	10 5	5646	(5/2, 7/2, 9/2 ⁺)		
		4571.7	14 7	4347.74	9/2 ⁻		
		5756.4	100 11	3162.87	7/2 ⁻		
		6273.5	46 12	2645.66	7/2 ⁺		
		7155.8	7 4	1763.18	5/2 ⁺		
8953.1	3/2 ⁺	7732.8		1219.39	1/2 ⁺	D+Q	δ : or +5.0 4.
		8951.9		0.0	3/2 ⁺	D+Q	δ : +8.2 7.
8984.2	5/2 ⁺	3260.5	1.3 7	5723.52	5/2 ⁺		
		4144.8	2.0 10	4839.10	(1/2 ⁺ , 3/2)		
		4214.1	0.8 5	4769.86	(5/2) ⁻		
		4359.6	0.8 5	4624.3	3/2, 5/2		
		4806.0	2.5 13	4177.9	3/2 ⁻		
		4810.4	2.5 13	4173.43	(5/2 ⁺)		
		5065.3	15 8	3918.47	3/2 ⁺		

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. [#]	$\delta^\#$	Comments
8984.2	5/2 ⁺	5820.8	10 5	3162.87	7/2 ⁻			
		6289.6	3.0 15	2694.00	3/2 ⁺			
		6337.9	2.0 10	2645.66	7/2 ⁺			
		7220.2	27 7	1763.18	5/2 ⁺			
		8983.0	100 10	0.0	3/2 ⁺			
9081.5	5/2 ⁺	3357.8	1.7	5723.52	5/2 ⁺			
		4200.3	1.7	4880.96	7/2 ⁽⁺⁾			
		4242.1		4839.10	(1/2 ⁺ ,3/2)			
		4311.4		4769.86	(5/2 ⁻)			
		4456.9		4624.3	3/2,5/2			
		4903.2	3.3	4177.9	3/2 ⁻			
		4907.7	3.3	4173.43	(5/2 ⁺)			
		5162.6	15	3918.47	3/2 ⁺	D(+Q)	-0.002 9	
		5918.1	10	3162.87	7/2 ⁻			
		6386.9	3.3	2694.00	3/2 ⁺			
		6435.2	1.7	2645.66	7/2 ⁺			
		7317.5	27	1763.18	5/2 ⁺	M1+E2	+0.11 2	B(M1)(W.u.)=1.16 +31-25; B(E2)(W.u.)=0.99 +50-38
		9080.2	100	0.0	3/2 ⁺	M1+E2	+0.089 9	B(M1)(W.u.)=2.26 +49-33; B(E2)(W.u.)=0.82 +27-19
9157.1	5/2 ⁻	3942.4	17	5214.5	(5/2 ⁺)			
		4386.9	16	4769.86	(5/2 ⁻)			
		4978.8	30	4177.9	3/2 ⁻			
		4983.3	30	4173.43	(5/2 ⁺)			
		5993.7	22	3162.87	7/2 ⁻	M1+E2	+0.10 2	
		6153.9	20	3002.64	5/2 ⁺			
		6462.5	3.0	2694.00	3/2 ⁺			
		6510.8	6.7	2645.66	7/2 ⁺			
		7393.1	17	1763.18	5/2 ⁺			
		9155.8	100	0.0	3/2 ⁺	(E1)		Mult.: D+Q with $\delta=+0.09$ <i>I</i> from $\gamma(\theta)$ in (p, γ); $\Delta\pi$ =yes from level scheme; M2 component is less likely.
10180.6	19/2 ⁻	1336.3 5	100.0 34	8844.3	17/2 ⁺	(E1)		B(E1)(W.u.)=0.00235 +21-18 <i>I_γ</i> : weighted average of 100 17 from (²⁸ Si, $\alpha\gamma$) and 100.0 34 from (¹⁶ O, $\alpha\gamma$). <i>E_γ</i> ,Mult.: from (¹⁶ O, $\alpha\gamma$), with Mult=D from $\gamma\gamma$ (ADO) with $\Delta J=1$; $\Delta\pi$ =(yes) from level scheme.
		1693.5 5	84 9	8487.0	15/2 ⁻	E2		B(E2)(W.u.)=44.4 46 <i>E_γ</i> , <i>I_γ</i> ,Mult.: from (¹⁶ O, $\alpha\gamma$), with Mult=Q, $\Delta J=2$ from $\gamma\gamma$ (ADO) and M2 ruled out by RUL.
		1861.3 5	83 6	8319.3	15/2 ⁻	E2		B(E2)(W.u.)=27.4 +26-24 <i>E_γ</i> , <i>I_γ</i> ,Mult.: from (¹⁶ O, $\alpha\gamma$), with Mult=Q, $\Delta J=2$ from $\gamma\gamma$ (ADO) and M2 ruled out by RUL.
10221.9	(17/2 ⁻)	1377.8 10	100	8844.3	17/2 ⁺			

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. [#]	Comments
10858.5	19/2 ⁺	2014.7 9	100 5	8844.3	17/2 ⁺	(M1)	$E_\gamma, I_\gamma, \text{Mult.}$: from (¹⁶ O, $\alpha\gamma$), with Mult=D from $\gamma\gamma(\text{ADO})$ with $\Delta J=1$; $\Delta\pi=\text{no}$ from level scheme.
11458.7	21/2 ⁺	2069.8 10 2614.5 5	32.5 22 100	8788.3 (15/2 ⁺) 8844.3 17/2 ⁺	E2 E2		$E_\gamma, I_\gamma, \text{Mult.}$: from (¹⁶ O, $\alpha\gamma$) with Mult=Q, $\Delta J=2$ from $\gamma\gamma(\text{ADO})$; M2 ruled out by RUL. B(E2)(W.u.)=15.9 8
12571.7	23/2 ⁻	1113.3 5	24.4 18	11458.7 21/2 ⁺	(E1)		$E_\gamma, \text{Mult.}$: from (¹⁶ O, $\alpha\gamma$), with Mult=Q, $\Delta J=2$ from $\gamma\gamma(\text{ADO})$; M2 ruled out by RUL. B(E1)(W.u.)=0.00272 +41-35
		2390.8 5	100 4	10180.6 19/2 ⁻	E2		$E_\gamma, I_\gamma, \text{Mult.}$: from (¹⁶ O, $\alpha\gamma$), with Mult=D from $\gamma\gamma(\text{ADO})$ with $\Delta J=1$; $\Delta\pi=\text{yes}$ from level scheme. B(E2)(W.u.)=26.0 +36-29
16306.4	(27/2 ⁻)	3734.5		12571.7 23/2 ⁻			$E_\gamma, I_\gamma, \text{Mult.}$: from (¹⁶ O, $\alpha\gamma$), with Mult=Q, $\Delta J=2$ from $\gamma\gamma(\text{ADO})$; M2 ruled out by RUL.

[†] Sources of values with uncertainties are listed under comments and those without uncertainties are deduced from level-energy differences, unless otherwise noted.

[‡] From (p, γ), unless otherwise noted.

[#] From $\gamma(\theta)$ and/or $\gamma\gamma(\theta)$ in (p, γ) for γ rays from low-spin levels ($J < 9/2$) levels and from $\gamma\gamma(\theta)(\text{DCO})$ and $\gamma(\text{lin pol})$ in (²⁸Si, $\alpha\gamma$) for γ rays from high-spin ($J \geq 9/2$) levels, unless otherwise noted. Magnetic/electric characters are determined based $\gamma(\text{lin pol})$ or RUL with measured $T_{1/2}$ where available, unless otherwise noted. For primary transitions from proton resonances where widths or resonance strengths are given in (p, γ), D+Q with definite $\delta(Q/D) > 0$ from $\gamma(\theta)$ is firmly assigned as M1+E2, since for those resonances the proton width Γ_p is dominant over Γ_γ and thus the total width Γ is much greater than Γ_γ which is roughly in the same scale as the resonance strength, typically a few eV in the ³⁴S(p, γ) dataset.

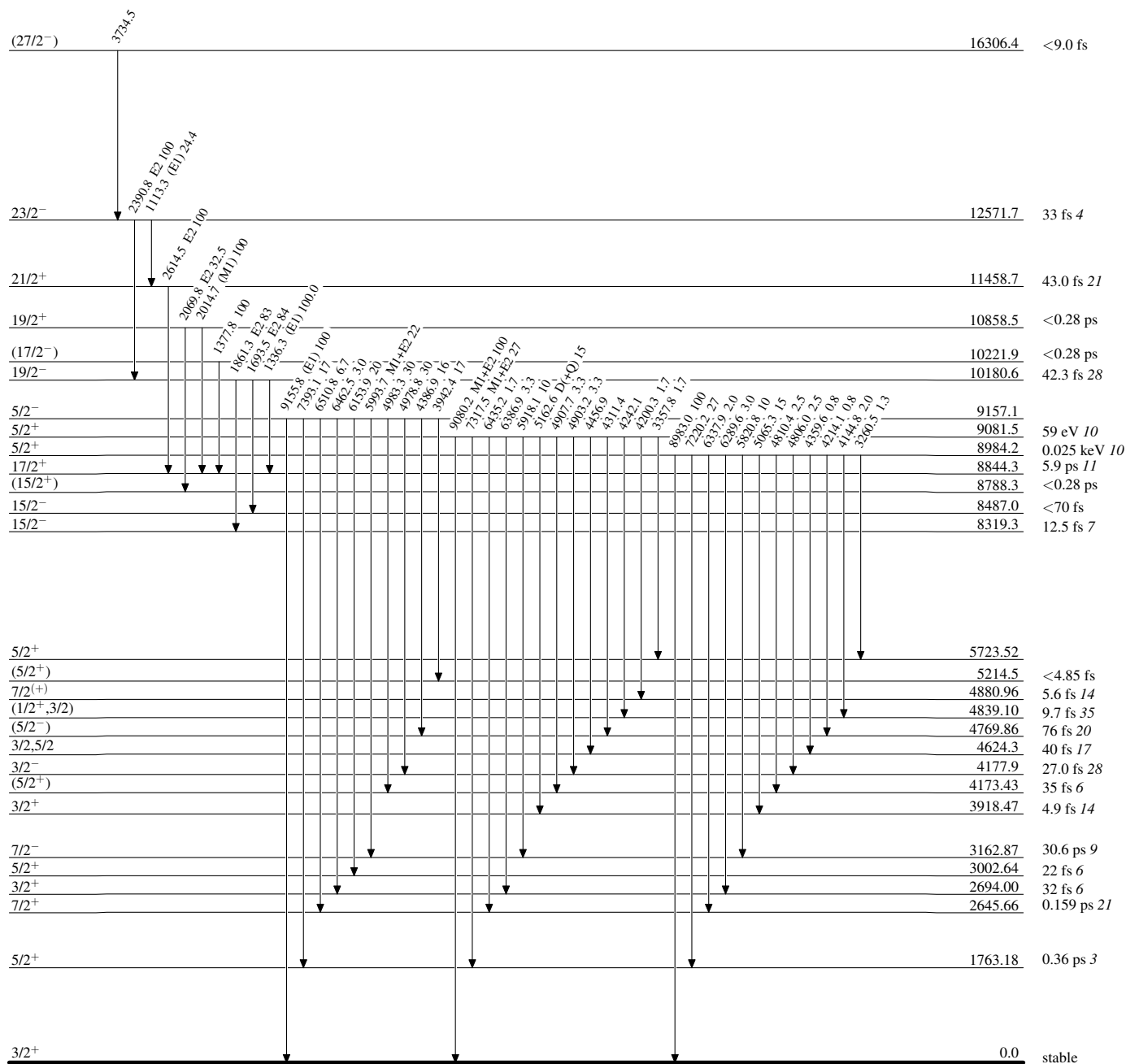
@ Primary transitions probably from the unresolved 7501+7503 doublet in 1976Me12 in (p, γ).

& Primary transitions probably from the unresolved 7837+7839 doublet in 1976Me12 in (p, γ).

^a Placement of transition in the level scheme is uncertain.

Adopted Levels, GammasLevel Scheme

Intensities: Relative photon branching from each level

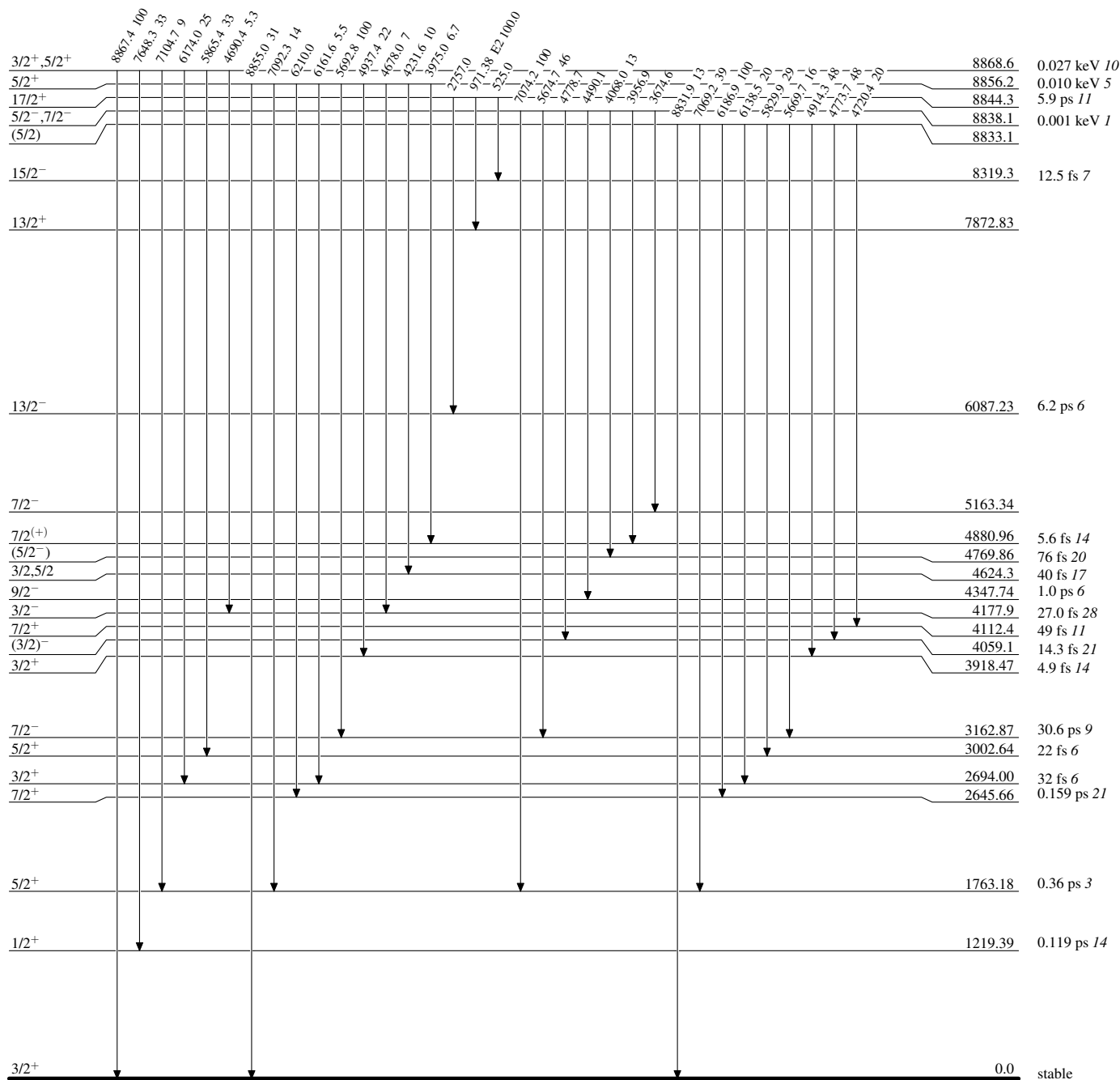
 $^{35}_{17}\text{Cl}_{18}$

Intensities: Relative photon branching from each level

Adopted Levels, Gammas

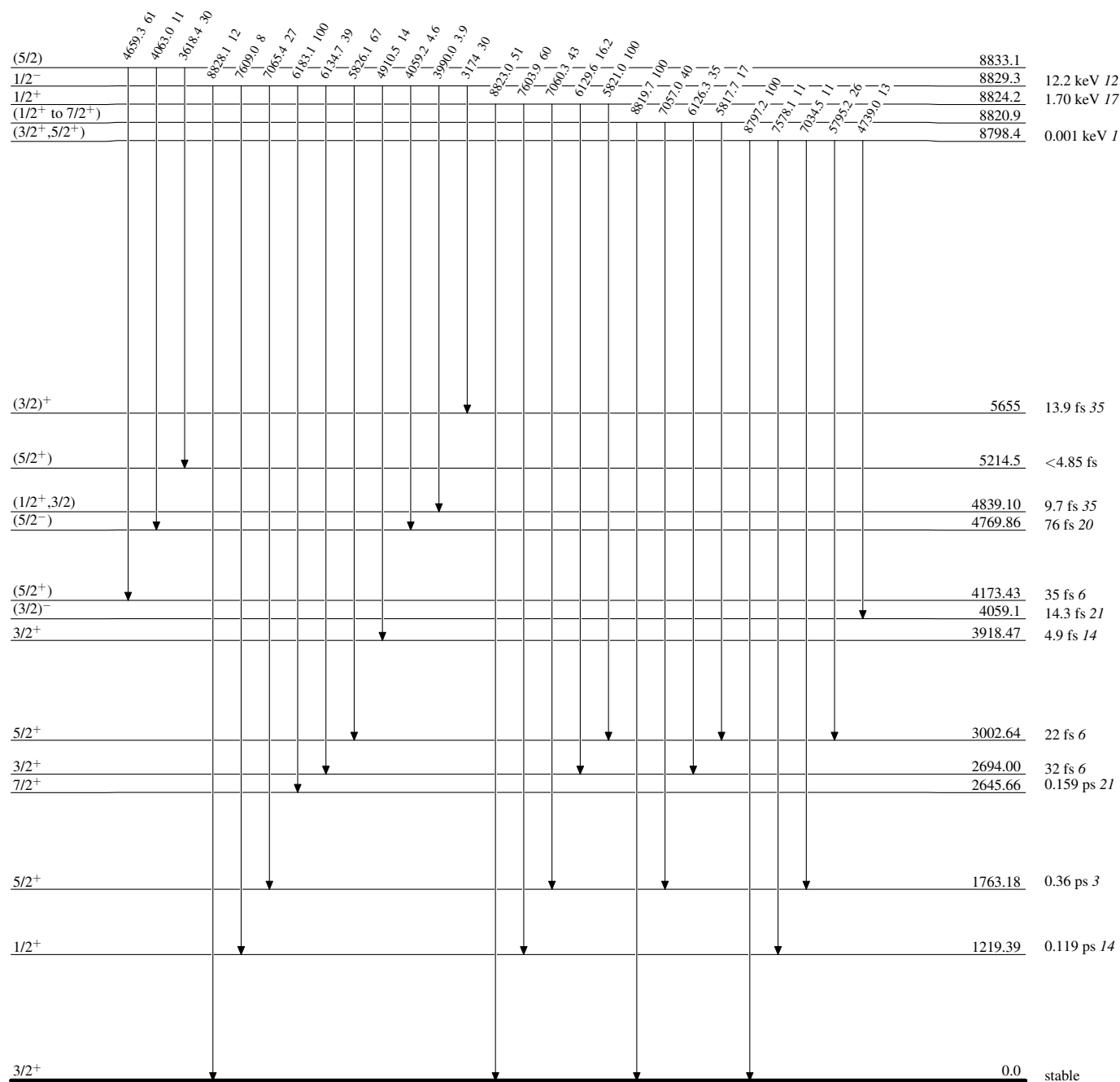
Level Scheme (continued)

Intensities: Relative photon branching from each level



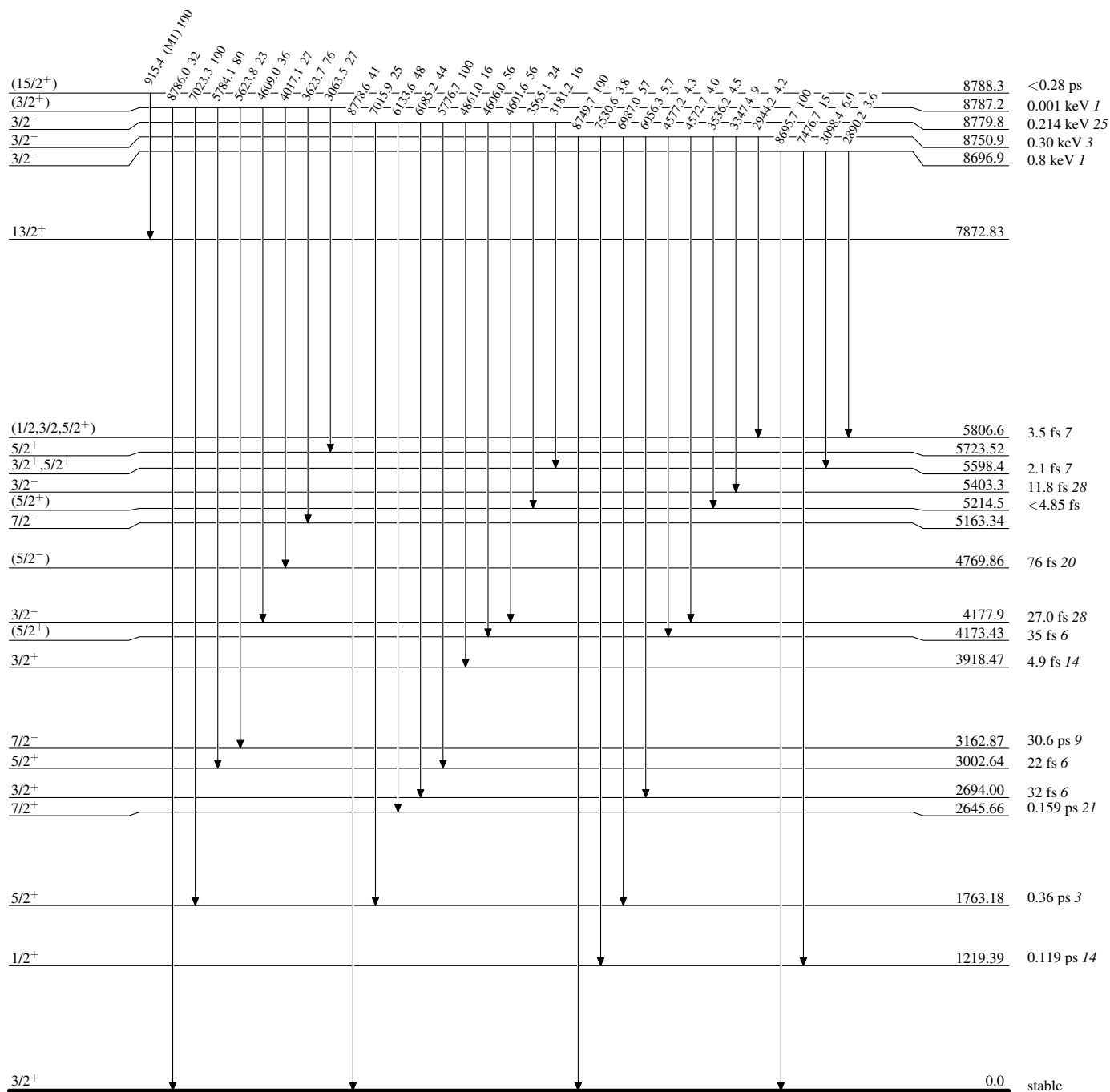
Adopted Levels, Gammas**Level Scheme (continued)**

Intensities: Relative photon branching from each level



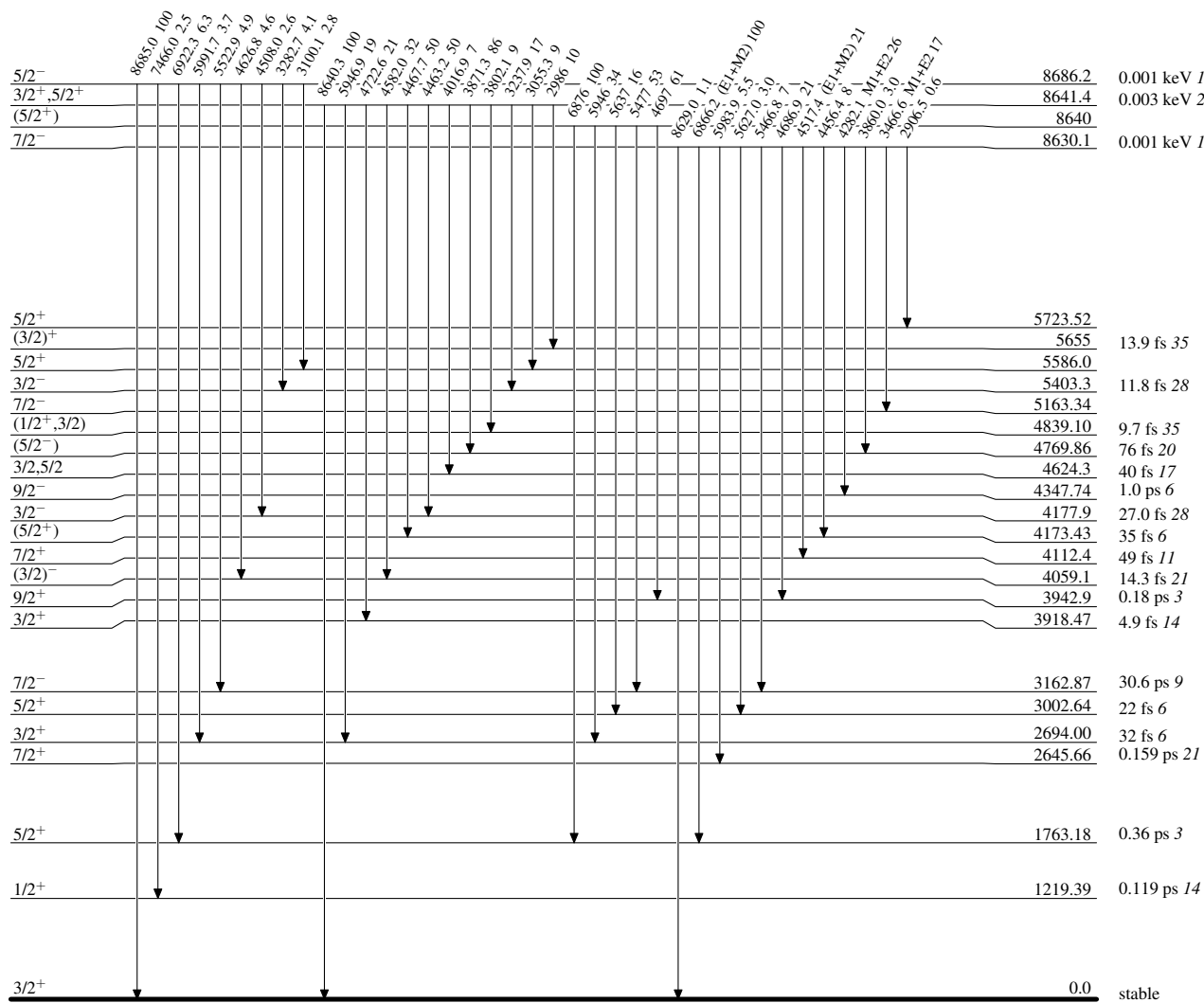
Adopted Levels, Gammas**Level Scheme (continued)**

Intensities: Relative photon branching from each level



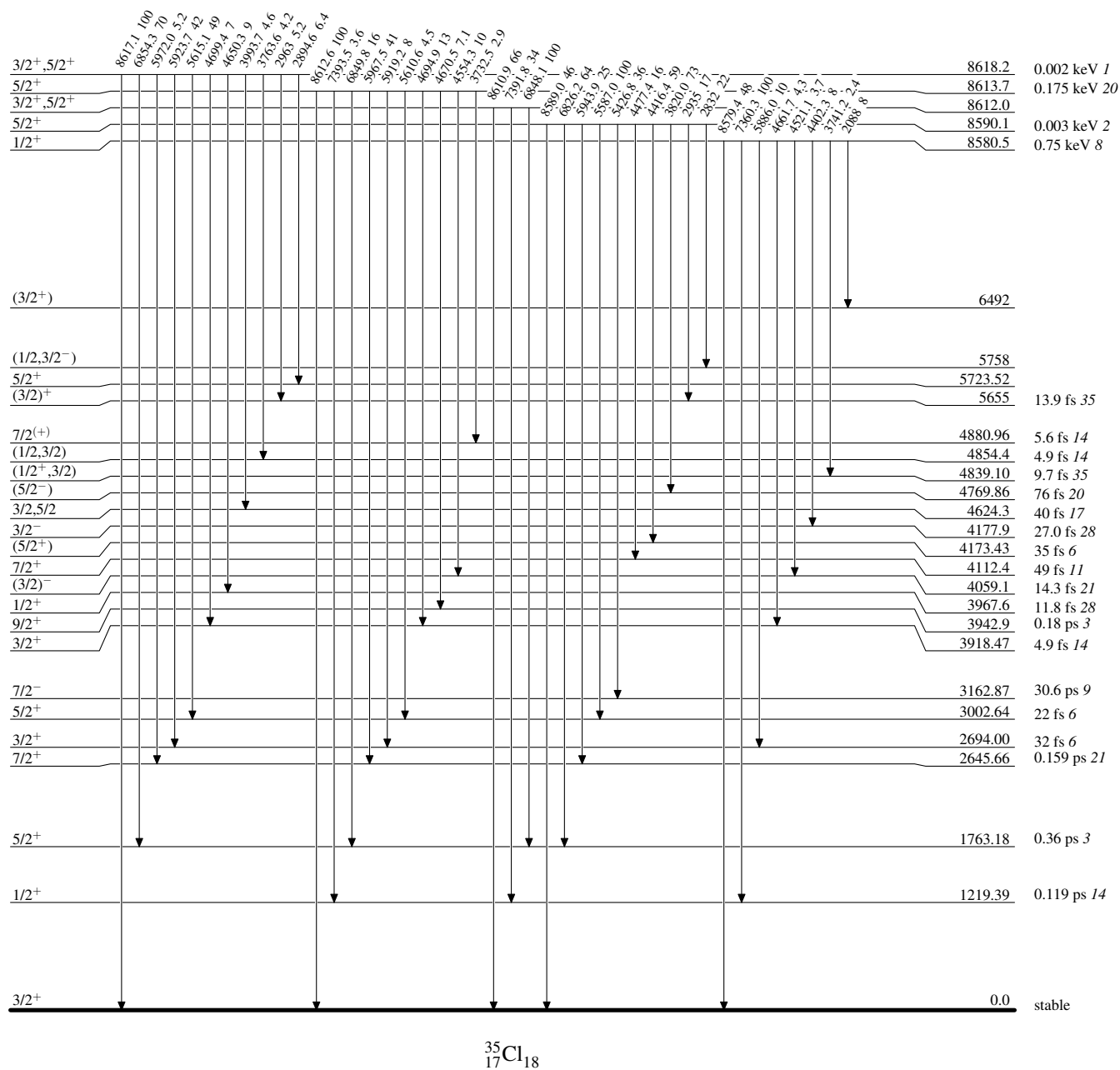
Adopted Levels, Gammas**Level Scheme (continued)**

Intensities: Relative photon branching from each level

 $^{35}_{17}\text{Cl}_{18}$

Adopted Levels, Gammas**Level Scheme (continued)**

Intensities: Relative photon branching from each level

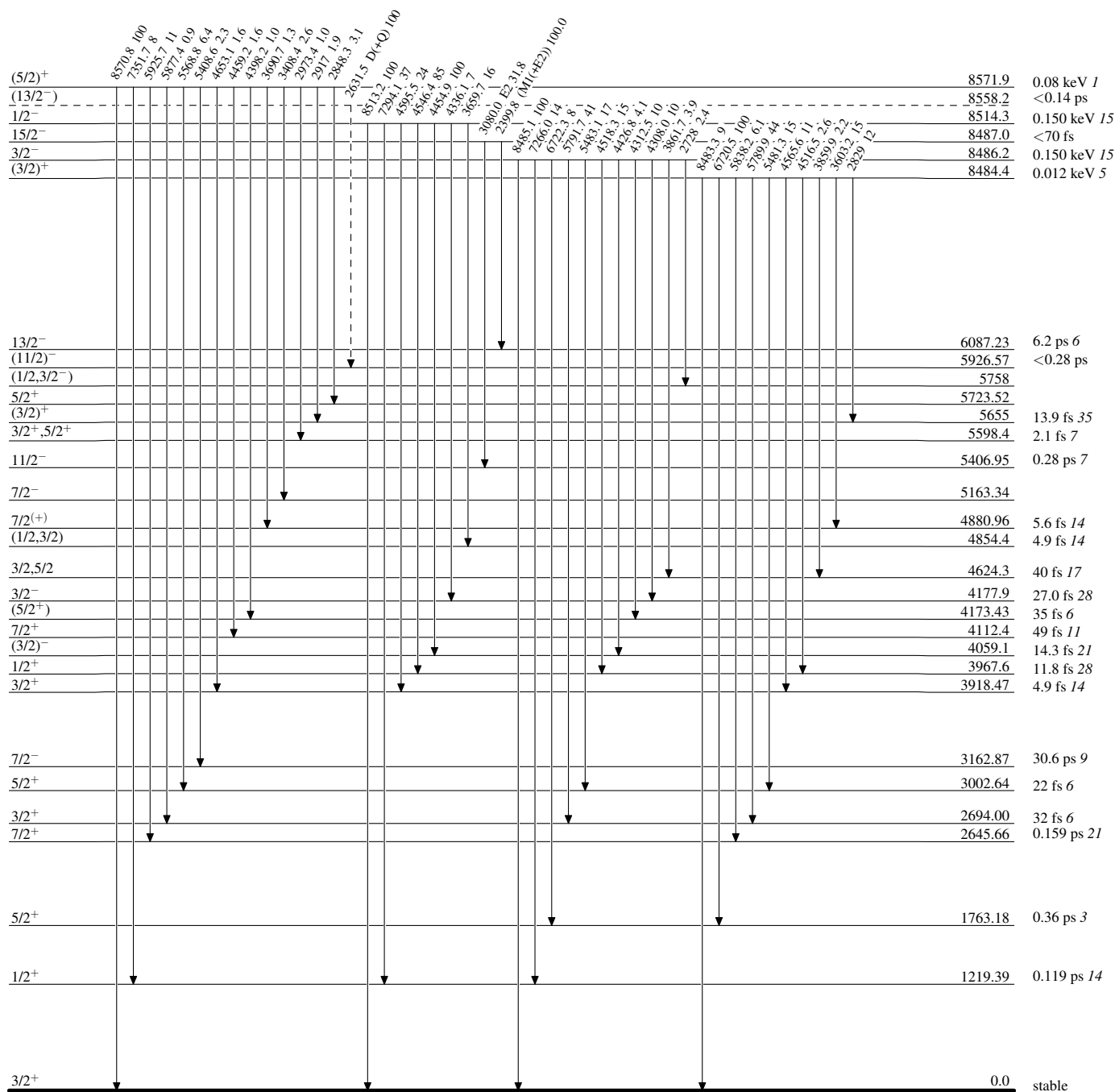


Adopted Levels, Gammas

Legend

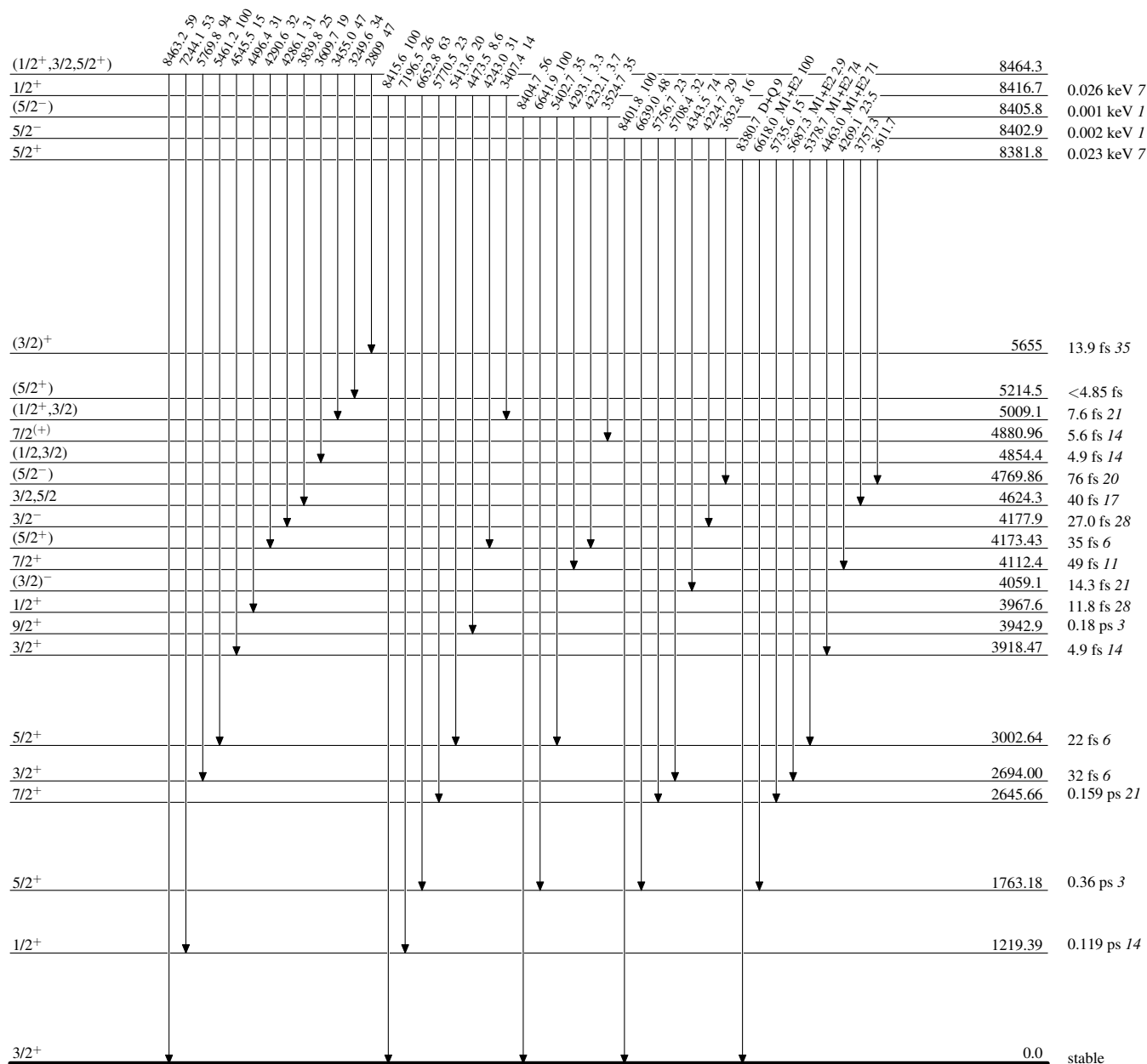
Level Scheme (continued)

Intensities: Relative photon branching from each level

-----► γ Decay (Uncertain)

Adopted Levels, GammasLevel Scheme (continued)

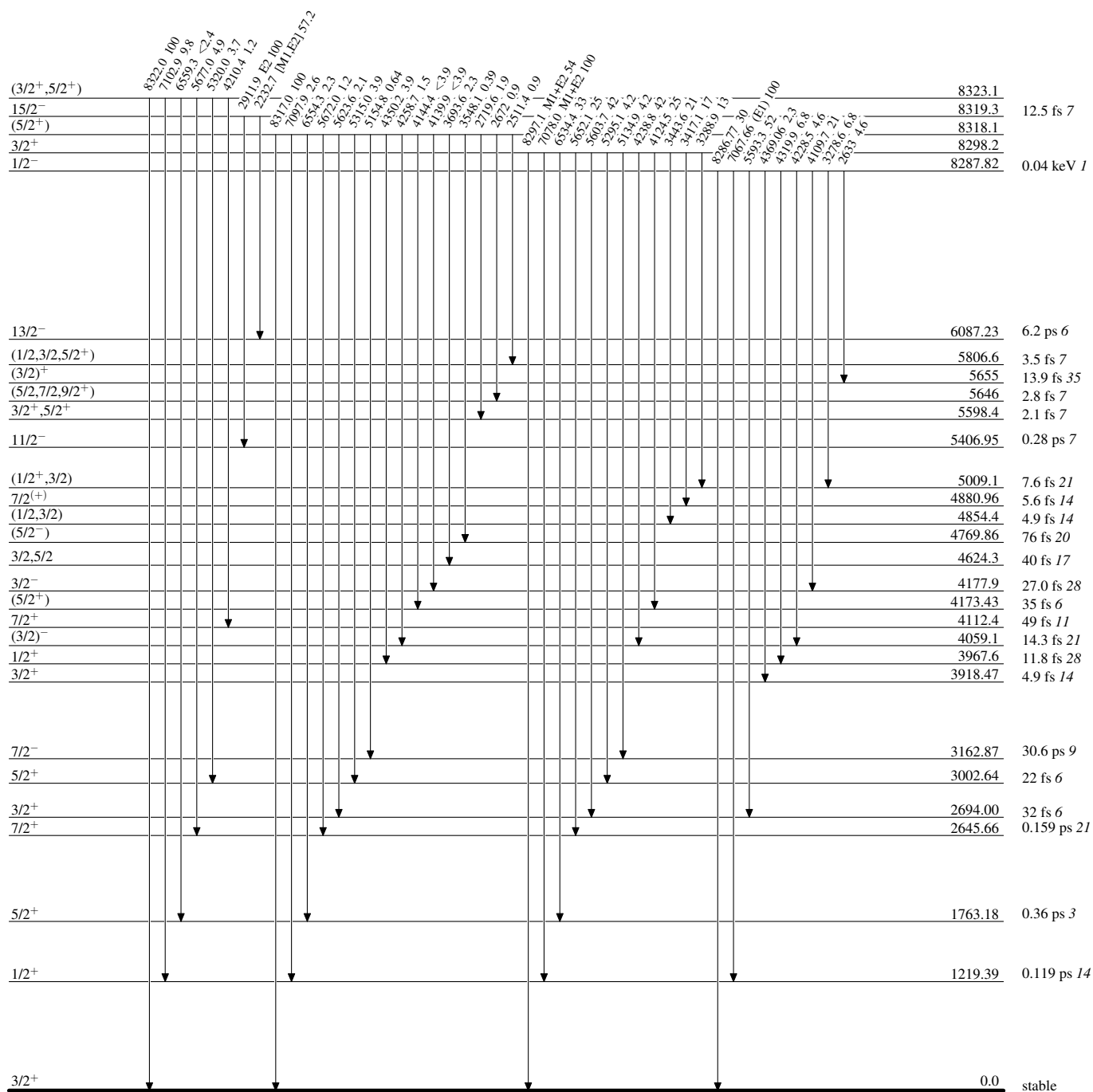
Intensities: Relative photon branching from each level

 $^{35}_{17}\text{Cl}_{18}$

Adopted Levels, Gammas

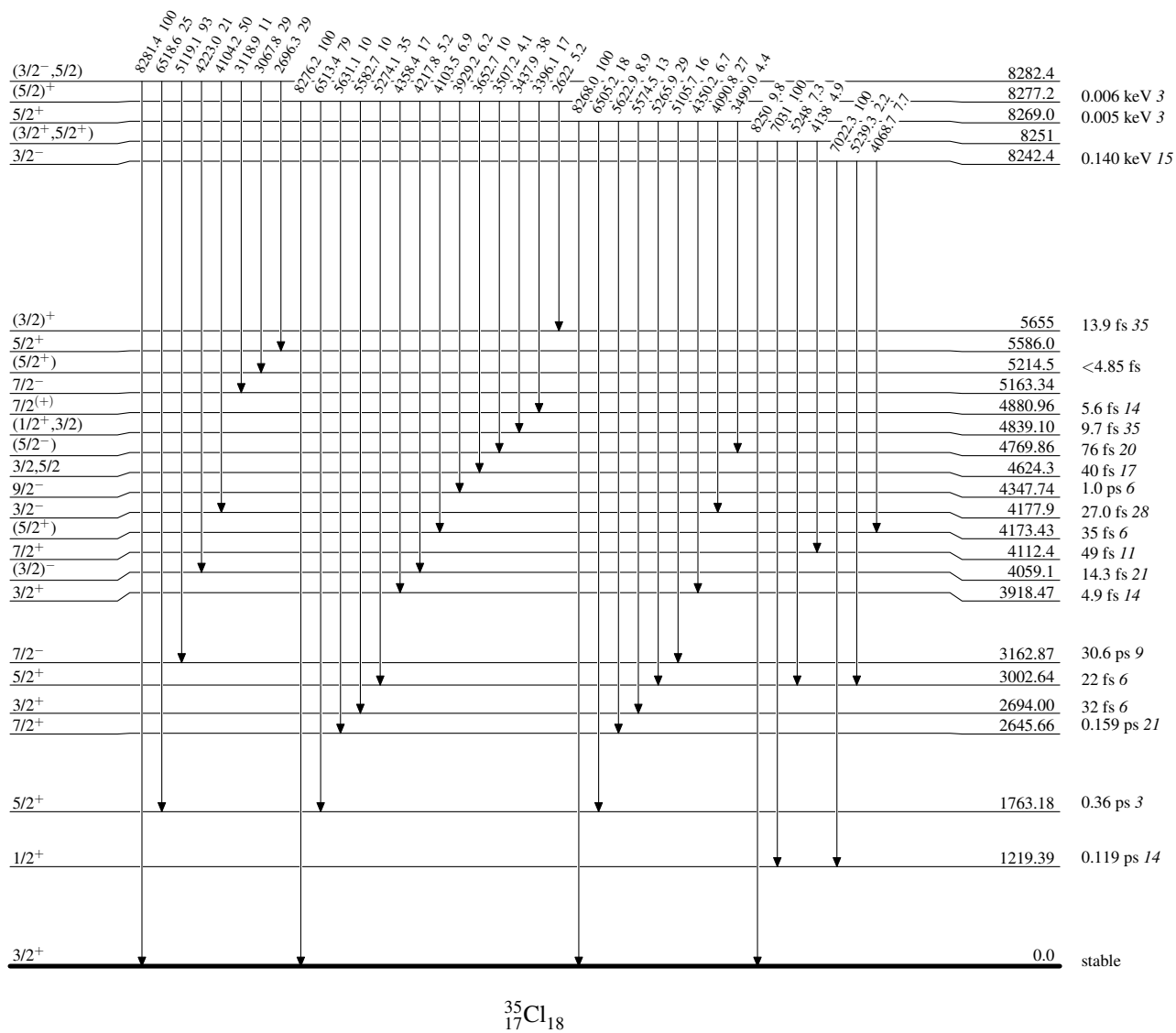
Level Scheme (continued)

Intensities: Relative photon branching from each level



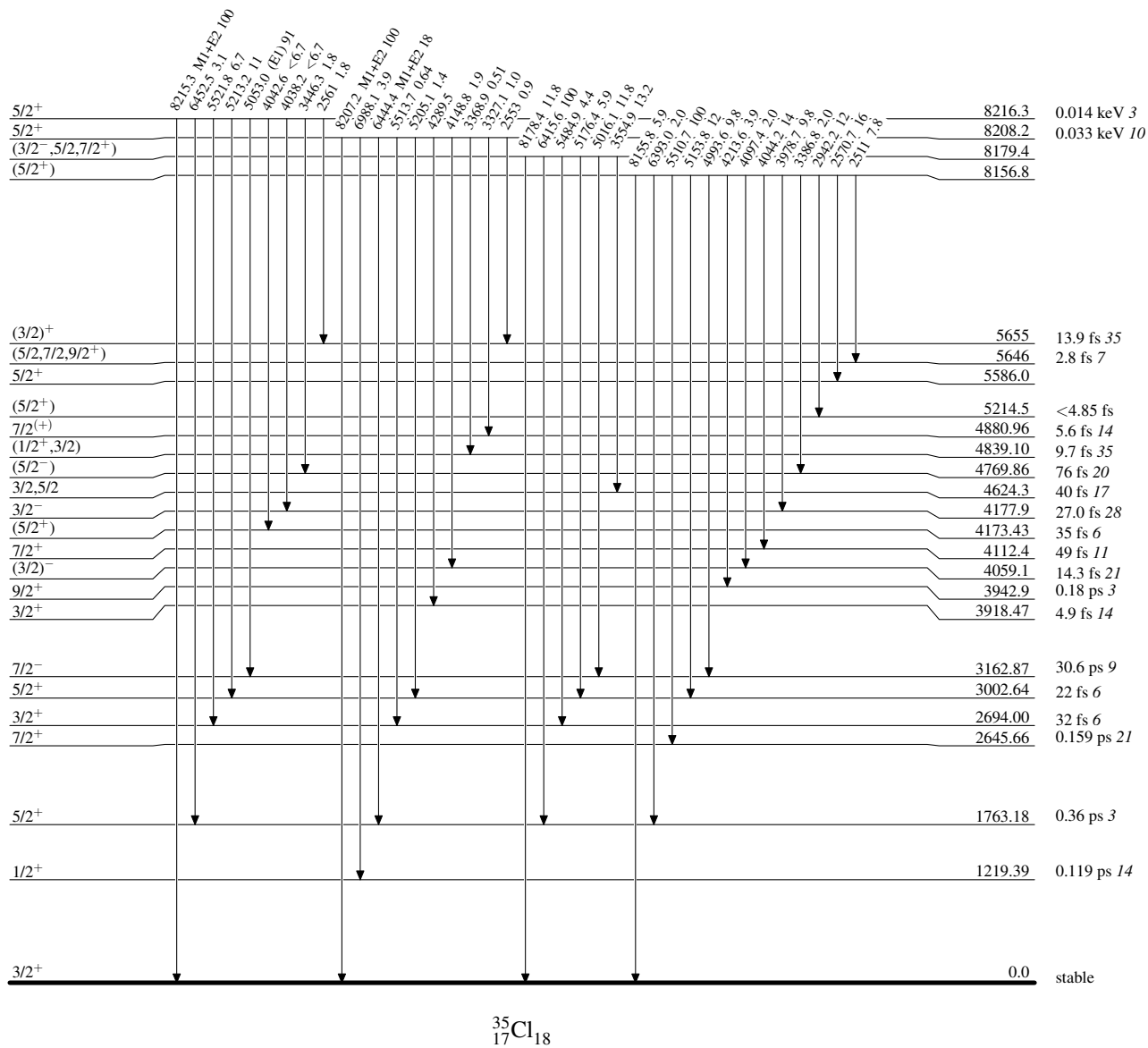
Adopted Levels, GammasLevel Scheme (continued)

Intensities: Relative photon branching from each level

 $^{35}_{17}\text{Cl}_{18}$

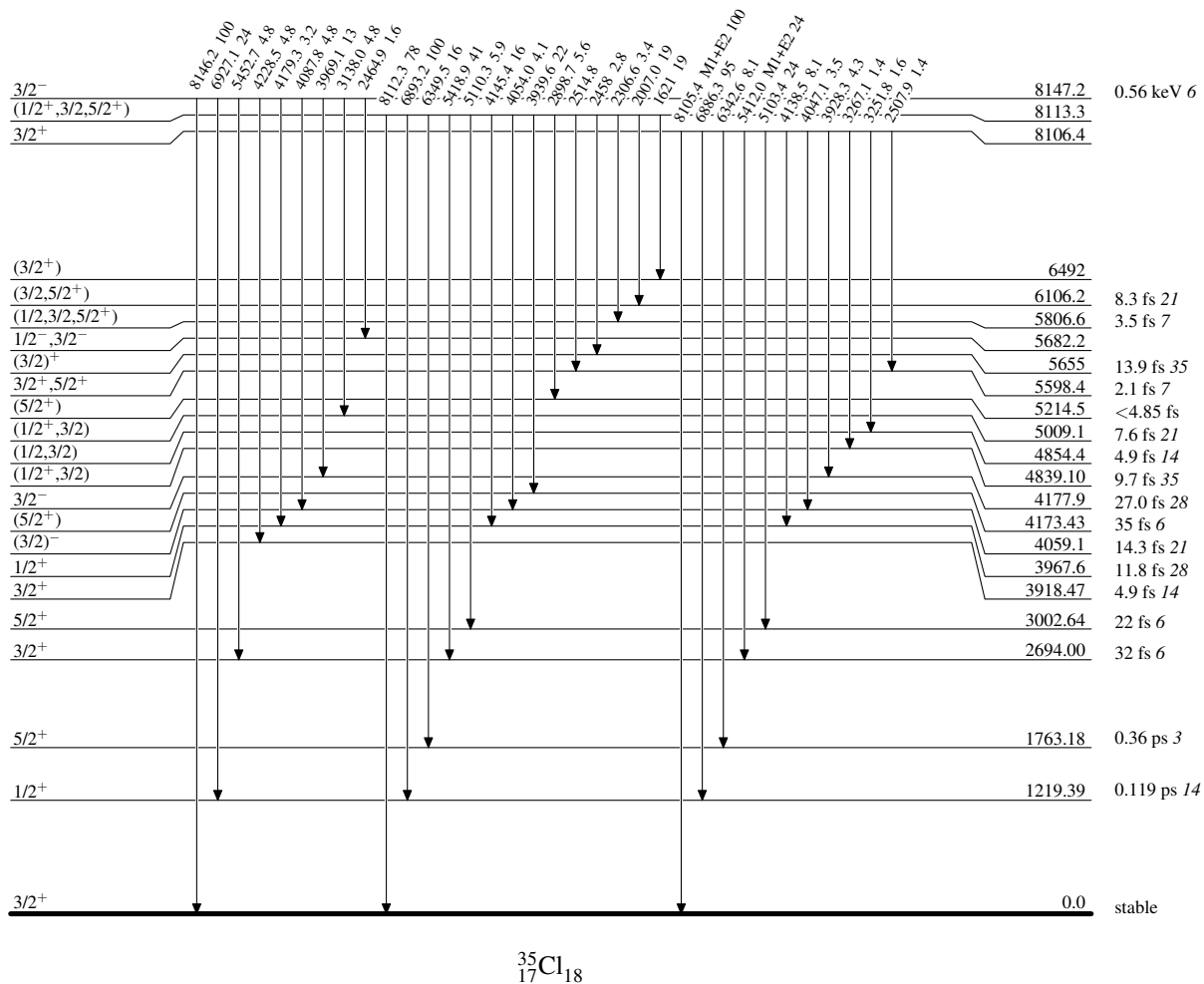
Adopted Levels, GammasLevel Scheme (continued)

Intensities: Relative photon branching from each level

 $^{35}_{17}\text{Cl}_{18}$

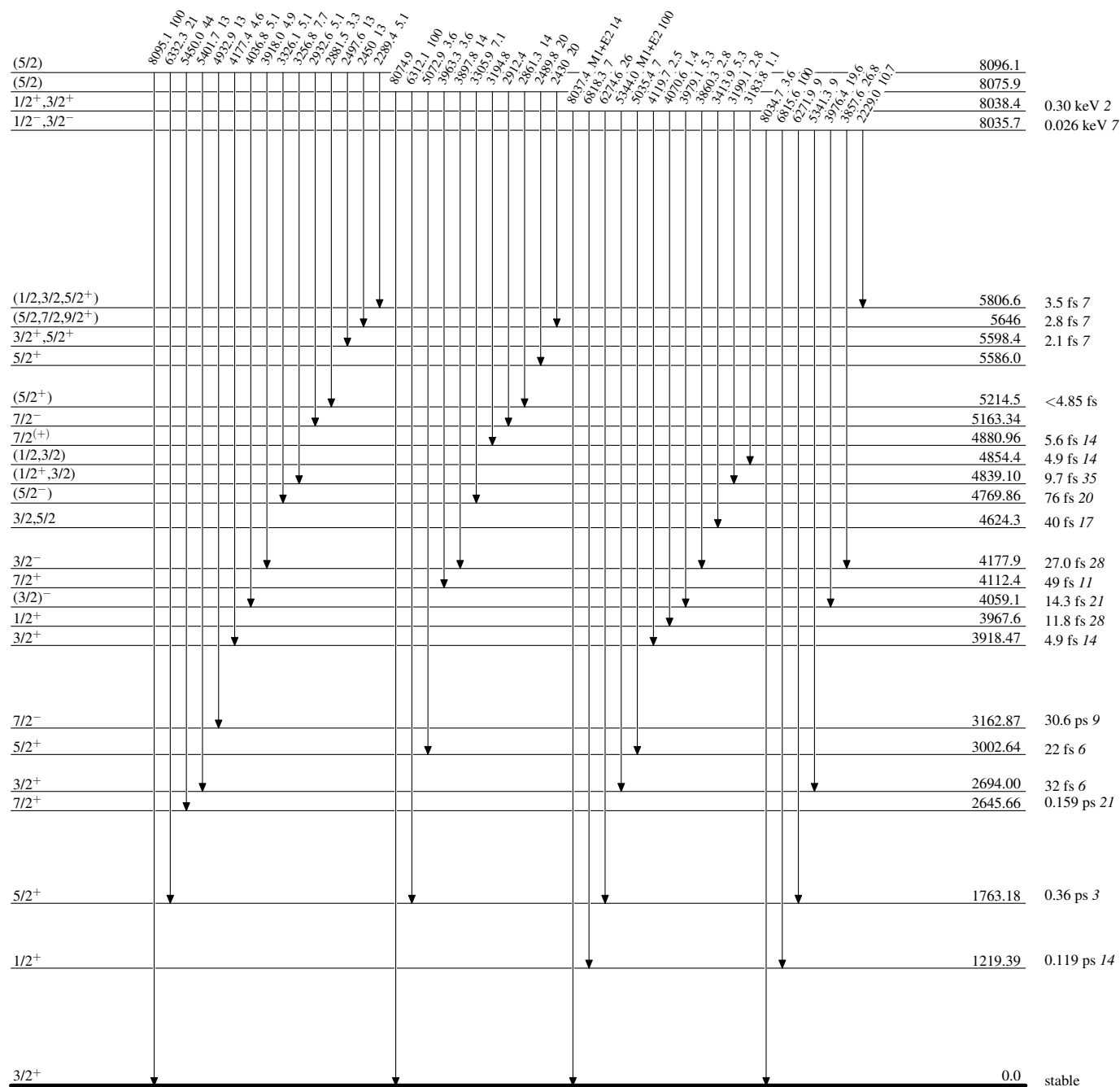
Adopted Levels, Gammas**Level Scheme (continued)**

Intensities: Relative photon branching from each level

 $^{35}_{17}\text{Cl}_{18}$

Adopted Levels, Gammas**Level Scheme (continued)**

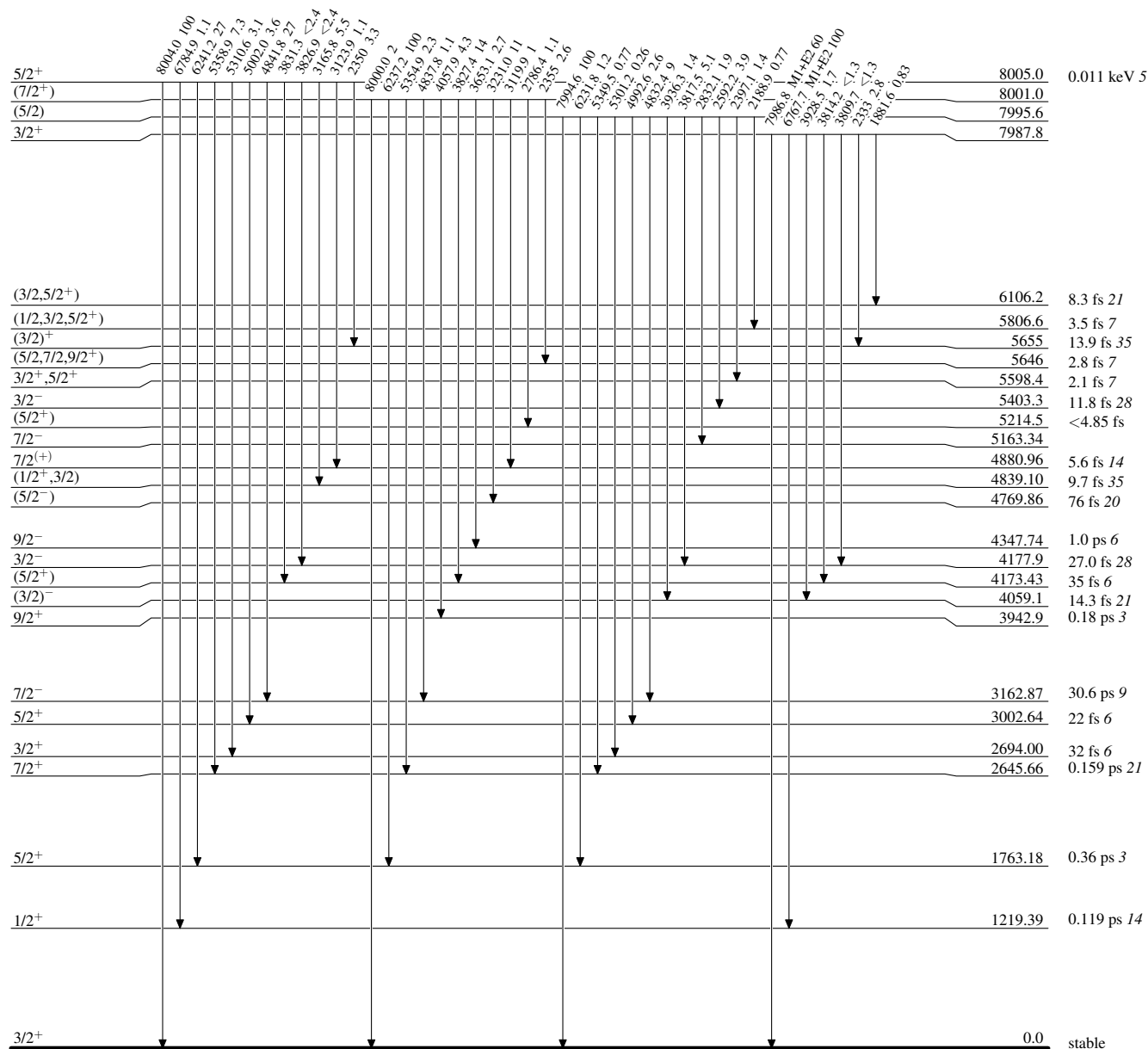
Intensities: Relative photon branching from each level



Adopted Levels, Gammas

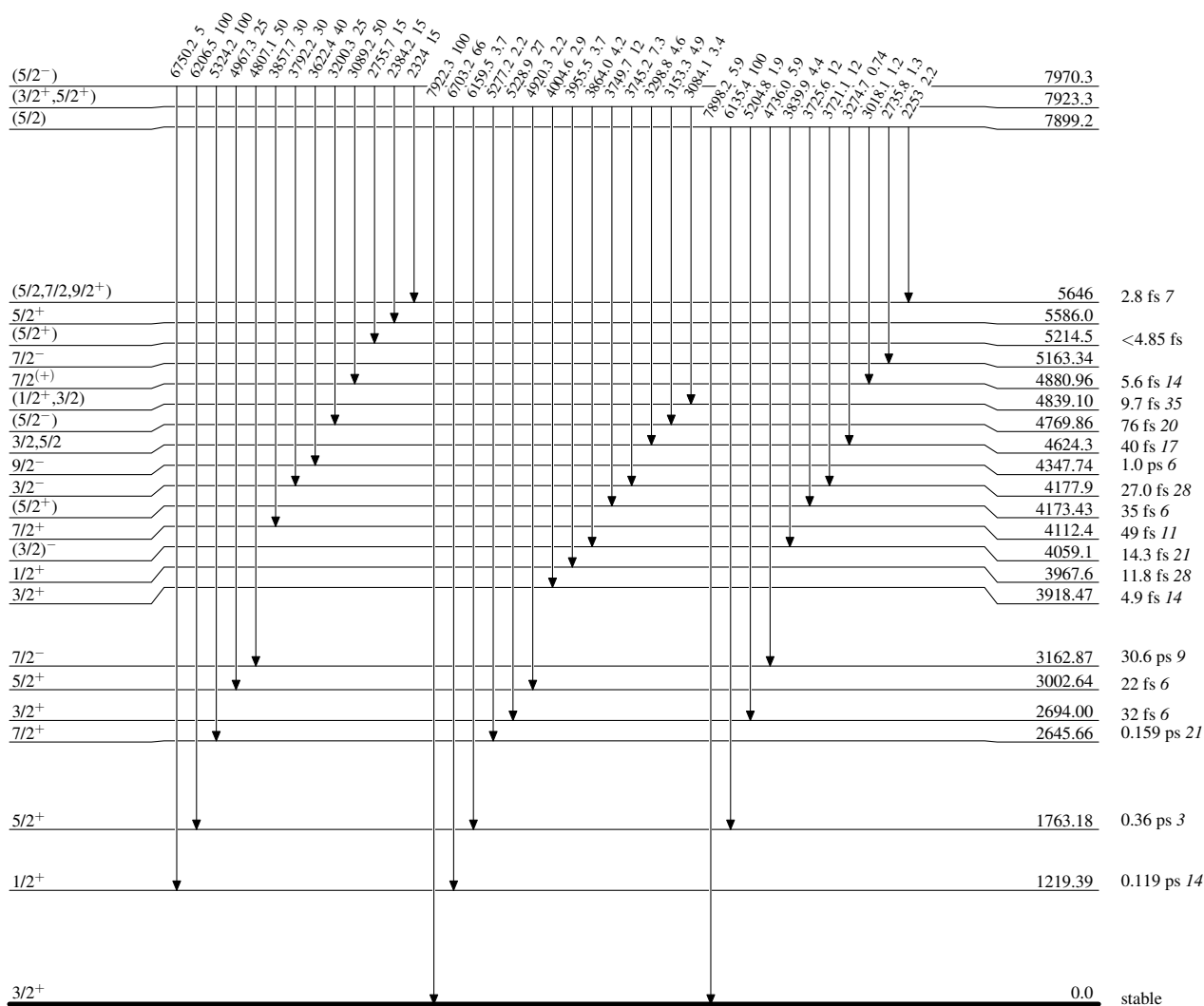
Level Scheme (continued)

Intensities: Relative photon branching from each level

 $^{35}_{17}\text{Cl}_{18}$

Adopted Levels, GammasLevel Scheme (continued)

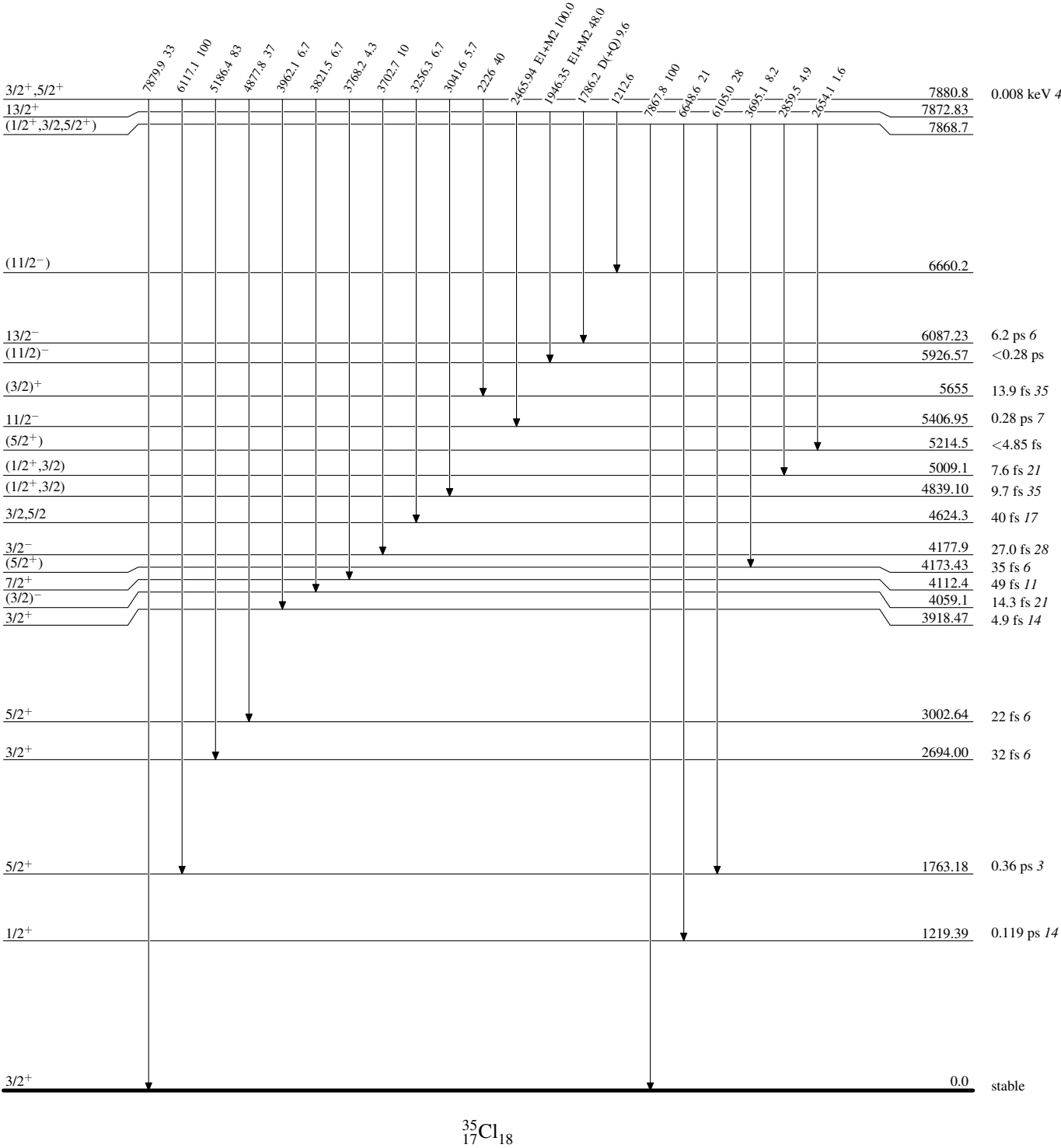
Intensities: Relative photon branching from each level

 $^{35}_{17}\text{Cl}_{18}$

Adopted Levels, Gammas

Level Scheme (continued)

Intensities: Relative photon branching from each level

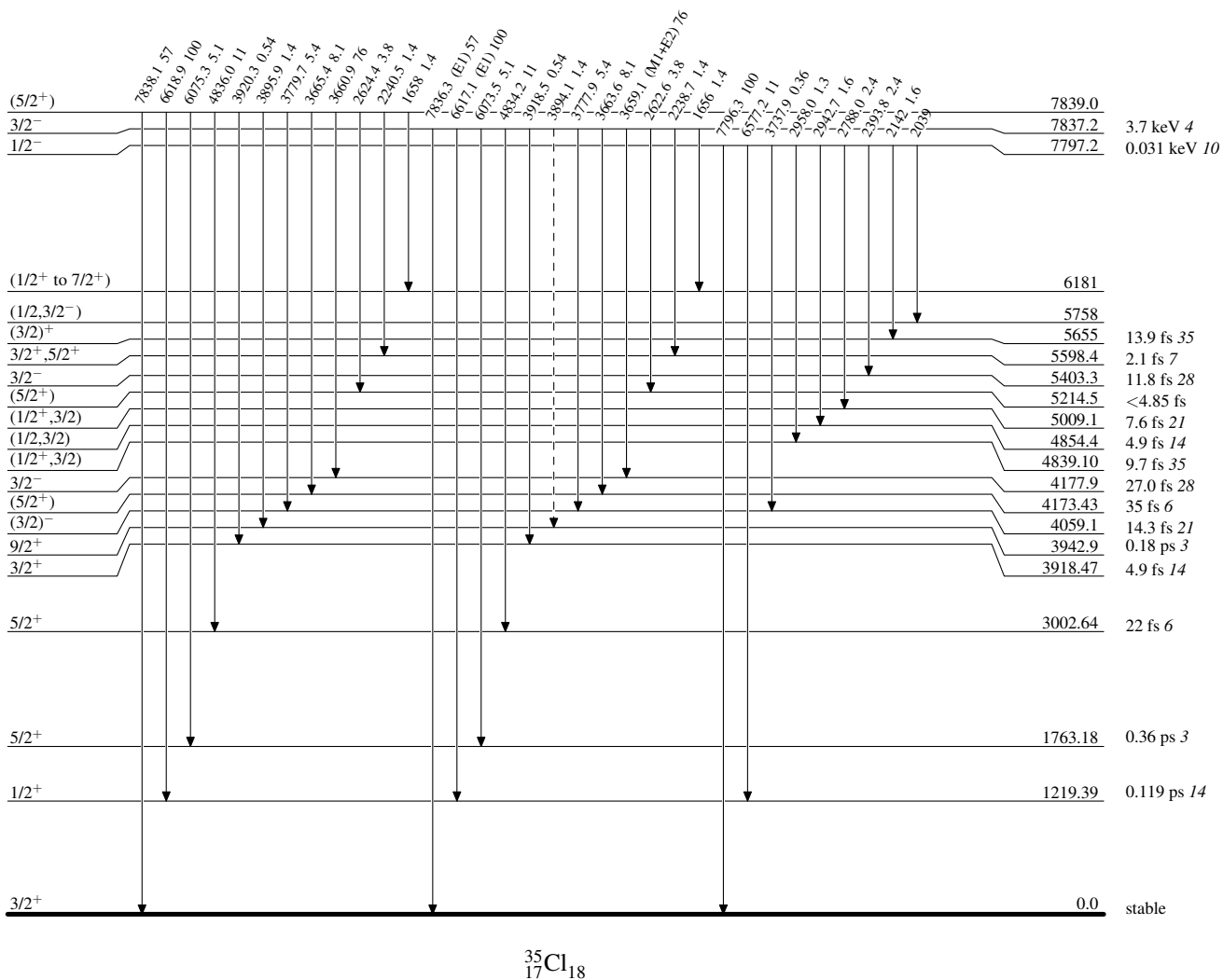


Adopted Levels, Gammas

Legend

Level Scheme (continued)

Intensities: Relative photon branching from each level

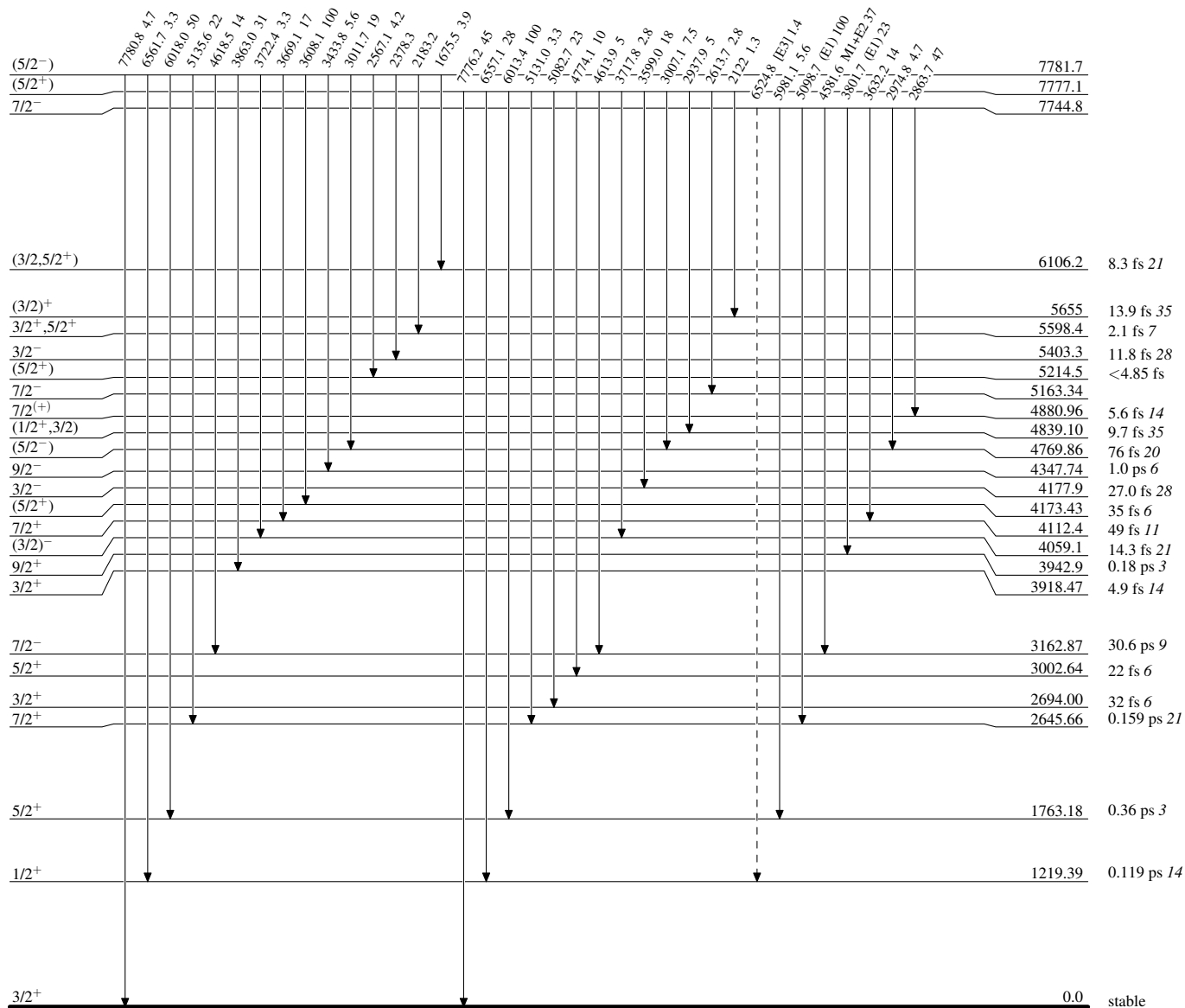
-----► γ Decay (Uncertain) $^{35}_{17}\text{Cl}_{18}$

Adopted Levels, Gammas

Legend

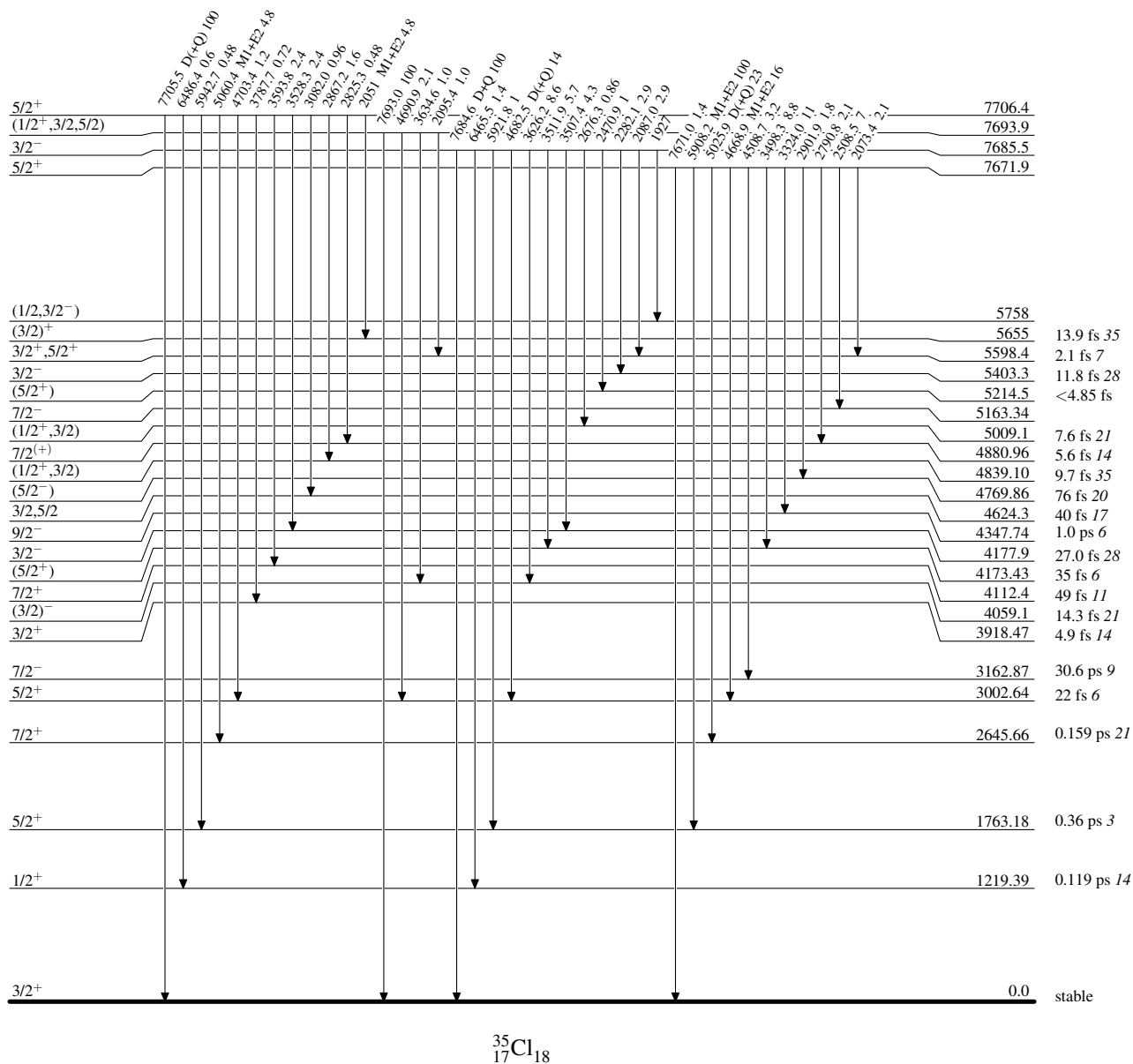
Level Scheme (continued)

Intensities: Relative photon branching from each level

-----► γ Decay (Uncertain) $^{35}_{17}\text{Cl}_{18}$

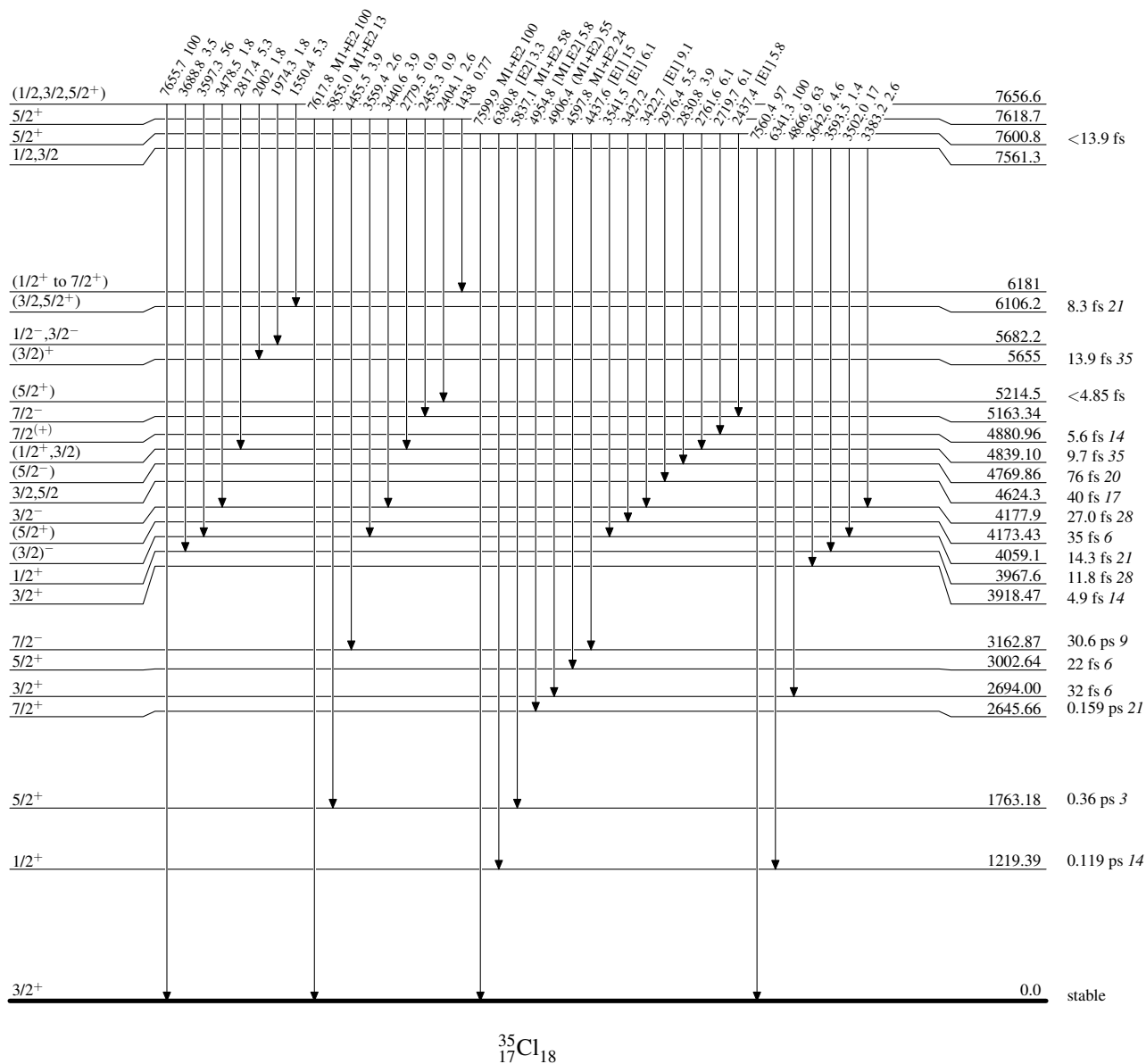
Adopted Levels, GammasLevel Scheme (continued)

Intensities: Relative photon branching from each level



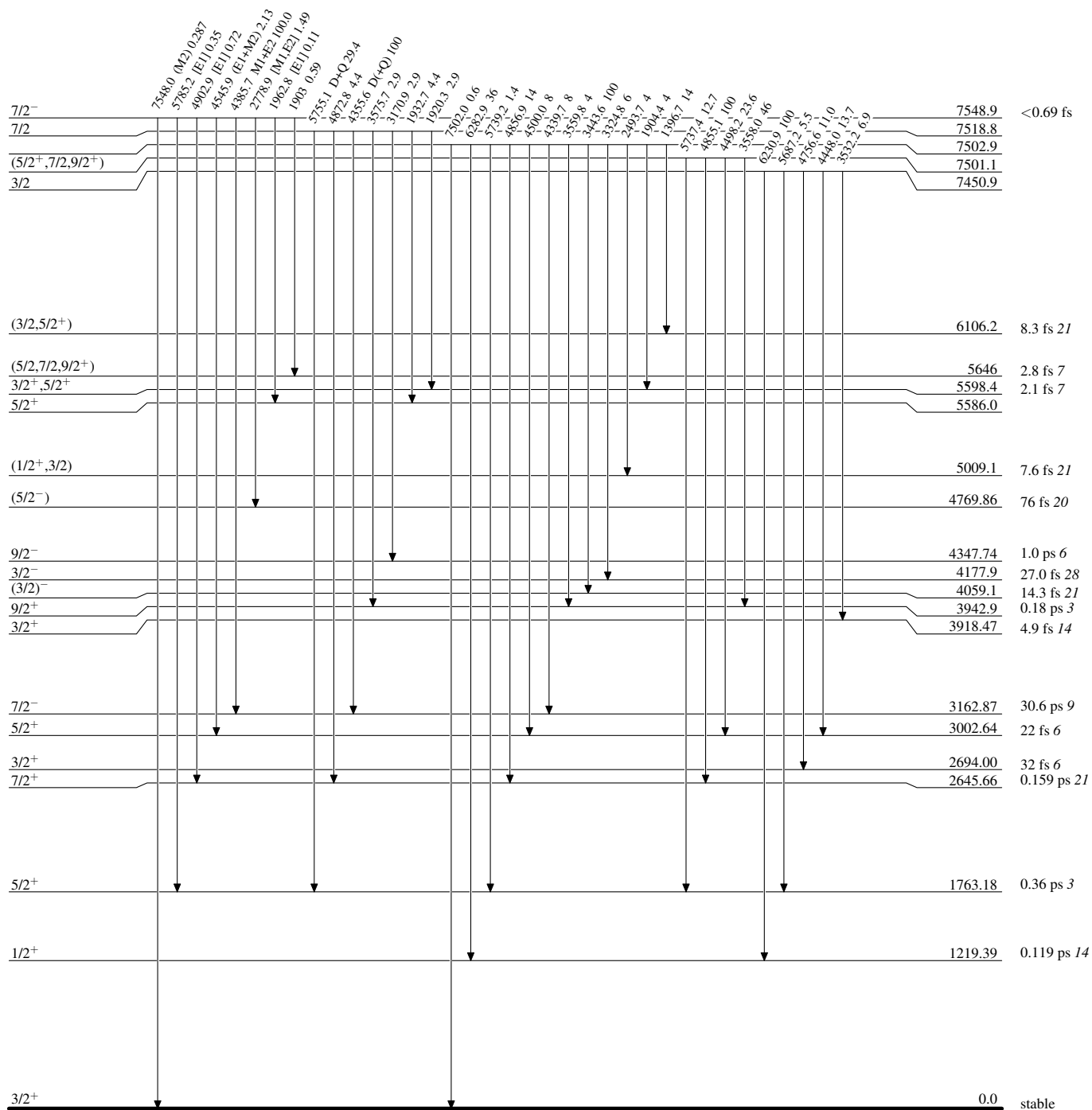
Adopted Levels, Gammas**Level Scheme (continued)**

Intensities: Relative photon branching from each level

 $^{35}_{17}\text{Cl}_{18}$

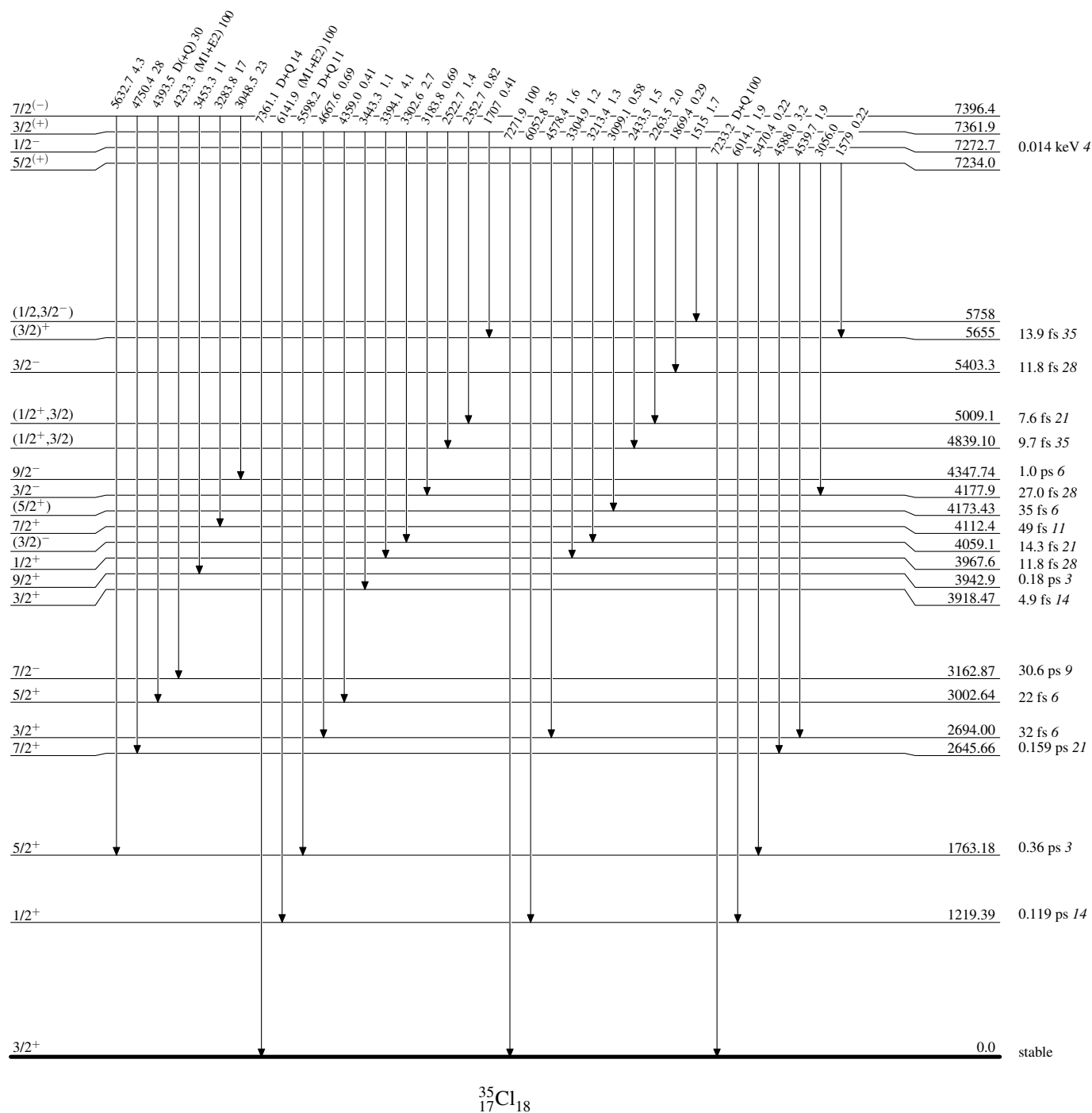
Adopted Levels, Gammas**Level Scheme (continued)**

Intensities: Relative photon branching from each level



Adopted Levels, Gammas**Level Scheme (continued)**

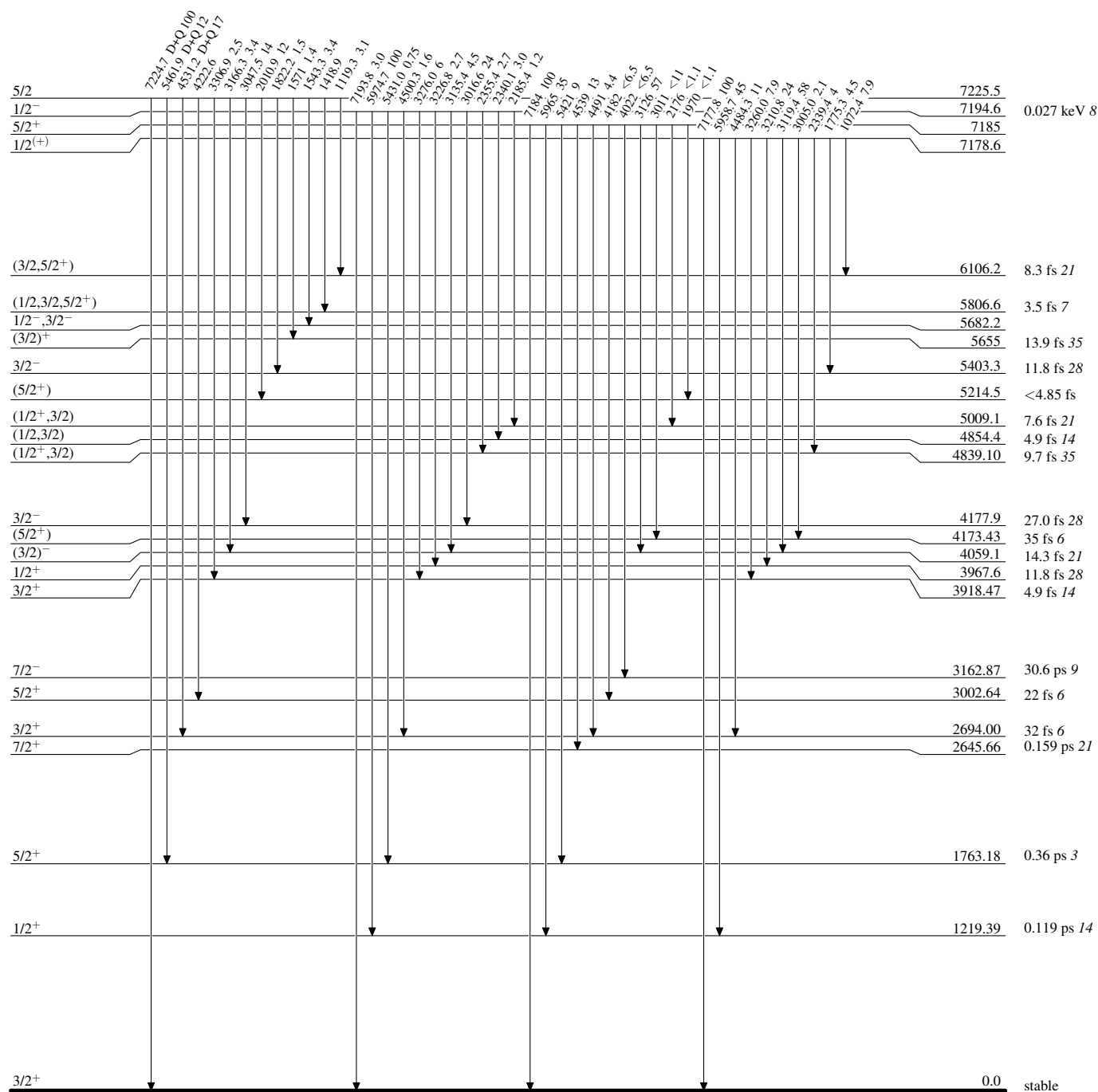
Intensities: Relative photon branching from each level



Adopted Levels, Gammas

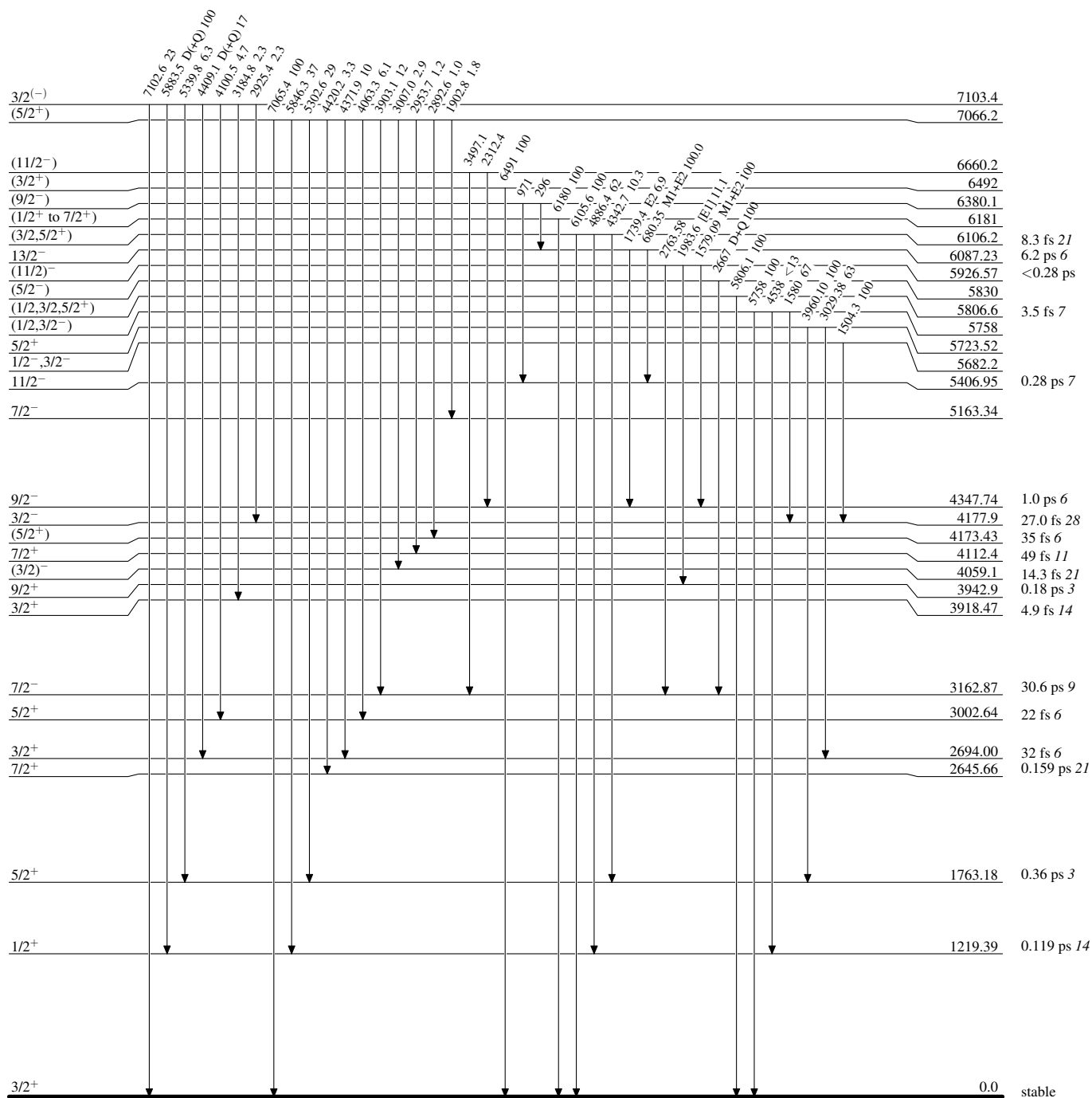
Level Scheme (continued)

Intensities: Relative photon branching from each level



Adopted Levels, Gammas**Level Scheme (continued)**

Intensities: Relative photon branching from each level

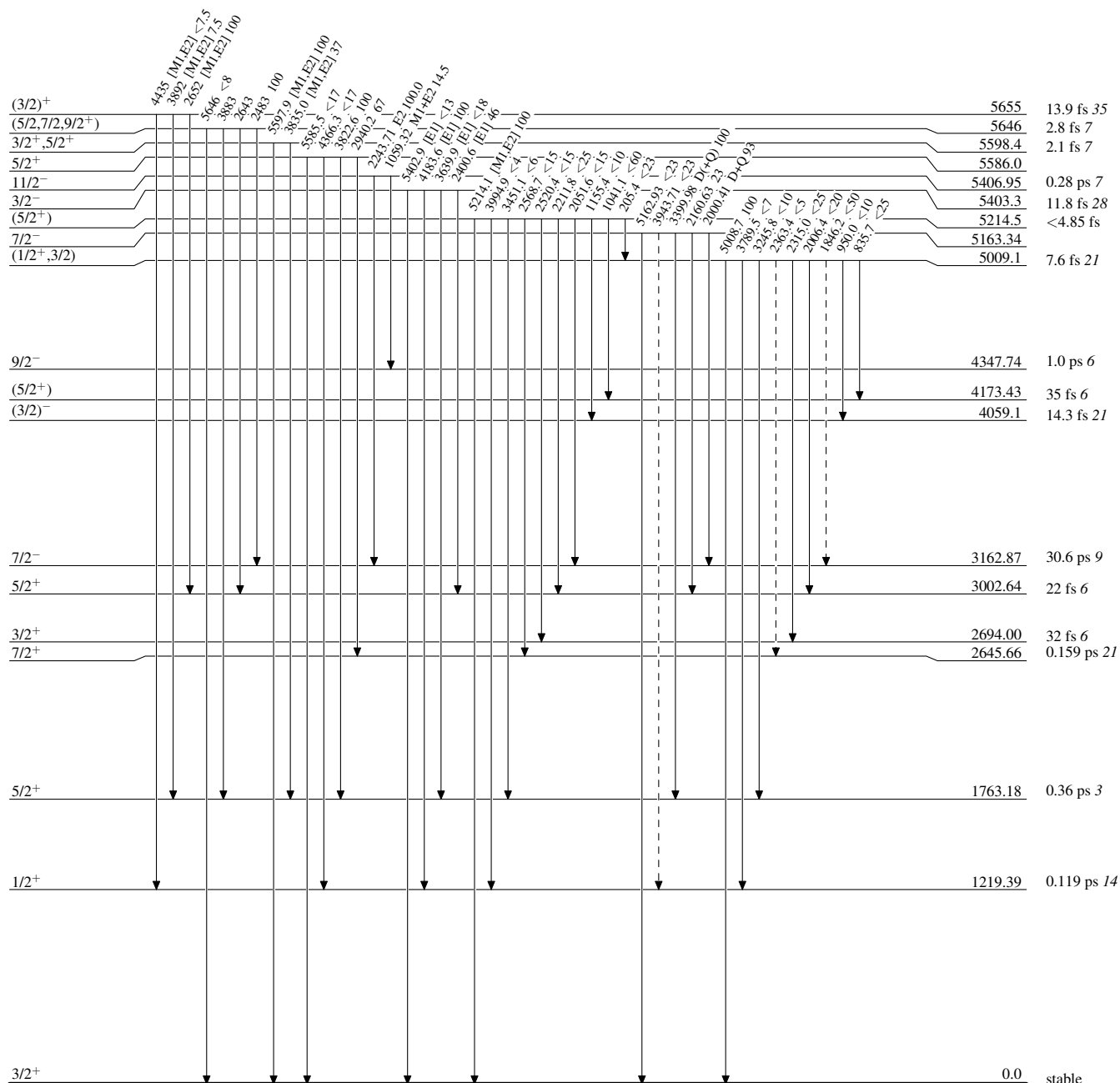


Adopted Levels, Gammas

Legend

Level Scheme (continued)

Intensities: Relative photon branching from each level

-----► γ Decay (Uncertain)


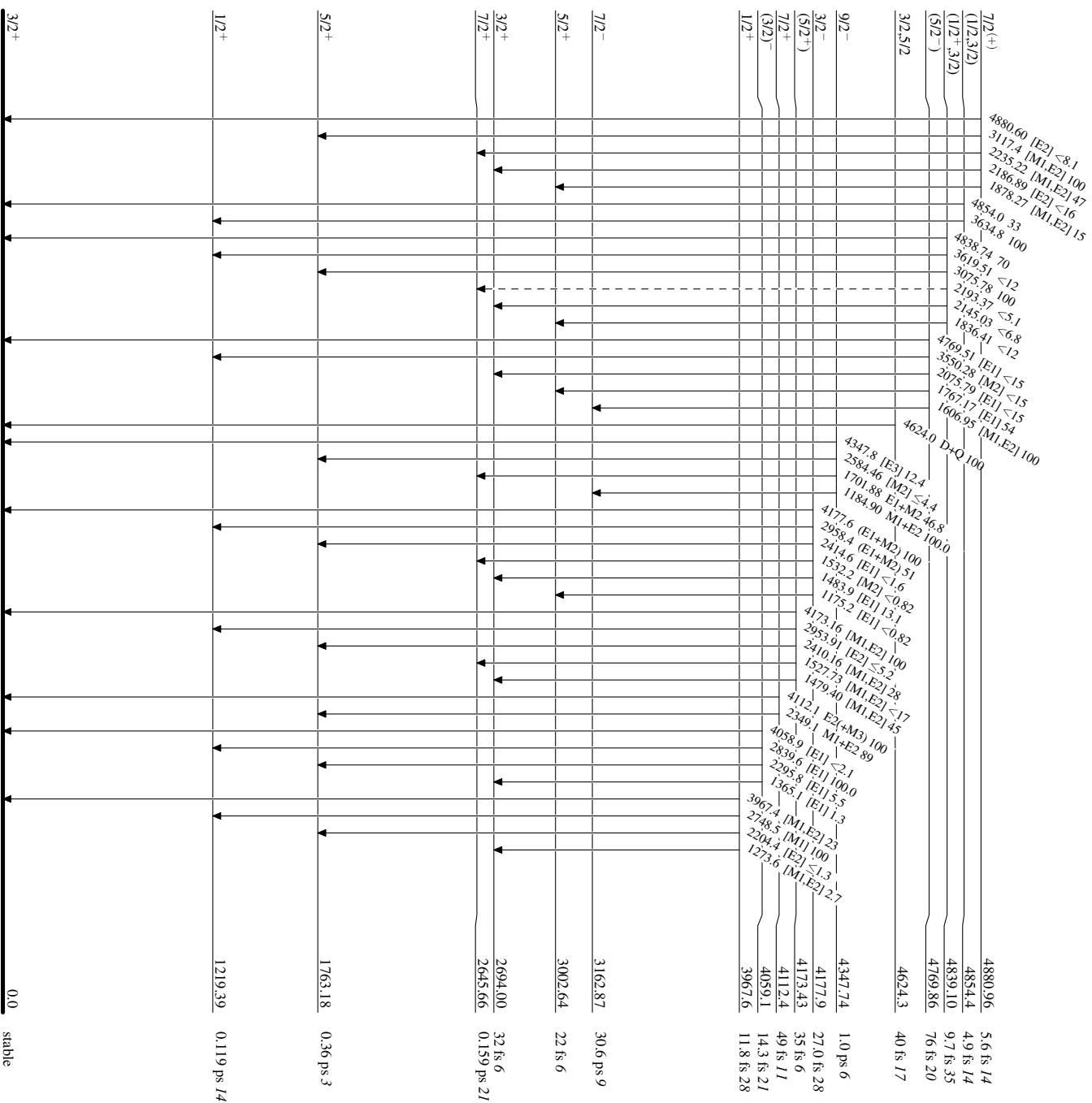
Adopted Levels, Gammas

Legend

Level Scheme (continued)

Intensities: Relative photon branching from each level

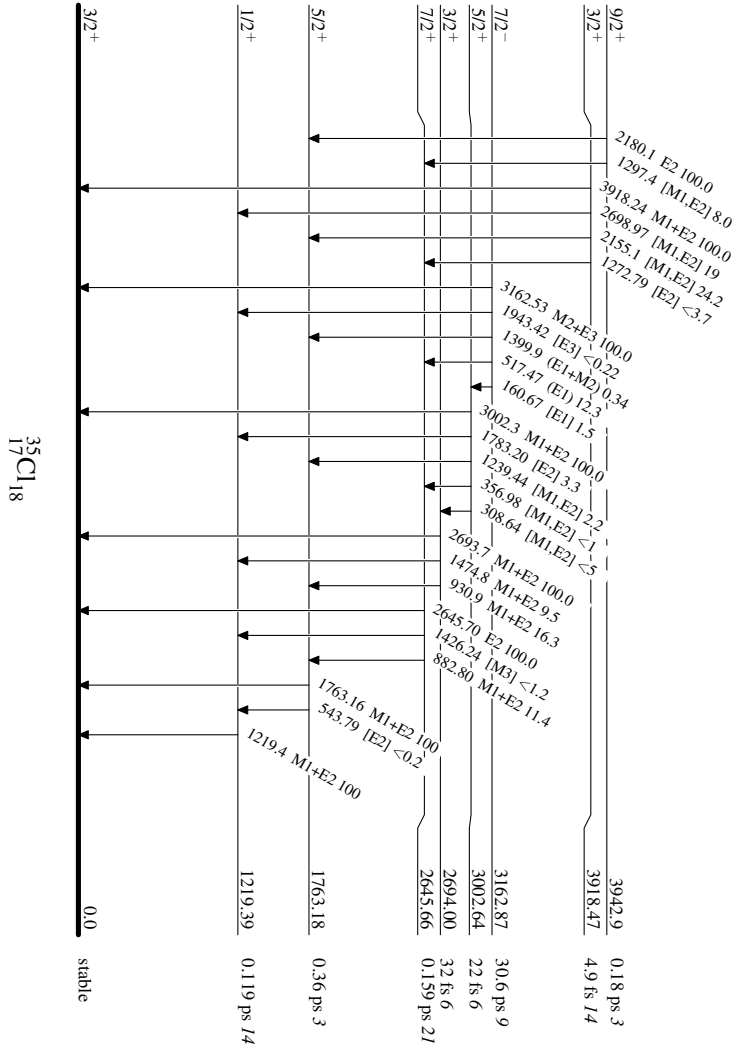
-----▶ γ Decay (Uncertain)



Adopted Levels, Gammas

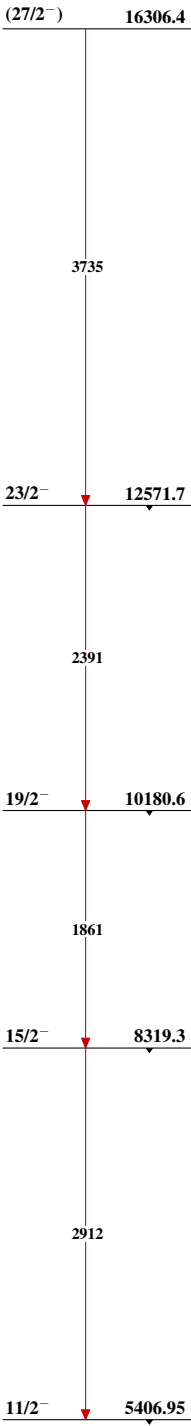
Level Scheme (continued)

Intensities: Relative photon branching from each level



Adopted Levels, Gammas

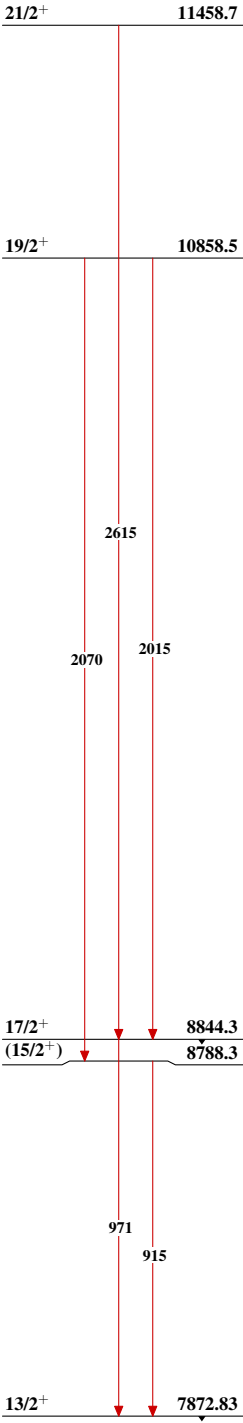
Band(A): Band based on f_{7/2}
orbital



³⁵₁₇Cl₁₈

Adopted Levels, Gammas (continued)

Band(B): Band based on 13/2⁺



³⁵₁₇Cl₁₈